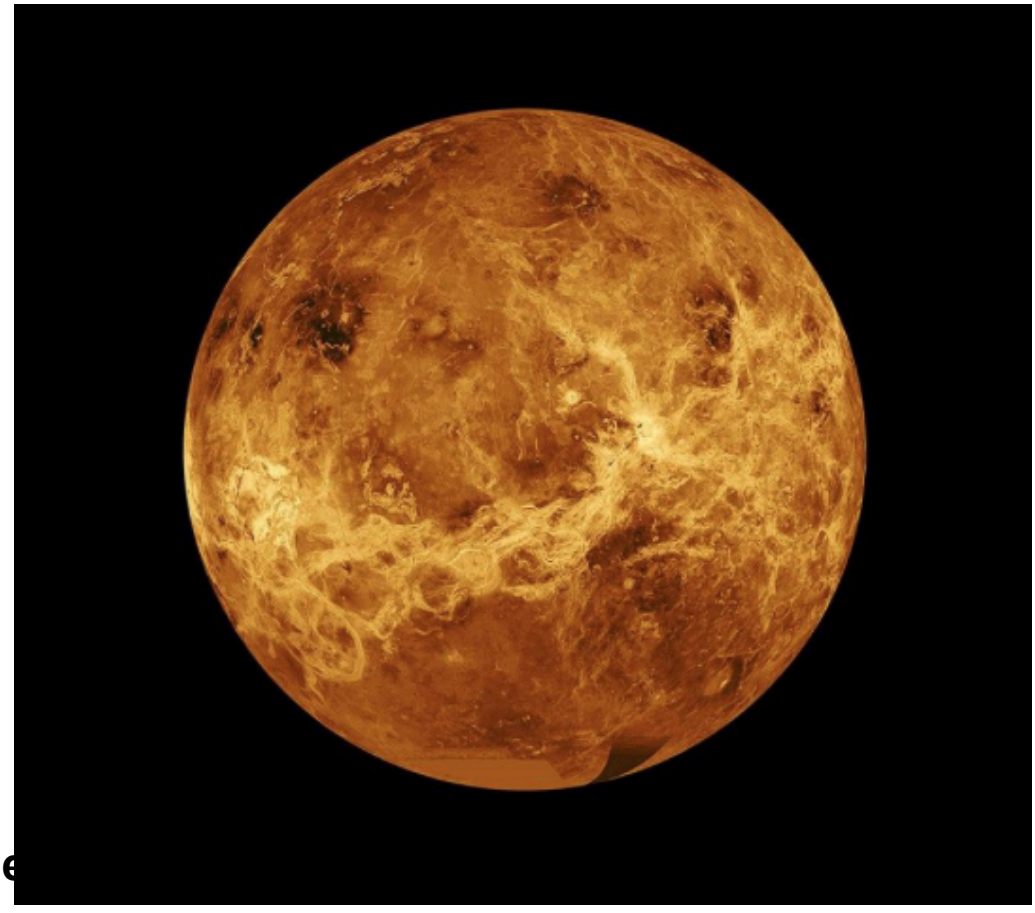
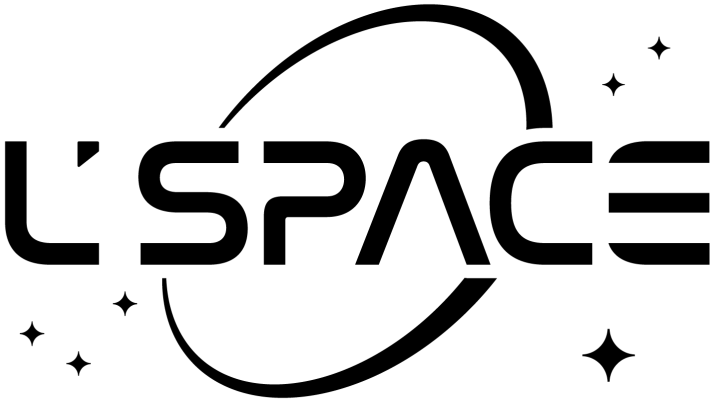
Preliminary Design Review (PDR)

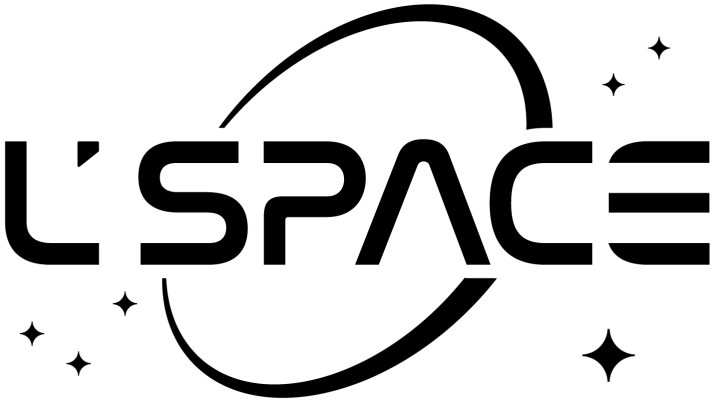
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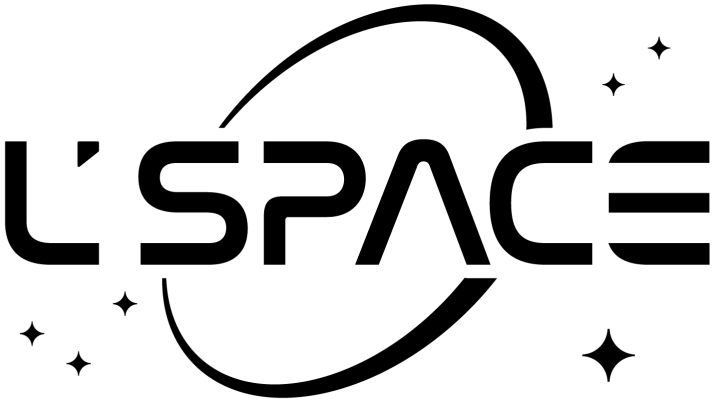
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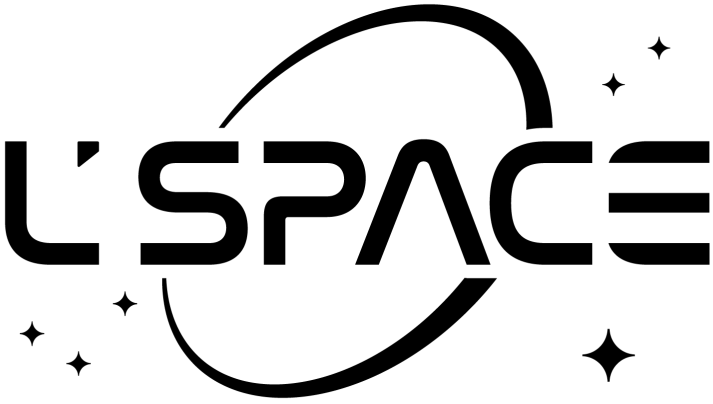
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Preliminary Design Review

1. Mission Overview

1.1. Mission Statement

The mission aims to deepen the understanding of how Venus' surface activity influences its atmosphere by exploring the complex interactions between the two. The team will achieve this by deploying an aerobot capable of collecting detailed data in the atmosphere and from high above the surface. These data include measurements of

trace gases like sulfur dioxide (SO_2), carbon monoxide (CO), and hydrogen chloride (HCl), heat signals that could indicate volcanic activity, and ultraviolet images that reveal cloud movement and structure.

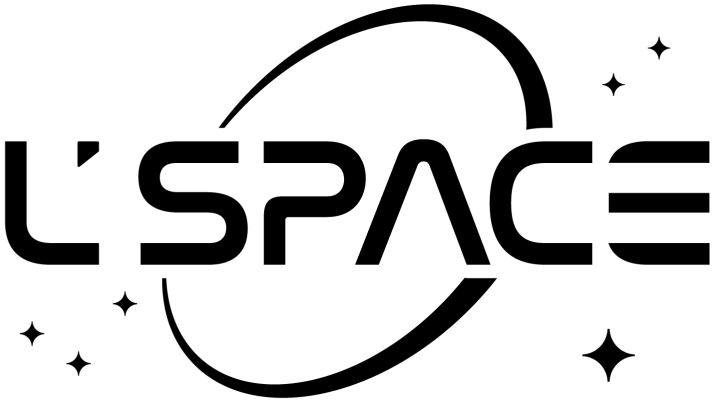
Although Venus is often referred to as Earth's "sister planet" due to its similar size and composition, its development took a drastically different path. Venus underwent intense volcanic resurfacing and developed a thick, toxic atmosphere and extremely high temperatures due to a runaway greenhouse effect. By identifying areas of active volcanism and studying how gases released from the surface change the atmosphere, the mission will help explain what drives the planet's current climate and whether it may have once been more Earth-like.

The aerobot's scientific tools will allow it to make direct measurements while flying through the atmosphere and to take detailed images of the surface from above. It will track changes in gas levels, like sulfur dioxide and carbon monoxide, and detect heat signals that may come from active volcanoes. By collecting this information over time and at different altitudes, the mission will reveal how Venus' surface and atmosphere interact and how its extreme climate has changed—or stayed the same—over time.

Ultimately, the mission's findings will help scientists better understand how rocky planets evolve and whether planets with similar conditions could still support life—or once did.

1.2. Science Traceability Matrix

The primary goal of this mission is to enhance our understanding of Venus' surface, interior, and atmosphere, focusing on geological activity, climate, and potential habitability. The mission is guided by two key objectives.



Objective One examines the interaction between Venus' surface and atmosphere through volcanic outgassing and chemical cycling. It aims to detect trace gases such as carbonyl sulfide (OCS), sulfur dioxide (SO_2), and hydrogen chloride (HCl), which indicate active volcanism. The aerobot will operate at altitudes of 30 to 50 kilometers, an area ideal for observing gas concentrations before they are altered by the upper atmosphere. A compact tunable laser spectrometer will be used for these measurements due to its high sensitivity and low power consumption.

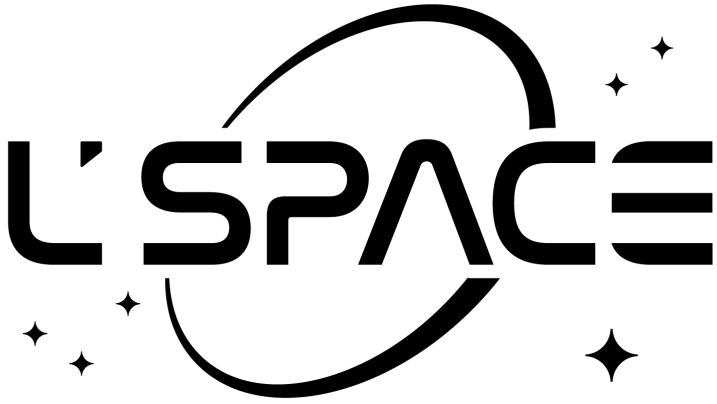
Objective Two investigates Venus' internal structure and how heat flow and crustal composition influence volcanic activity. The team will study pyroxene minerals in the crust and map thermal anomalies using a long-wave infrared (LWIR) imager to identify volcanic regions. This instrument is effective in detecting surface variations despite Venus' harsh conditions.

The selection of instruments was influenced by strict constraints on mass, volume, and power, with all sensors being miniaturized and proven to withstand high temperatures. Redundancy in critical measurements, such as overlapping sensors, will help mitigate the effects of potential instrument failures.

By targeting both atmospheric and surface processes, this mission will provide new insights into Venus' evolution and support future exploration of extreme rocky planets.

Science Goals	Science Objectives	Science Measurement Requirements		Instrument Performance Requirements		Predicted Instrument Performance	Instrument	Mission Requirements
		Physical Parameters	Observables					
Acquire new knowledge about Venus' surface, interior, and atmosphere to strengthen the scientific community's understanding geological activity, climate, and potential past habitability	Investigate the composition, dynamics, and chemical interactions within Venus' cloud layers	Define the percent concentration of Carbonyl Sulfide (OCS) in Venus' atmosphere at an altitude range of 30-50 km	Collect absorbance spectra and fluorescence peaks of OCS in the 185-300 nm range	Wavelength range:	185-300 nm (Ultra Violet)	180-320 nm	UV-Vis Spectrometer	The aerobot must maintain a stable altitude and orientation within 30-50 km of the Venusian atmosphere to ensure precise spectral measurements.
				Integration time:	1-10s	5s		
				Sensitivity:	10^{-6} to 10^{-9} mol fraction	10^{-7} to 10^{-9}		
				Spectral resolution:	≤ 1 nm	0.5-1.0 nm		The aerobot must shield instruments from Venus' harsh atmospheric conditions, including high temperatures, corrosive sulfuric acid clouds, and intense solar radiation.
	Determine the presence of ongoing volcanic activity and	Detect pyroxene minerals (clinopyroxene & orthopyroxene)	Detect FTIR absorbance spectra features of the pyroxene	Wavelength Range	8-12 μm	8-12 μm	Planetary Hyperspectral Thermal Emission	
				Temperature Sensitivity:	± 1 K	± 1 K		

	resurfacing events by detecting thermal anomalies and volcanic gas emissions.	in volcanic rocks using thermal infrared (TIR) spectroscopy	minerals in the 8-12µm range	Spectral Resolution:	≤ 10 cm^{-1}	≤ 10 cm^{-1}	Spectrometer (pHyTES)	The aerobot must generate sufficient power for continuous instrument operation and efficiently transmit collected data back to Earth.
				Spatial resolution:	≤ 1 km per pixel	500 m/pixel		



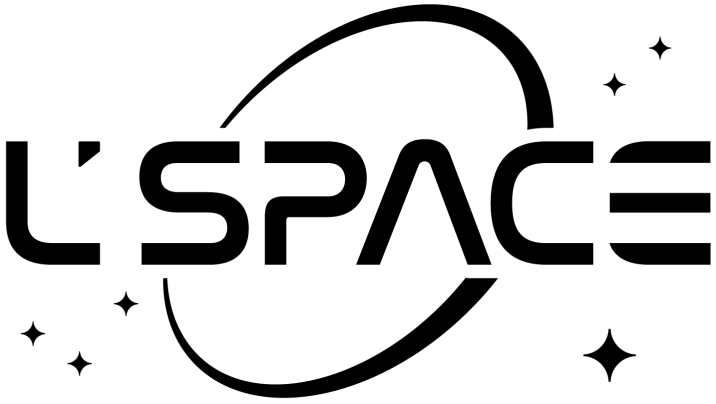
1.3. Summary of Mission Location

This mission focuses on exploring Venus, a planet of significant scientific interest due to its hostile surface conditions, dense atmosphere, and history of extreme volcanic activity. The objective is to investigate the interactions between Venus' surface and atmosphere by collecting high-resolution atmospheric and surface data using an aerobot. Understanding these interactions will provide crucial insights into planetary evolution, climate dynamics, and the potential for past habitability on rocky exoplanets with similar conditions.

The primary region of interest is the middle atmosphere of Venus, specifically the cloud layer located between 30 and 50 kilometers in altitude. This zone has been selected because it is where chemical exchange between the surface and atmosphere is most active and where the aerobot can safely operate without being damaged by the planet's extreme surface heat or pressure. The 30–50 km range features stable atmospheric dynamics and lower temperatures (approximately 50–70°C), creating a relatively hospitable environment for aerial exploration (Seiff et al., 1985).

One of the main targets in this region is carbonyl sulfide (OCS), a trace gas produced by volcanic outgassing. At these altitudes, OCS concentrations are expected to be more stable and measurable before being broken down in the upper atmosphere. Studying this "OCS sink" zone could help reconstruct Venus' volcanic history and provide insights into how volcanism shapes the planet's atmospheric chemistry (Yung et al., 2009).

The second key location of interest is the composition of Venus' surface, particularly in areas suspected of recent volcanic activity. Although direct surface operations are not feasible, the aerobot will collect data remotely using longwave infrared (LWIR) imaging and ultraviolet spectral analysis during its lowest altitude passes. These instruments will target volcanic regions such as Idunn Mons and Themis Regio, which have been identified in previous missions as geologically active. A key mineral of interest is olivine, which weathers quickly under Venusian conditions. The detection of unweathered

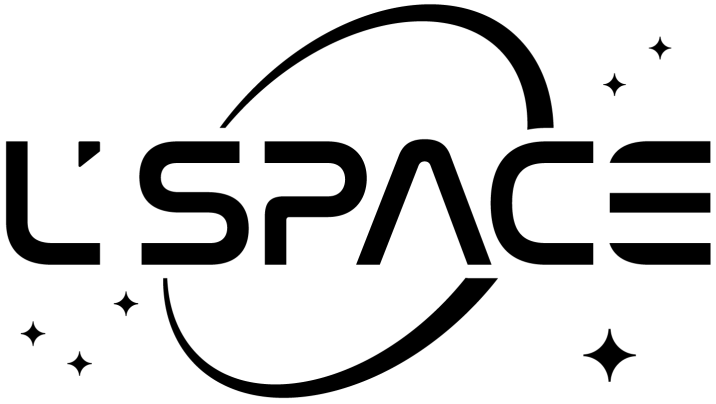


olivine would suggest recent lava flows and help determine magma composition and interior evolution (Filiberto et al., 2020).

By flying through different layers of the atmosphere and collecting data above volcanic features, the aerobot will provide a comprehensive dataset that links atmospheric gas signatures with specific surface sources. This multi-altitude, multi-target approach will facilitate the correlation between gas concentrations and surface activity, thereby supporting the mission's core objectives.This mission focuses on exploring Venus, a planet of significant scientific interest due to its hostile surface conditions, dense atmosphere, and history of extreme volcanic activity. The objective is to investigate the interactions between Venus' surface and atmosphere by collecting high-resolution atmospheric and surface data using an aerobot. Understanding these interactions will provide crucial insights into planetary evolution, climate dynamics, and the potential for past habitability on rocky exoplanets with similar conditions.

1.4. Mission Requirements

The development and implementation of the Aerobot spacecraft must align with a set of customer constraints, scientific objectives, and mission-critical requirements to ensure



its successful completion. These include specifications related to mass, dimensions, power sources, budget, and schedule, all designed to meet technical and operational standards while ensuring the system is ready for integration and launch.

One key requirement is the mass constraint, which stipulates that the Aerobot system must not exceed 50 kg. This is necessary for safe transport and to minimize power consumption. Compliance with this requirement will be verified through an inspection prior to launch from Cape Canaveral. Similarly, the spacecraft's dimensions must not exceed 1m x 1m x 1m in its stowed configuration. While the system can expand once deployed, it must remain within these limits before launch, as verified through inspection.

The use of Radioisotope Thermoelectric Generators (RTGs) or any derivative is strictly prohibited due to safety and regulatory concerns. Additionally, any radioactive material included in the spacecraft must not exceed a total mass of 5g. These restrictions will also be enforced through pre-launch inspection.

To maintain financial discipline, the total mission cost must not exceed $200 million. Funds will be carefully allocated to the most critical systems, with ongoing financial analysis to ensure budget compliance. Furthermore, the Aerobot must be fully integrated with other mission systems by October 1, 2028, allowing sufficient time for testing and validation. Integration will occur at the Goddard Space Flight Facility in Greenbelt, Maryland, and progress will be monitored through the project schedule.

A critical milestone for the mission is achieving Launch Readiness Date (LRD) by March 1, 2029. By this date, the Aerobot must be fully operational and prepared for deployment from Cape Canaveral, Florida. Compliance with this deadline will be assessed through schedule reviews and milestone verification.

In addition to customer-defined constraints, further system requirements have been established to ensure mission success. The Aerobot must be capable of maintaining

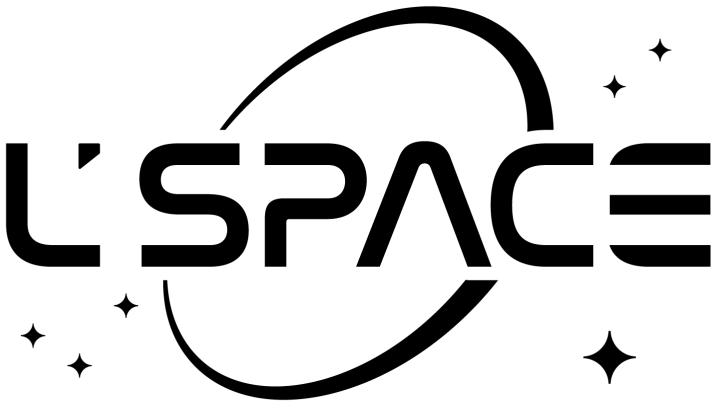
operational functionality for at least 48 hours under mission conditions. It must also demonstrate maneuverability within Venus' atmosphere at altitudes ranging from 70 km to 58 km above the surface, where environmental conditions present significant challenges.

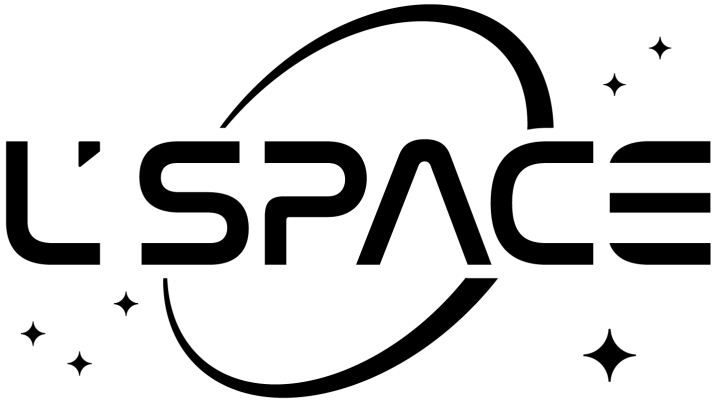
To meet these operational requirements, the system will be designed using lightweight materials and a compact configuration, in alignment with mass and volume constraints. Moreover, the Aerobot must be constructed with materials resistant to Venus' sulfuric acid and other chemically reactive atmospheric components to ensure long-term durability.

Each of these requirements will be verified through a combination of inspection, testing, demonstration, and analysis to confirm compliance before integration and launch. By adhering to these constraints, the Aerobot will be positioned for a successful mission while meeting all technical, financial, and regulatory requirements.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem
CS-1	Aerobot system shall not exceed 50 kg	Customer Constraint	N/A	SYS.3.0	Inspection	All
CS-2	Dimensions shall not exceed 1mx1mx1m	Customer Constraint	N/A	SYS.3.0	Inspection	All
CS-3	Shall not have a RTG or any derivative thereof	Customer Constraint	N/A	N/A	Inspection	All
CS-4	Shall not exceed a cost of $200 M	Customer Constraint	N/A	N/A	Inspection	All
CS-5	Ready for integration by October 1st, 2028	Customer Constraint	N/A	SYS.1.0, SYS.2.0, SYS.4.0	Inspection	All

CS-6	Must be ready for launch by March 1st, 2029	Customer Constraint	N/A	N/A	Inspection	All
SYS.1.0	The aerobot system must survive and maintain operating conditions for 48 hours	System Constraint	Customer.5	N/A	Demonstration	All
SYS.2.0	The aerobot system shall be able to maneuver throughout the Venusian surface at an altitude of 70-58 km above surface	System Constraint	Customer.5	N/A	Demonstration	All
SYS.3.0	The aerobot system shall use lightweight materials and compact design	Fit within the defined mass and volume constraints	Customer.1, Customer.2, Customer.4	N/A	Test	All
SYS.4.0	The aerobot system shall be resistant to sulfuric acid and other chemically reactive atmospheric components	Venus' clouds contain sulfuric acid, which can degrade materials	Customer.5, Customer.4	N/A	Demonstration	All


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1.5. Concept of Operations (ConOps)

The Aeris mission is designed to explore the atmosphere and surface of Venus using a balloon-deployed platform. The primary goal is to gather data on Venus' atmospheric composition, surface features, and volcanic activity potential. Due to the harsh environmental conditions on Venus, the mission will unfold in several operational phases.

The mission begins with the atmospheric entry phase, during which the spacecraft separates from its primary carrier and enters Venus' dense atmosphere. As descent occurs, a parachute will deploy to slow the fall and ensure a controlled release of the balloon-based aerobot. Upon reaching a stable altitude within the upper cloud deck, roughly 70–75 km in the stratosphere, the aerobot will begin its operational phase with system checkouts and initial stabilization.

Once stabilized, the aerobot will begin collecting ultraviolet and infrared spectral data to analyze the chemical makeup of Venus’ cloud layers. These initial measurements at high altitude will help characterize the stratospheric cloud composition, contributing to a deeper understanding of the planet’s upper atmosphere.

As the aerobot continues its descent into the upper troposphere, between 60 and 70 km, it will perform a detailed chemical analysis of cloud particles, focusing on sulfuric acid concentrations and trace gases. Simultaneously, temperature and pressure profiling will be conducted to improve models of atmospheric dynamics. During this phase, collected data will be transmitted in real-time to an orbiter for relay to Earth.

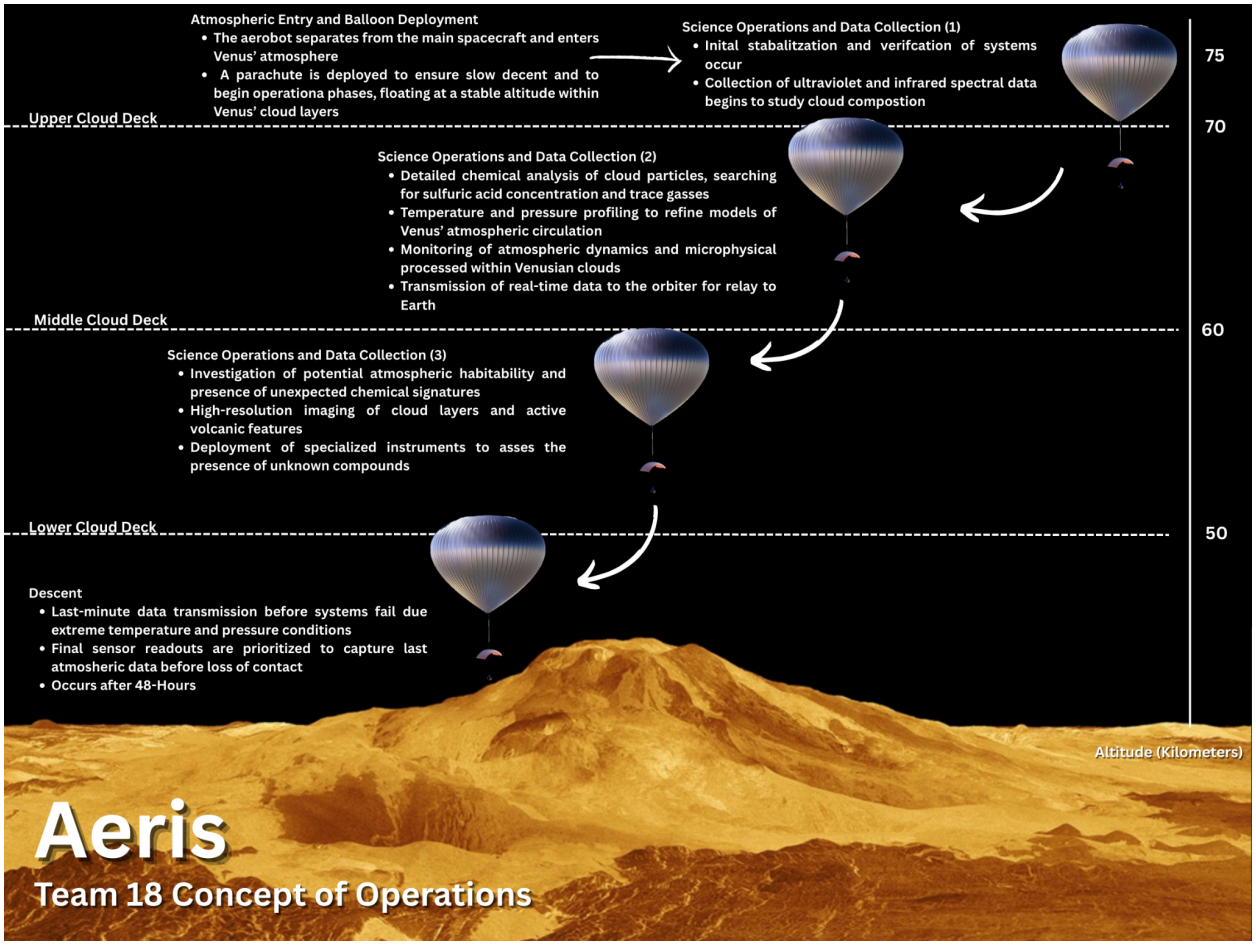
At altitudes between 50 and 60 km, within the lower troposphere, the aerobot will examine the potential for atmospheric habitability by searching for unexpected chemical signatures. High-resolution imaging systems will capture detailed views of the lower cloud layers and observe surface features visible through the clouds. Specialized

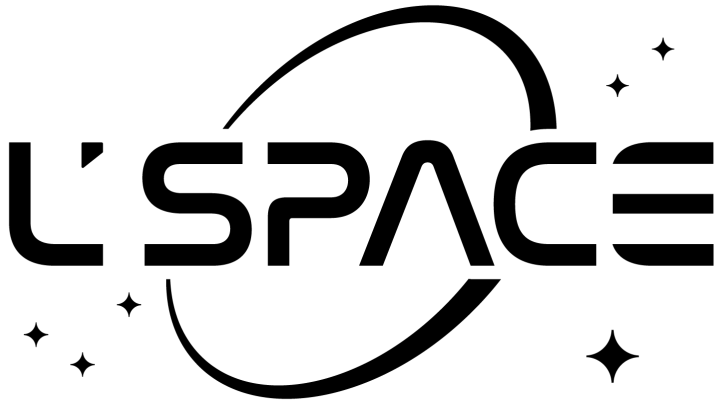
instruments will also monitor for unknown compounds that may indicate ongoing chemical or geological processes.

As the aerobot descends into the dense transition zone between 30 and 50 km, it will conduct targeted measurements of carbonyl sulfide (OCS), a gas linked to volcanic outgassing. This region is ideal for capturing OCS concentrations before the gas is destroyed in the upper atmosphere by photolysis. Measurements collected in this layer will provide critical insight into Venus' geologic activity and its impact on atmospheric chemistry.

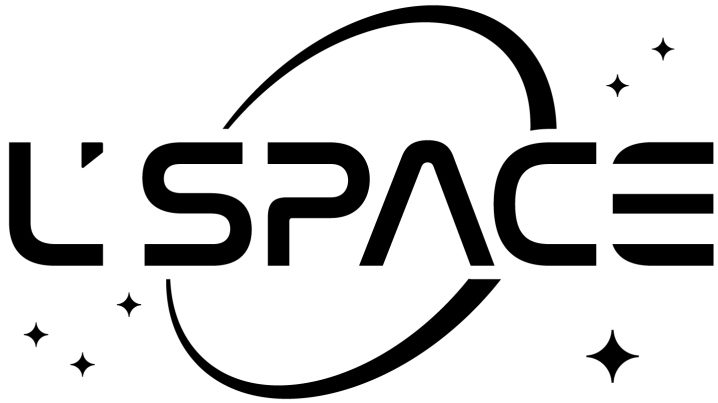
Below 30 km, the extreme environmental conditions—temperatures exceeding 450°C and high atmospheric pressure—will ultimately cause system failure. Instruments will begin shutting down sequentially, and final data packets will be transmitted before the probe becomes non-operational. This will mark the end of the mission's scientific operations.

Despite its limited lifetime, the Aeris mission is expected to significantly advance the understanding of Venus by characterizing atmospheric composition, detecting potential volcanic activity, and contributing to long-term planetary science and habitability studies. The phased descent strategy ensures targeted data collection within the narrow operational window at scientifically valuable altitudes.


Atmospheric Entry and Balloon Deployment
• The aerobot separates from the main spacecraft and enters Venus' atmosphere
• A parachute is deployed to ensure slow decent and to begin operationa phases, floating at a stable altitude within Venus' cloud layers
Science Operations and Data Collection (1)
• Inital stabalitzation and verifcation of systems occur
• Collection of ultraviolet and infrared spectral data begins to study cloud compostion
Upper Cloud Deck
Science Operations and Data Collection (2)
• Detailed chemical analysis of cloud particles, searching for sulfuric acid concentration and trace gasses
• Temperature and pressure profiling to refine models of Venus' atmospheric circulation
• Monitoring of atmospheric dynamics and microphysical processed within Venusian clouds
• Transmission of real-time data to the orbiter for relay to Earth
Middle Cloud Deck
Science Operations and Data Collection (3)
• Investigation of potential atmospheric habitability and presence of unexpected chemical signatures
• High-resolution imaging of cloud layers and active volcanic features
• Deployment of specialized instruments to asses the presence of unknown compounds
Lower Cloud Deck
Descent
• Last-minute data transmission before systems fail due extreme temperature and pressure conditions
• Final sensor readouts are prioritized to capture last atmosheric data before loss of contact
• Occurs after 48-Hours
75
70
60
50
Altitude (Kilometers)
Aeris
Team 18 Concept of Operations


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1.6. Vehicle Design Summary


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1.7. Science Instrumentation Summary

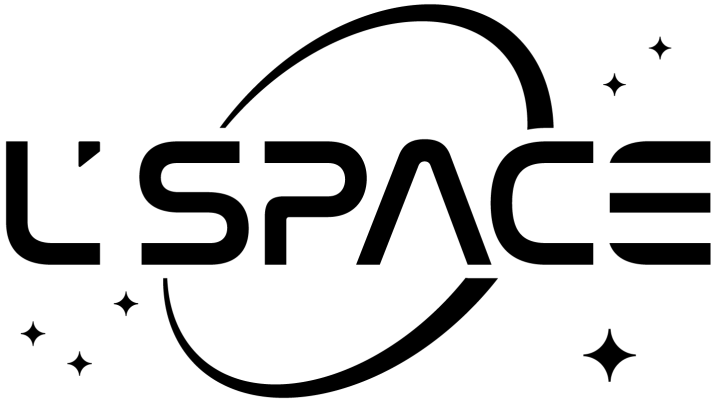
The Venus Aerobot payload includes two key science instruments designed to address core objectives related to atmospheric composition and surface mineralogy: the UV-Visible (UV-Vis) Spectrometer and the Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES). Together, these instruments enable high-resolution remote sensing of Venus's atmosphere and surface, offering valuable data to improve understanding of the planet's climate history, geological activity, and potential habitability.

UV-Visible (UV-Vis) Spectrometer
The UV-Vis Spectrometer is designed to detect electromagnetic radiation in the 200–500 nm range, covering ultraviolet and visible wavelengths. It operates by collecting sunlight scattered or absorbed by Venus's atmosphere and measuring the resulting spectral intensity across its detection band. The instrument identifies specific absorption or fluorescence peaks associated with chemical species such as sulfur dioxide (SO_2), carbonyl sulfide (OCS), and other trace gases critical to understanding Venus's climate dynamics and photochemical processes.

This instrument directly supports the science objective of analyzing atmospheric composition and structure, offering insights into how gases interact with solar radiation and possibly identifying signs of active volcanism or chemical disequilibrium. With a spectral resolution of 0.1 nm, the UV-Vis Spectrometer provides the precision needed to detect narrow spectral features of minor constituents, helping researchers characterize the upper and middle atmospheric layers.

Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES)
pHyTES is a thermal infrared spectrometer that operates in the 8–12 μm range, optimized for detecting thermal emission features associated with silicate minerals. It functions by measuring the emitted infrared radiation from the Venusian surface as the aerobot floats at varying altitudes. These emissions are influenced by surface composition and temperature, allowing pHyTES to infer the presence of minerals such as pyroxene or feldspar that may indicate volcanic activity or resurfacing events.

With a spectral resolution of 10 cm^{-1}, pHyTES enables the mapping of subtle differences in surface emissivity, helping to determine whether certain regions exhibit spectral signatures



consistent with recent lava flows or geologic uplift. This directly contributes to the science objective of characterizing Venus's geologic activity and surface-atmosphere interactions.

Both instruments are mounted on a stabilized platform to maintain a pointing accuracy of ±0.5°, ensuring reliable and repeatable measurements during flight. The UV-Vis Spectrometer primarily looks outward toward the atmosphere and solar zenith angle, while pHyTES is oriented toward the nadir to capture surface emissions.

By operating concurrently, the instruments provide multi-layered observations of Venus's environment—one from the top-down (atmosphere) and the other from the bottom-up (surface). This complementary data set enhances our ability to correlate atmospheric changes with surface processes, a key aspect of understanding Venus as a dynamic planetary system.

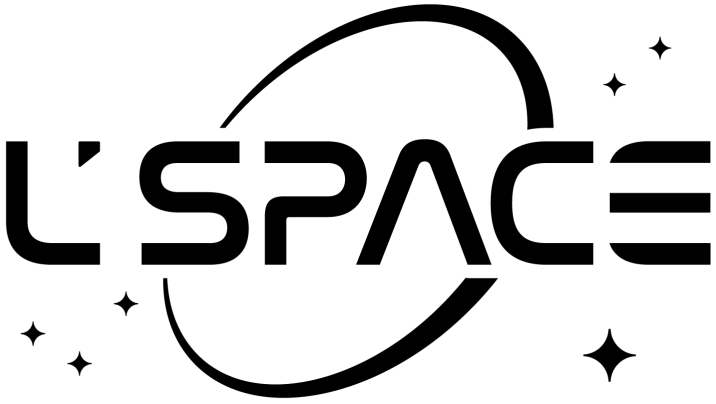
1.8. Programmatic Summary

1.8.1. Team Introduction

Perry Patterson, Team 18's Project Manager, is studying at Oklahoma City Community College in Oklahoma. As a Project Manager, Perry oversees the planning and execution of the mission as well as the coordination between the different disciplines. Perry has relevant experience in communication and education through teaching.

Keioko Haynes, Team 18's Deputy Project Manager of Resources, is studying at University of Illinois at Chicago. As a Deputy Project Manager of Resources, Keioko is in charge of ensuring that the mission has the proper budget, personnel, and equipment. Keioko has relevant experience through serving as the secretary and interim treasurer for UIC's AIAA club.

Celso Cervantes, Team 18's Lead Systems Engineer, is studying at University of Wisconsin Madison, in Wisconsin. As a Lead Systems Engineer, Celso is in charge of



ensuring that all subsystems work together flawlessly. Celso has relevant experience through his involvement in organizations like SHPE, ALPFA, and Diverse Leaders of Tomorrow. Through these organizations Celso has learned to collaborate with diverse groups, communicate ideas clearly, and lead with empathy.

Sydney Schultz, Team 18's Chief Scientist, is studying at Iowa State University. As a Chief Scientist Sydney ensures that the mission's research goals are prioritized. Sydney has relevant experience through being the operations lead for her cubesat team.

Palash Pawar, Team 18's Program Analyst, is studying at University of Texas at Austin. As a Program Analyst, Palash analyzes budgets, schedules, risks, and program performance. Palash has relevant experience through data analysis and technical writing skills.

Roufieda Nabil, Team 18's Scientist 1, is studying at the University of Texas at Tyler. As a scientist, Roufieda is in charge of mission planning and data analysis. Roufiedas relevant experience includes having a strong work ethic, as well as technical skills.

Nevin Shiju, Team 18's Mechanical Engineer, is studying at the University of Texas at Dallas. As a Mechanical Engineer Nevin is in charge of spacecraft design and operations. Nevins relevant experience includes a strong foundation in system logic, data flow, and hardware-software interaction.

Jonathan Pena, Team 18's Mechanical Engineer, is studying at Texas A&M University. As a mechanical engineer Johnathan contributes to the designing, testing, and launch phases of the mission. Jonathan has relevant experience through taking mechanical design classes.

Milton Rosa, Team 18's Mission Assurance Specialist, is studying at University of Houston, in Texas. As a Mission Assurance Specialist, Milton ensures that each and

every component and decision meets that standard of safety and reliability for mission success. Milton has relevant experience through completed coursework in thermal systems, electronics, and systems engineering which provided a strong foundation for analyzing risks in extreme environments like Venus, Milton also has hands-on experience with cybersecurity and penetration testing, using tools such as Python and Nmap to analyze system vulnerabilities.

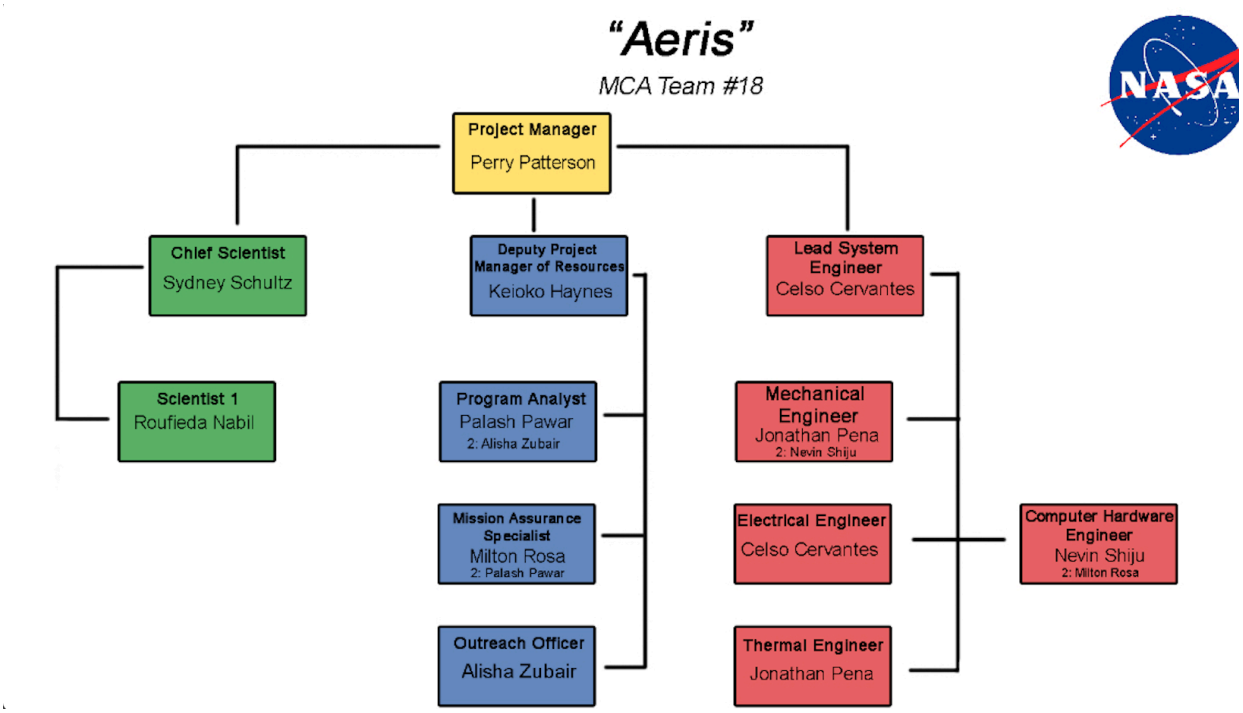
Alisha Zubair, Team 18’s Outreach Officer, is studying at Binghamton University in New York. As an Outreach Officer, Alisha is in charge of interacting with the public, educators, and students to increase awareness about the mission's goals and discoveries. Alisha has relevant experience through working on multiple campaigning teams for various organizations such as Luv Micheal and Leukemia and Lymphoma Society.

1.8.2. Team Management Overview

Team organization is still split up between Programmatics, Science and Engineering disciplines. The team handles delegation of the workload by collaborating with each discipline. Regarding time sensitive decisions, members of Programmatic, Science, and Engineering disciplines work together to make decisions. No changes have been made to this style. Technical and scientific decisions are made in a timely manner and handled by specific disciplines.

There are weekly team meetings on Saturdays, where members can voice their concerns, ask important questions about the deliverables, and stay up to date. Weekly team meetings have helped keep the team organized and professional. Most conflicts have been resolved through following this structure. If a meeting is missed by a member, there are meeting notes that are provided so that everyone is given the same information.

The team has shown a good amount of collaboration and communication despite varying time zones. Issues regarding certain disciplines have been resolved smoothly by referring to our team leaders. This has led to the timely completion of our deliverables and tasks.



1.8.3. Schedule Overview

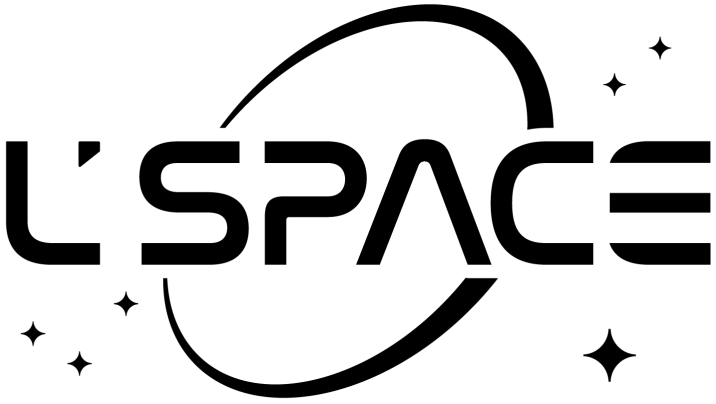
To clarify the structure and timeline of the mission, the Major Milestone Chart below outlines all key life-cycle reviews, Standing Review Board (SRB) events, customer-imposed deadlines, and Key Decision Points (KDPs). Milestones are

presented in chronological order to support a logical and efficient progression through each phase.

SRB milestones are particularly critical and are required for the Critical Design Review (CDR), Production Readiness Review (PRR), and Operations and Readiness Review (ORR). The SRB plays an advisory role, with the overarching goal of supporting the successful execution of the Agency's mission (NASA, 2023).

Phase	Milestone
C	Critical Design Review (CDR)
C	Standing Review Board
C	Production Readiness Review (PRR)
C	Standing Review Board
C	System Integration Review (SIR)
C	Standing Review Board
Intermission	Key Date Point D
D	System Acceptance Review (SAR)
D	**System Integration Date (Oct 1st, 2028)**
D	Operation and Readiness Review (ORR)
D	Standing Review Board
D	Mission Readiness Review (MRR)
D	**Launch (March 1st, 2029)**

Intermission	Key Date Point E
E	Critical Events Readiness Review (CERR)
E	Decommissioning Review (DR)
E	End of Mission
Intermission	Key Date Point F
F	Disposal Readiness Review (DRR)
F	Closeout

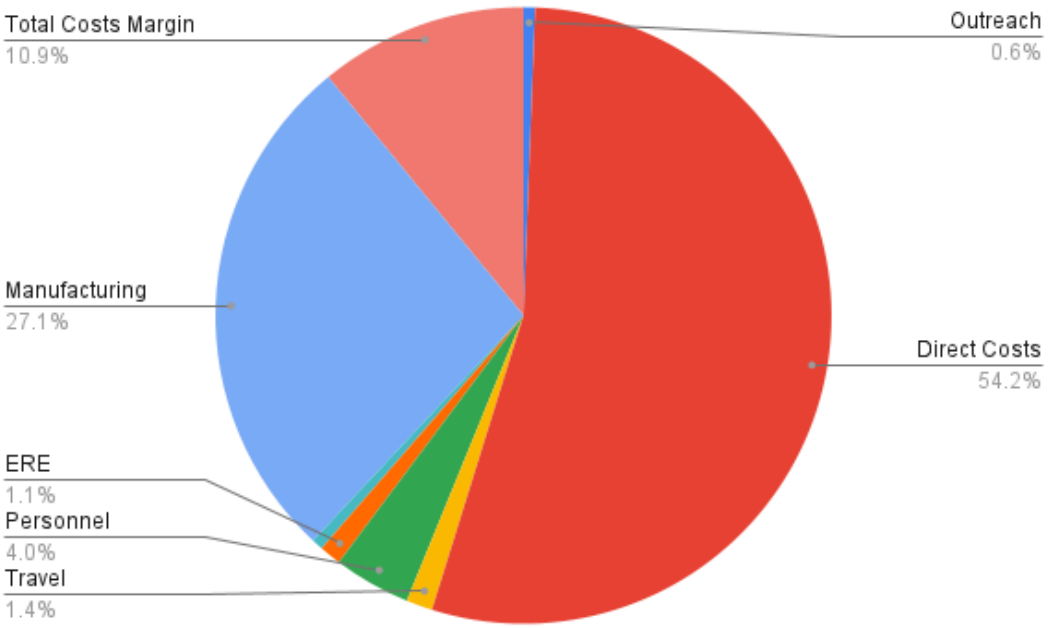


1.8.4. Cost Overview

As defined as a customer constraint, the total mission cost shall not exceed $200 million. The estimated total cost for Team Aeris' mission is $199,744,240, incorporating personnel expenses, direct costs, travel, and outreach. A margin of 5-15%, as suggested by the stakeholders, has been applied to account for uncertainties and potential cost fluctuations, ensuring financial flexibility throughout the mission lifecycle.

Factoring in the appropriate margin, Personnel costs constitute $23,442,447, covering salaries, benefits, and contractor fees essential for technical and administrative support. Direct costs, estimated at $164,381,906 using a method of analogy with NASA's DAVINCI Mission, encompass manufacturing procurement and testing resources necessary for the mission execution. Travel expenses, counted at $341,121, include conference expenses, KDP meetings, and integration and launch site visits. Lastly, Outreach efforts, allocated at $1,389,298, on educational engagement, public communications, and stakeholder involvement to maximize the mission's impact. The fiscal year 2025 will be used rather than the respective calendar years for each phase.

The Cost Overview structure proposes cost efficiency while maintaining an operational effectiveness, with margins for mitigation. A more detailed graphic listing the cost breakdown of the budget will be provided in the next deliverable, the Preliminary Design Review (PDR). Additionally, a Rough Budget Estimate is located in the Appendix for further, generalized clarification of the budget breakdown.


Total Costs Margin
10.9%
Outreach
0.6%
Manufacturing
27.1%
Direct Costs
54.2%
ERE
1.1%
Personnel
4.0%
Travel
1.4%

2. Overall Vehicle and System Design

2.1. Spacecraft Overview

Spacecraft Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem
CC.1	Aerobot system shall not exceed 50 kg	Customer Constraint		SYS.3.0	Inspection	All
CC.2	Dimensions shall not exceed 1mx1mx1m	Customer Constraint		SYS.3.0	Inspection	All
CC.3	Shall not have a RTG or any derivative thereof	Customer Constraint			Inspection	All
CC.4	Shall not exceed a cost of $200 M	Customer Constraint			Inspection	All
CC.5	Ready for integration by October 1st, 2028	Customer Constraint		SYS.1.0 , SYS.2.0 , SYS.4.0	Inspection	All
CC.6	Must be ready for launch by March 1st, 2029	Customer Constraint			Inspection	All
CC.7	Radioactive material should not exceed a mass of 5 g	Customer Constraint			Inspection	All
SYS.1.0	The aerobot system must survive and maintain operating conditions for 48 hours	System Constraint	CC.5		Demonstration	ME, Power
SYS.2.0	The aerobot system shall be able to maneuver throughout the Venusian surface at an altitude of	System Constraint	CC.5		Demonstration	ME, Power

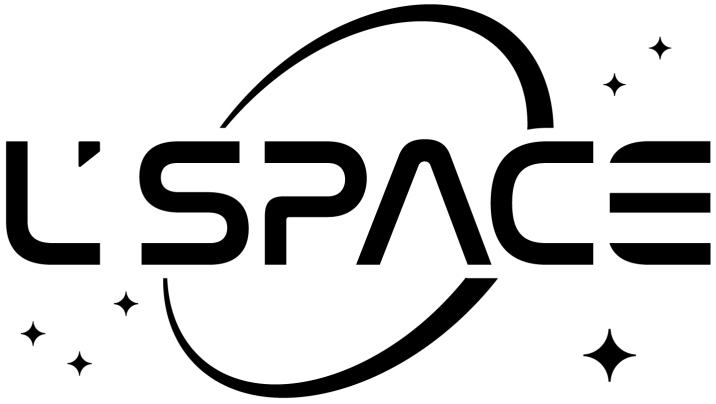
	70-58 km above surface					
SYS.3.0	The aerobot system shall use lightweight materials and compact design	Fit within the defined mass and volume constraints	CC.1, CC.2, CC.4		Test	All
SYS.4.0	The aerobot system shall be resistant to sulfuric acid and other chemically reactive/radioactive atmospheric components	Venus' clouds contain sulfuric acid, which can degrade materials	CC.5, CC.7		Test	All
SYS.5.0	The aerobot system					All

The spacecraft is a Venus aerobot—a balloon-based aerial vehicle designed to navigate the planet's upper and mid-atmosphere. It enables extended-duration science observations while floating at varying altitudes. The vehicle is optimized for mass, power, and thermal constraints unique to the Venusian environment.

2.1.1. Mechanical Subsystem Overview

The mechanical subsystem serves as the backbone of the entire mission architecture, providing structural support, mechanical stability, and environmental protection for all critical payloads and components. It is responsible for housing and integrating the payload systems, including the science instrumentation, communications hardware, and power units, while ensuring that these systems remain secure and operational throughout the mission's duration. Every component of the mechanical subsystem was developed with careful consideration of the extreme environmental conditions expected on Venus, including temperature fluctuations that can exceed 200°C, high atmospheric pressure, and corrosive chemical exposure.

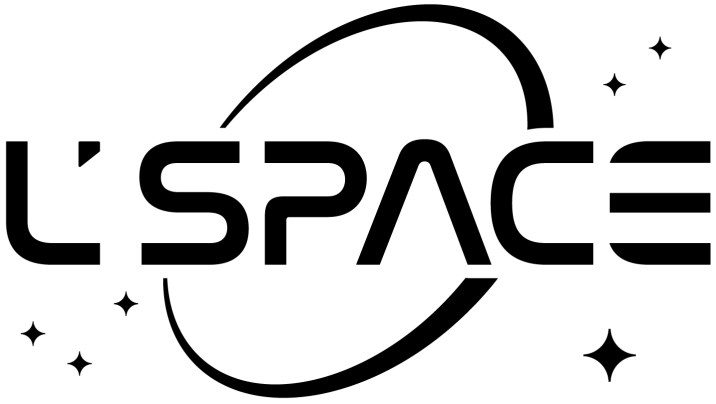
To meet these challenges, the mechanical design was tailored to comply not only with customer-provided requirements, but also with derived constraints established through detailed



systems engineering analysis. The choice of materials was a crucial aspect of the subsystem's development. High-temperature-resistant alloys and corrosion-resistant coatings were selected to withstand Venus' harsh climate. Structural members were designed with thermal expansion and stress tolerances in mind to prevent warping or failure during the mission.

The primary subassemblies of the mechanical subsystem include the chassis, Descent Subsystem (DSS) interface, and Guidance, Navigation, and Control (GNC) structural mount. The chassis forms the main load-bearing structure, distributing mechanical loads evenly across the body while supporting internal payloads. The DSS interface ensures that the contractors system and the NASA made systems can function together. Meanwhile, the GNC sub assembly provides sensors and controllers, enabling accurate data collection and vehicle control.

The entire mechanical subsystem was designed not only for strength and reliability, but also with mass-efficiency and modularity in mind, ensuring it fits within launch constraints while allowing for ease of assembly, maintenance, and testing. In essence, the mechanical subsystem integrates form and function into a resilient platform that supports the full range of mission operations. Its capability to endure Venus' extreme environment while supporting delicate instrumentation is critical to the success of the mission.



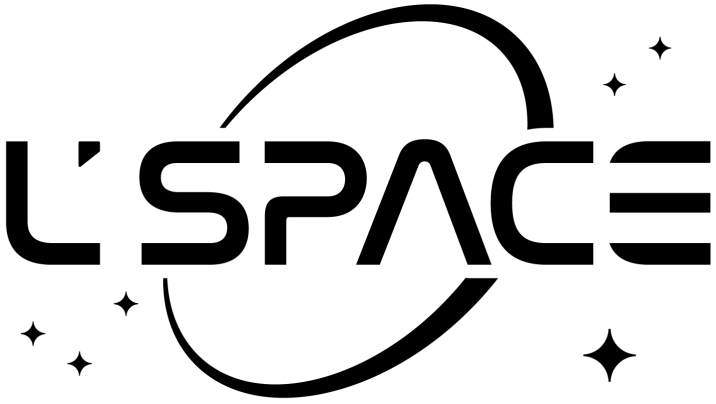
2.1.1.1 Mechanical Subsystem Requirements

The requirements for the mechanical subsystem are listed below. These requirements were derived from the system and customer requirements that this subsystem must comply with.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
SYS 1.0	The aerobot system must survive and maintain operating conditions for 48 hours	System Constraint	CC 5	ME 1.0	Test	Met
SYS 2.0	The aerobot system shall be able to maneuver throughout the Venusian surface at an altitude of 70-58 km above surface	System Constraint	CC 5	N/A	Demonstration	Met
SYS 3.0	The aerobot system shall use lightweight materials and compact design	Customer Constraint	CC 1, CC 2	ME 2.0	Inspection	Met
SYS 4.0	The aerobot system shall be resistant to sulfuric acid and other chemically reactive atmospheric components	The environment on Venus could damage the aerobot	N/A	ME 2.0	Test	Met
ME 1.0	The aerobot shall have a central structure to house the components and payload	Must store all components safely for the duration of the mission	SYS 1.0	ME 1.1, ME 1.2	Inspection	Met
ME 2.0	The aerobot shall have a guidance and navigation system to control and estimate location	The aerobot must know where it is to acquire the correct data	SYS 2.0	ME 2.1	Inspection	Met
ME 3.0	The Mechanical Subsystem shall be comprised of heat and corrosion resistant materials	Venus rains sulfuric acid, which corrodes most metals	SYS 4.0	N/A	Test	Met
ME 1.1	The Mechanical Subsystem shall have a chassis that maintains stability in strong winds	Venus has strong winds that could disorient the hardware and payload	ME 1.0	N/A	Analysis	Met
ME 1.2	The Mechanical Subsystem shall interface with all other subsystems	It shall provide the skeleton, and not interfere with other subsystems	ME 1.0	N/A	Analysis	Met

ME 2.1	The Guidance and Navigation system shall be precise, and estimate the altitude with 5% error	The altitude estimation must be really precise in order for the science instruments to collect good data	ME 2.0	N/A	Analysis	Met

Table 2.1.1.1 A



2.1.1.2. Mechanical Sub-Assembly Overview

The mechanical subsystem consists of two main subassemblies: the chassis and the guidance, navigation, and control (GNC) mount. It includes three key components: the chassis structure, the Inertial Measurement Unit (IMU), and the radar altimeter. These elements work together to provide structure, mechanical stability, and navigation data throughout the mission. Each subassembly was designed with Venus' extreme environmental constraints in mind, including high temperatures, dense atmosphere, and corrosive clouds.

For the chassis subassembly, the design was directly influenced by system constraints and component interfaces. The overall form is inspired by the Venus Express Aerobot configuration used by the European Space Agency (ESA). The chassis takes a box-like shape with specific portions removed to reduce mass while still maintaining strength and functionality. This includes carefully placed mass-reduction holes in the upper surface. These holes were designed in a way that preserves the structural integrity around the Descent Subsystem (DSS) mounting area, ensuring that the interface remains rigid and secure. The DSS interface follows an 8-bolt pattern with exact spacing constraints outlined in the Interface Control Document (ICD) provided by L'SPACE. As a result, the top face of the chassis was constrained to a 1m x 1m surface, which drove the overall geometry.

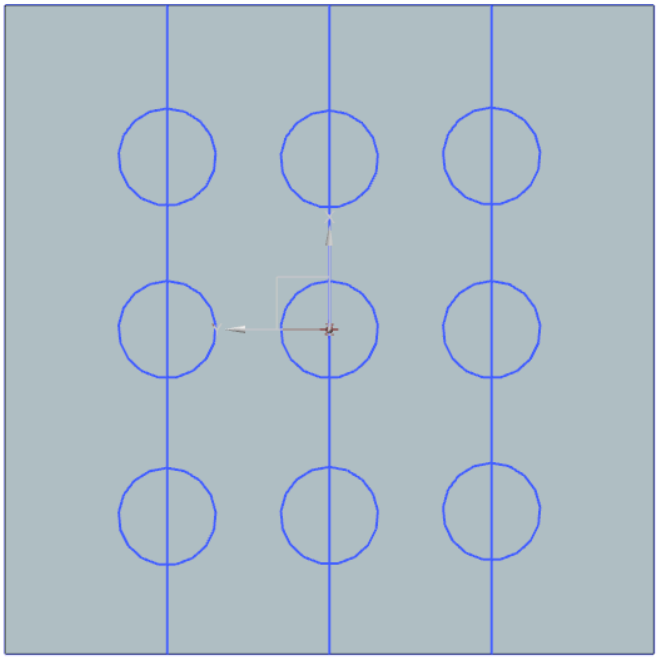


Figure 2.1.1.2 A

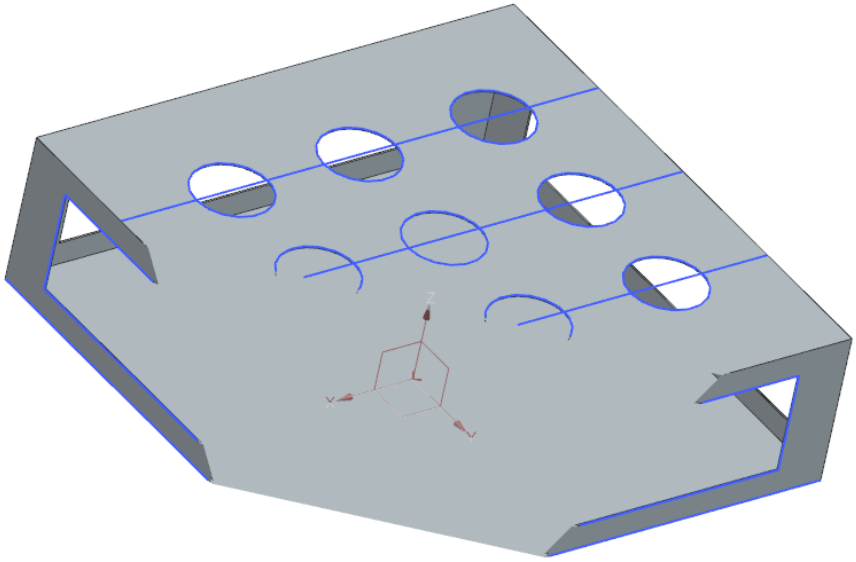


Figure 2.1.1.2 B

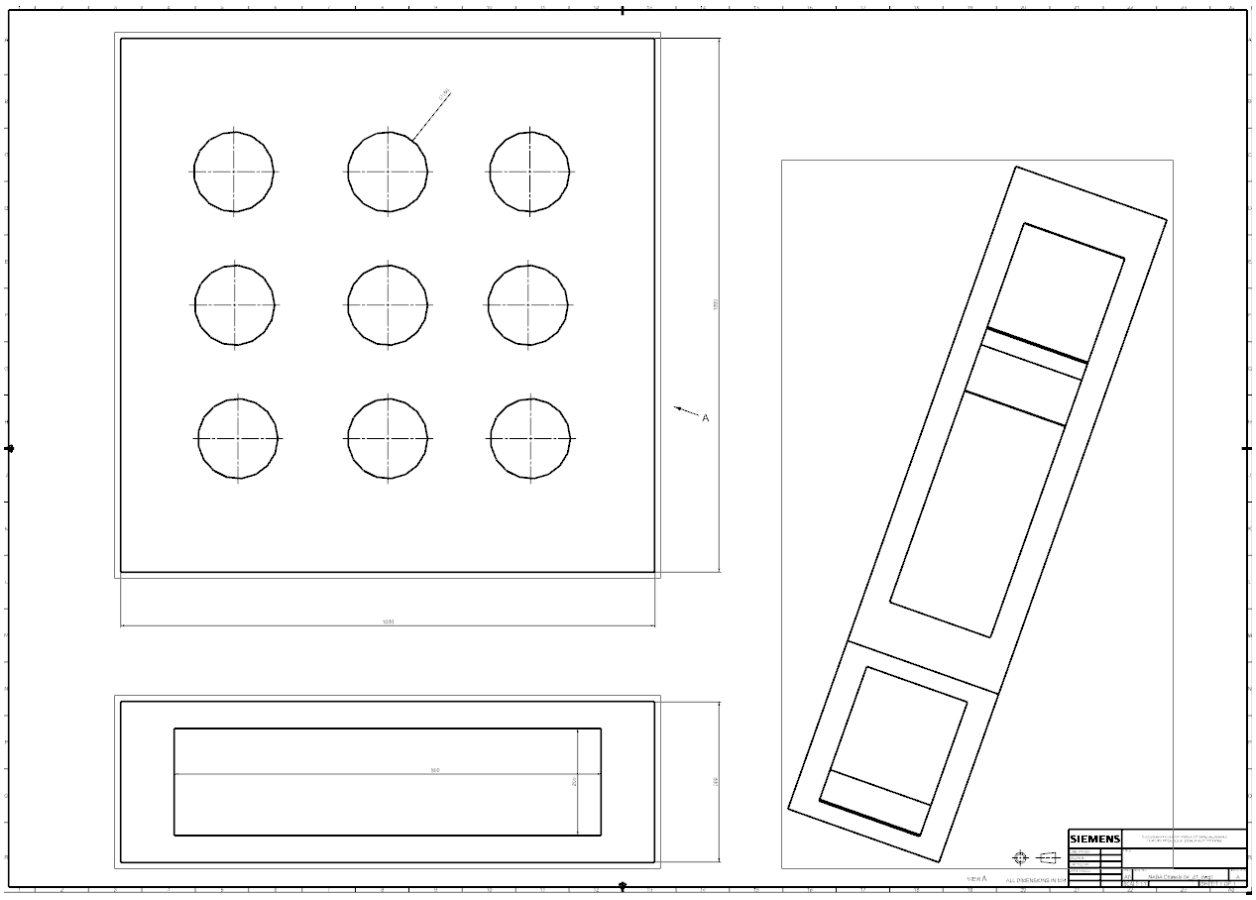
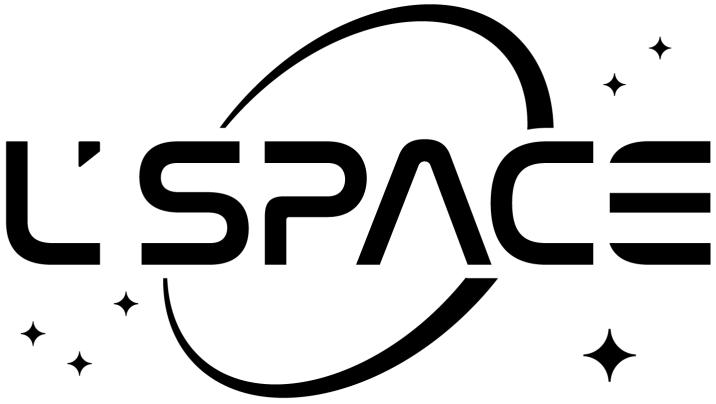


Figure 2.1.1.2 C

Narrative: The figures shown above show the preliminary design of the chassis, with the open areas exposing the inner housed components. This is simply to house the payload and provide structure to the Aerobot. The dimensions for the length, width, and height are given. The dimensions for the holes to reduce mass are also given. The thickness of the structure, and length of the openings are specified.

The material used for the chassis is Ti-6Al-4V, a titanium alloy containing 6% aluminum and 4% vanadium. This alloy is well-known for its use in aerospace applications and has flown on NASA



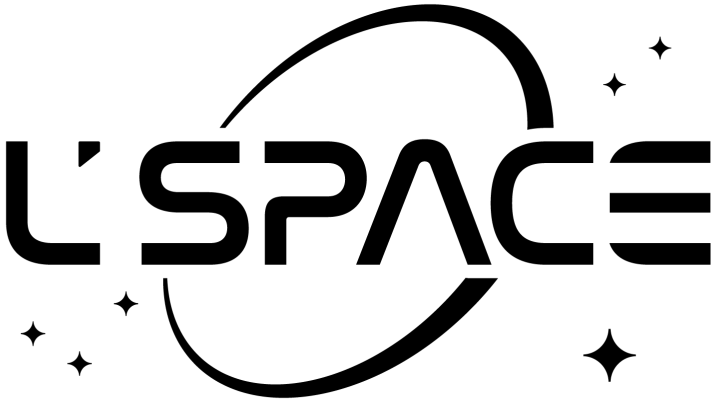
missions such as the Parker Solar Probe. It was chosen for its high strength-to-weight ratio, resistance to corrosion, and thermal tolerance up to 800K. The aluminum improves corrosion resistance, which is essential in Venus' sulfuric acid environment, and the vanadium improves fatigue resistance and high-temperature durability. Testing referenced through *Aerospace Research Central* reported that samples of Ti-6Al-4V were exposed to Venus-like conditions for 60 days without oxidizing or sulfiding, confirming its suitability for this mission.

The titanium sheets will be sourced from Prismier as the primary supplier, with Titanium Industries listed as the secondary supplier. Both provide aerospace-grade Ti-6Al-4V with the required specifications. Manufacturing of the chassis will be completed in-house using sheet metal forming and Gas Tungsten Arc Welding (GTAW), along with argon shielding gas to prevent oxidation during welding. These techniques ensure the welds remain strong and defect-free under Venusian stress conditions.

To protect the IMU and radar altimeter from external temperatures and pressure, both will be housed inside a sealed titanium sleeve within the chassis. This sleeve acts as both mechanical support and thermal insulation, designed to maintain safe operating temperatures for sensitive components. The sealed environment allows internal temperatures to remain within operational limits, even when external conditions exceed 450°C.

The GNC sub assembly includes the IMU, radar altimeter, and the DSS mechanical interface. The IMU is the Northrop Grumman LN-200S, which uses fiber optic gyros and electro-mechanical accelerometers to track velocity and angle changes. It provides full six degrees of freedom data, essential for tracking motion and orientation across pitch, roll, yaw, and linear motion. Since Venus has no GPS, the IMU is key to dead-reckoning position estimation, especially during descent or when other sensors fail. This IMU is hermetically sealed, lightweight, and has been used on missions like Curiosity and Coriolis, making it a solid fit for this mission. It also features low-noise and low-voltage inertial sensors, reducing interference with science instruments.

The radar altimeter selected is the RFbeam MR3003_RD, which was chosen as the primary option based on size, power efficiency, and availability. While the MR3003_RD was not originally designed for Venus, its capabilities are promising. However, it will need modifications to operate reliably in Venus' dense and corrosive atmosphere. Planned improvements include



strengthening the housing, and thermal shielding. These adjustments allow us to use a commercially available model while ensuring mission reliability. The radar altimeter will provide real-time height data, allowing the system to maintain stable altitude during science operations and terrain tracking. Used in combination with the IMU, the two systems offer redundant navigation, meaning if one sensor becomes noisy or drifts, the other can compensate, ensuring continuous and accurate data for safe operation.

The Descent Subsystem (DSS) mounts directly to the top face of the chassis via the 8-bolt pattern defined in the ICD. It was ensured that the design could handle the DSS load by considering both the total mass and the shifting center of gravity during descent. While the internal mechanisms of the DSS are managed by a separate contractor, the mechanical structure was designed to maintain stability and withstand the mechanical stresses transferred through the interface.

The Radar Altimeter will work in unison with the IMU to provide the CDH system with much needed altitude estimation for appropriate usage of data collection. The Radar Altimeter will be housed in the titanium sleeve meant to provide insulation for the Aerobot, due to it needing to be within operating temperatures of 223K ro 273K. The Radar Altimeter provides precise altitude estimation by transmitting radio signals downwards and measuring the time it takes for the reflected signals to return. Using this device in combination with the IMU ensures a stable altitude estimate even when the radar altimeter encounters noise. The radar altimeter will also be useful by following the terrain profile, which is important for low flight and maintaining consistent clearance above the ground. The fusion of these devices will act as a failsafe for the other in the case of poor data. For example, if the IMU drifts over time because of noise, or if the Radar Altimeter is corrupted by dust, the other can compensate. This particular combination is crucial for the demanding conditions of Venus. The radar altimeter gives you direct surface-relative height, while the IMU gives you motion and attitude tracking. Together, they provide high-fidelity situational awareness. The radar altimeter will work through the dense cloud layers, and regardless of atmospheric pressure. Since during the period of data collection the aerobot will encounter the worst of the dense and turbulent atmosphere, it is important to have a device that functions. This device functions independently of weather, unlike barometers or other similar devices. The radar altimeter gives your aerobot a real-time, accurate

measurement of its height above the surface, which is critical for stable hovering, drift correction, and safe operation in the upper atmosphere.

For the altitude control, also known as the Descent Subsystem (DSS), proper mechanical interfacing had to be acknowledged. The DSS would be mounted on top of the chassis, via an 8 bolt pattern specified in the Interface Control Document (ICD) provided by L'SPACE. This constrained the top face of the chassis to be 1mx1m. The scope and details of this subsystem are covered by the contractor. The other requirement derived from this subsystem is bearing the load. The mass and center of mass were considered in order to design the mechanical structure, ensuring stability.

TRL of Mechanical Subsystem: 4

TRL	Subassembly	Description
7	Chassis	The chassis CAD is preliminary. The configuration of this structure has been decided. It includes the dimensions, thickness, and materials. The assigned Technology Readiness Level corresponds to its heritage on space missions such as Mars Curiosity [ ], although for this mission it is unclear whether the metal can perform in the environment.
6	Northrop Grumman LN-200S IMU	The Northrop Grumman LN-200S has been used on many missions before, and it is very versatile. It has been used on rovers, such as Curiosity, and orbiters like Coriolis. It provides state-of-the-art fiber optics and gyros that provide velocity and angle changes to the operator. [#]
4	RF-Beam MR 3003_RD Radar Altimeter	This component will be sourced from RFBeam. However, this design does not quite meet the environment requirements, but meets the mass constraint, and has all of the capabilities. NASA could also

		possibly need to develop something specifically for this mission, while also staying within the constraints.

Table 2.1.1.2 A

Narrative: The TRL table discusses the three components that were decided on, and provides a brief justification for each, including specific information on the design and mission heritage.

	Chassis	**IMU**	**Radar Altimeter**	**Totals**
Mass (kg)	37.8	0.748	0.15	38.69
Power Draw (W)	0	12	7	19
Dimensions (m)	1x1x0.5	0.0889x0.0851	0.076 × 0.044 × 0.0385	1x1x0.5

Table 2.1.1.2 B

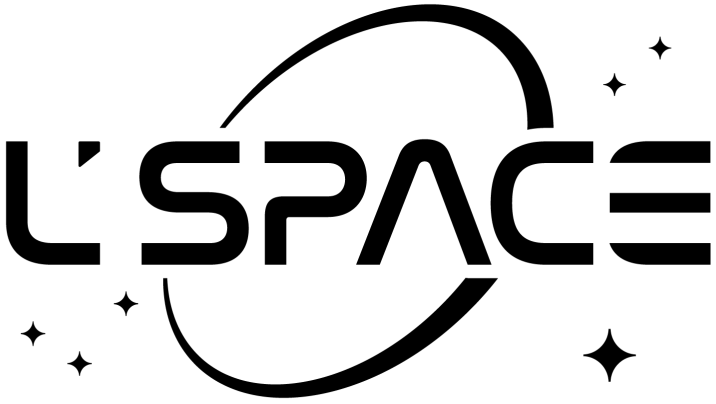
Narrative: The mass, dimensions, and power breakdown of the entire mechanical subsystem are provided. The totals are listed in the column to the right. While the mechanical subsystem design is progressing well, it still requires expert review and refinement, especially for components being adapted to Venus. Continued focus will be placed on finalizing the thermal protection strategy for the internal hardware. Additional ideas like oscillation damping may be explored in future iterations to improve the stability of onboard science instruments, and possibly the GNC system as well.

Below are the trade studies conducted for each subassembly/component that comprises the mechanical subsystem. At least three options were considered for each component, which provided a well rounded overall subsystem.

Materials for Mechanical SS	

Criteria	Explanation	Grade .	Weight	Ti-6Al-4V	Titanium	Molybdenum
Can withstand harsh environment	Must be able to withstand temperatures of 450 °C, sulfuric acid corrosion, and pressures of 90 atm	10 = high, 5 = medium 1 = low 0 = Fail	35%	9	7	9
Lightweight material	The materials should not be too heavy, as it may require more power from the DSS, and lower cost	10 = high, 5 = medium 1 = low 0 = Fail	25%	8	7	6
Cost effective	The mechanical SS should not make up most of the budget, as to reserve more money for hardware and software	10 = high, 5 = medium 1 = low 0 = Fail	25%	9	9	8
Ease of manufacturability	Shall be easily manufacturable and readily available, reducing cost and time	10 = high, 5 = medium 1 = low 0 = Fail	15%	8	7	4
		TOTALS :	**100%**	86.00%	75.00%	55.00%

Table 2.1.1.2 C



Narrative: This trade study was conducted for the materials of the Chassis. The requirements derived earlier gave a great backbone, and it came down to three materials. Two are based on Titanium, with the other being Molybdenum. After testing however, the alloy handled oxidation and sulfidization better. The weights were chosen to ensure that the material could handle the demanding environment of Venus, be within mass constraints, and stay within budget. As shown in the scoring, the alloy can handle the environment better, and is lighter and easier to manufacture than Titanium.

GNC orientation						
Criteria	**Explanation**	**Grade**	**Weight**	**NG LN 200S**	**HG 99000**	**KVH 1750**
Accuracy and Precision	Measurement accuracy, bias stability, and noise performance	10 = high, 5 = medium 1 = low 0 = Fail	35%	9	8	7
Space Heritage	Has it flown on other space missions or planetary environments?	10 = high, 5 = medium 1 = low 0 = Fail	25%	8	8	5
Power Consumption	Lower power consumption preferred to save subsystem resources	10 = high, 5 = medium 1 = low 0 = Fail	20%	9	6	7
Size and Mass	Smaller and lighter IMUs reduce mass and improve packaging	10 = high, 5 = medium	20%	8	6	8

		1 = low 0 = Fail				
		TOTALS:	**100%**	85.50%	72.00%	67.00 %

Table 2.1.1.2 D

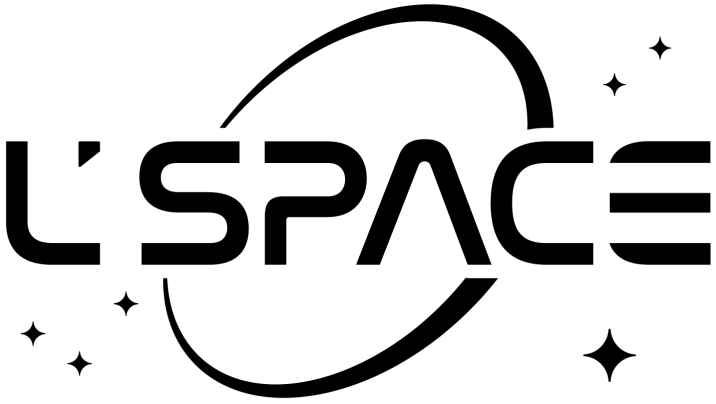
Narrative: This trade study evaluated three IMU options based on mission-specific needs: accuracy, flight heritage, power efficiency, and compactness. The Northrop Grumman LN-200S, Honeywell HG9900, and KVH 1750 were compared. The LN-200S ranked highest due to its superior accuracy, low noise, and strong space heritage. It also met size and power constraints better than its competitors. Weights were assigned to emphasize precision and proven performance in extreme environments. While the HG9900 had similar heritage, it consumed more power. The KVH 1750 was compact but lacked the accuracy and legacy needed for Venus navigation. Scoring confirmed the LN-200S as the most balanced and reliable option for the mission.

GNC altimeter						
Criteria	**Explanation**	**Grade**	**Weight**	**RFBeam**	**Thales Alt**	**Lockheed AURORA**
Penetration in Dense Atmosphere	Ability to provide altitude in Venus-like cloudy and corrosive atmosphere	10 = high, 5 = medium 1 = low 0 = Fail	40%	9	7	8
Range Resolution	Vertical resolution accuracy, needed for science payload targeting	10 = high, 5 = medium 1 = low 0 = Fail	25%	8	6	7

Heritage and Customizability	Previously flown or designed for space and customizable by vendors	10 = high, 5 = medium 1 = low 0 = Fail	20%	9	6	5
Power Efficiency	Low power draw while maintaining signal strength	10 = high, 5 = medium 1 = low 0 = Fail	15%	8	5	6
		TOTALS:	**100%**	86.00%	62.50%	68.50%

Table 2.1.1.2 E

Narrative: This trade study compared three radar altimeters: RFbeam MR3003_RD, Thales Altimeter 1, and Lockheed AURORA-2. Evaluation focused on atmospheric penetration, resolution, heritage, and power efficiency. The RFbeam MR3003_RD emerged as the best choice, with strong performance in dense atmospheric conditions, good resolution, and customization potential. The weighting prioritized accuracy and adaptability for Venus' environment. Thales' option lacked needed resolution, and Lockheed's offered less flexibility and heritage. Overall, the MR3003_RD offered the best performance-to-constraint balance, making it the top candidate for precise altitude estimation.



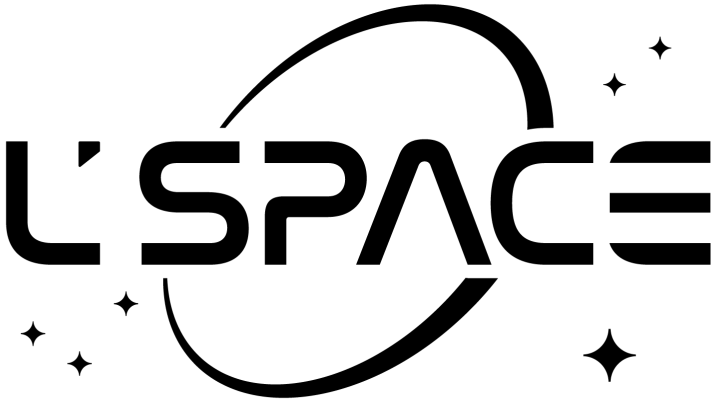
2.1.1.3. Mechanical Subsystem Recovery and Redundancy Plans

The mechanical subsystem, by its nature, is a hardware heavy system and does not include any embedded software components within its structural assemblies. Because of this, it does not require traditional fault recovery mechanisms typically associated with software-based or electromechanical systems. The function of the mechanical subsystem is to provide stability, protection, and integration of the other systems that rely on it for support. As a result the emphasis is placed on durability, fail-safe design principles, and selective redundancy where it is most critical to mission success.

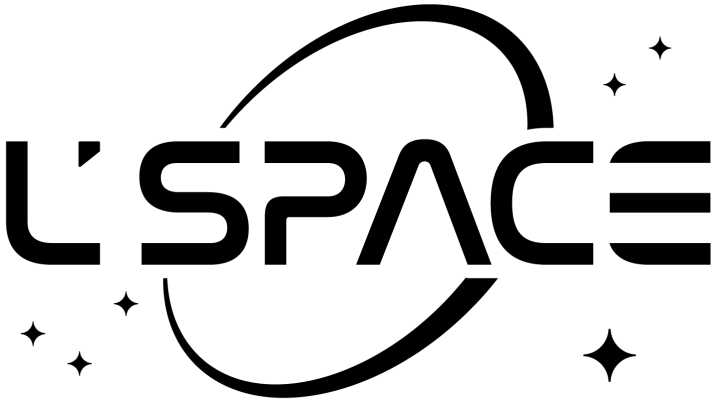
While the mechanical subsystem itself does not require active recovery protocols, redundancy is implemented in select critical sensing interfaces where mechanical integrity directly impacts system performance — specifically in the Inertial Measurement Unit (IMU) and the Radar Altimeter. These two instruments serve as key navigation and positioning sensors for the Guidance, Navigation, and Control (GNC) system. Their mechanical integration and physical positioning are directly managed through the mechanical subsystem.

In the event that one of these sensors experiences degraded performance due to environmental noise, or mechanical vibration, the other sensor is capable of serving as a failsafe, ensuring continued operation. For example, should the radar altimeter fail or produce noisy data due to interference or environmental factors, the IMU can provide essential positional information to estimate altitude through integrated acceleration data. Conversely, the radar altimeter can serve as a real-time altitude verification tool if the IMU drifts due to sensor bias or cumulative error. This cross-compensation mechanism is critical in mission scenarios involving terrain navigation, stability control, and precise instrument deployment. Especially for this environment, where the likelihood of Venus' thick clouds and ferocious winds altering the readings is very high.

From a structural standpoint, redundancy in the chassis or structural framework was deliberately excluded. The rationale behind this decision is: (1) the chassis was designed with sufficient strength and environmental tolerance such that a backup structure would provide unnecessary added benefit, and (2) inclusion of redundant chassis components would impose a significant mass penalty, violating mass constraints. The design of the chassis is simple, and reinforced to handle the structural loads and distribute them across the surface.



In conclusion, the mechanical subsystem maintains mission resilience not through reliable recovery methods, but through fail safe design, and robust engineering practices. Redundancy is applied only where it meaningfully contributes to mission assurance — namely in the sensing systems critical to GNC operations. For the chassis and other static structures, confidence is built into the system through material science, and structural design. This approach ensures that the mechanical subsystem remains both lightweight and reliable, fulfilling its role as the foundation for the spacecraft's overall integrity and functionality.



2.1.1.4. Mechanical Subsystem Manufacturing and Procurement Plans

The mechanical subsystem consists of two main subassemblies and three components that need to interface directly with other systems such as thermal control, power, and science payloads. These mechanical elements work together to provide the structural support, stability, and mobility needed for the mission. Every decision—whether material selection or component layout—was made based on Venus' extreme conditions, and balanced against limits in mass, time, and cost.

For the chassis, we chose a Titanium alloy (Ti-6Al-4V), which has proven reliability on missions like Curiosity and Perseverance. This alloy offers the strength and corrosion resistance needed for Venus while still being lightweight. It's also suitable for manufacturing techniques like alloying, sheet metal forming, and welding. We plan to source the Ti-6Al-4V sheets from Prismier, a contractor known for fast turnaround — typically 1 to 2 business days — which helps us stay on schedule. Our second supplier is Titanium Industries which is known for offering aerospace-grade titanium products and global distribution. We chose Prismier because they have proven heritage in the industry, and a more credible information source in their website. The lead times are also preferable.

Welding titanium in harsh conditions brings its own challenges, mainly because the metal is highly reactive at high temperatures. To make sure the welds are strong and safe from oxidation, we're doing the welding in-house using Gas Tungsten Arc Welding (GTAW), combined with Argon shielding gas. This method has high precision and will help us avoid issues like porosity or cracking in the weld joints, which could compromise the structural integrity under Venus' extreme environment.

The second subassembly is the Guidance, Navigation, and Control (GNC) mount, which supports two critical components: an Inertial Measurement Unit (IMU) and a Radar Altimeter. These two sensors work together to handle position tracking and altitude control—especially important given that Venus has no GPS and extremely dense cloud cover.

For the IMU, we considered two main suppliers: Northrop Grumman and Honeywell. Our current preference is the Northrop Grumman LN-200S, mainly due to its lower weight compared to Honeywell's HG9900. Both models have flight heritage from missions like Perseverance and

Coriolis, which gives us confidence in their long-term reliability. Both companies offer some level of customization for environmental conditions, but specific lead times and details are only available through direct supplier contact.

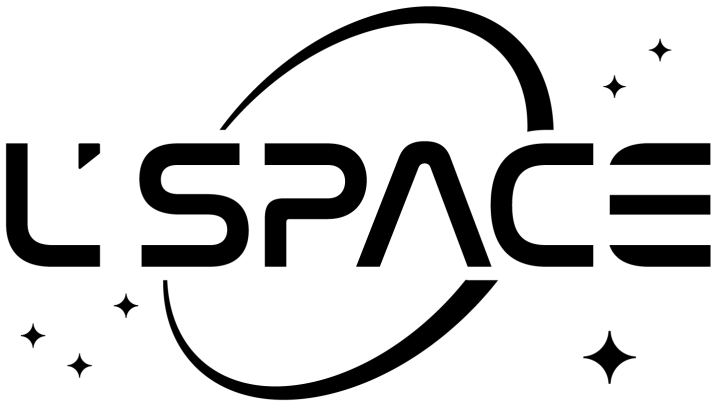
The Radar Altimeter works with the IMU to provide reliable altitude data, especially during descent and surface operations. After review, RFbeam's MR3003_RD was selected as the preferred option due to its compact size, low power use, and availability. However, to handle Venus' extreme atmosphere, the sensor will require modifications, including improved housing, added thermal shielding, and adjustments to signal processing. These upgrades make it a practical and efficient choice, balancing performance with mission constraints.

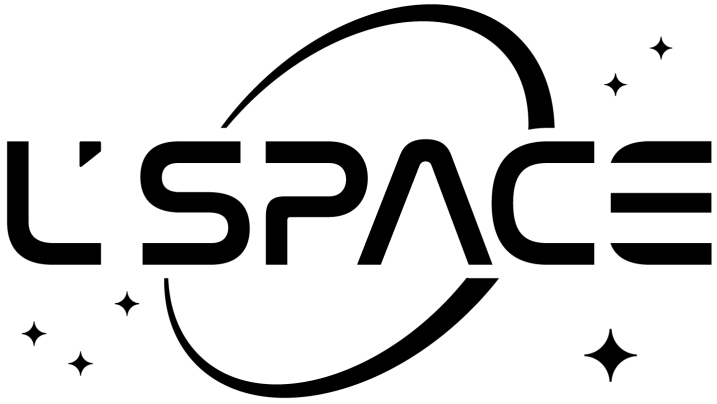
We took inspiration from the Venera 16 mission, which used a radar altimeter designed for Venus. However, that unit was too large and power-heavy for our platform. Our version will be smaller and lighter but still accurate enough to function in low-visibility environments. This level of precision is worth the extra effort since it directly affects mission success.

For both subassemblies, proper integration with the rest of the system is essential. The chassis is designed to support the overall structure and interface directly with key hardware components, including the IMU and radar altimeter. These components will be housed inside a titanium sleeve, which provides a controlled thermal environment to protect sensitive electronics from Venus' extreme surface temperatures. The titanium sleeve offers both structural support and thermal insulation (more information included in the thermal control section), taking advantage of titanium's high-temperature tolerance and strength.

While there are no active vibration isolation systems, the chassis and titanium sleeve combination provides a rigid and stable platform to reduce unnecessary motion during operation. This setup keeps the mechanical design simple and reliable while ensuring that critical components remain protected and functional throughout the mission.

Overall, the mechanical subsystem is built around the principles of strength, simplicity, and mission-specific optimization. Each component has a clear role and is designed to handle Venus' environment while staying within mission constraints. With tight integration and tested materials, we expect this setup to perform reliably through the mission's full lifecycle.


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2.1.1.5. Mechanical Subsystem Verification Plans

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
ME 1.0	The aerobot shall have a central structure to house the components and payload.	Inspection	Visual inspection confirms all components are securely mounted inside the structure.	Inspect the assembled aerobot to ensure the payload and components are housed correctly and remain secure under simulated conditions.
ME 2.0	The aerobot shall have a central structure to house the components and payload.	Inspection	System presence and physical installation can be verified through visual inspection.	Inspect the installed navigation system and verify integration with control software and sensors.
ME 3.0	The aerobot shall have a central structure to house the components and payload.	Test	Material resistance must be validated under simulated corrosive and thermal conditions.	Test material samples in a chamber simulating Venusian temperatures and acidic atmosphere to confirm resistance.
ME 1.1	The Mechanical Subsystem shall have a chassis that maintains stability in strong winds.	Analysis	Simulations can model wind loads and assess structural response before physical testing.	Perform CFD and FEA simulations to assess structural performance under expected wind conditions.
ME 1.2	The Mechanical Subsystem shall interface with all other subsystems.	Analysis	Subsystem integration is best validated using interface control documentation and layout models.	Review CAD models and ICDs to ensure physical and functional interface compatibility.

ME 2.1	The Guidance and Navigation system shall be precise, and estimate the altitude with 5% error.	Analysis	Precision and accuracy are validated through simulations and performance modeling.	Analyze sensor data and run navigation algorithms under test conditions to verify error margins.

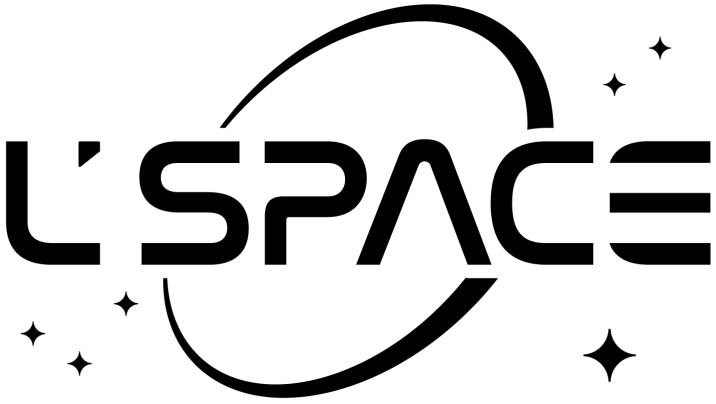
Table 2.1.1.5 A

The Mechanical Subsystem is responsible for providing the structural integrity, physical housing, and interface framework for all other subsystems on the Aerobot. It must also support the proper installation and operation of the guidance, navigation, and control systems. The verification of this subsystem is critical to ensure it performs under the extreme environmental conditions of Venus while maintaining system alignment, payload stability, and structural survivability.

The verification process begins with visual inspections for requirements related to housing, interface, and component installation. Requirement ME 1.0 states that the Aerobot shall include a central structure to house all components and payload. This is verified by inspecting the integrated chassis and confirming that each component is properly seated and secured within its mounting brackets. Clearances, fasteners, and material interfaces will be verified visually and documented through an assembly checklist. For ME 2.0, which requires the presence of a guidance and navigation system, visual inspection confirms installation of both the Inertial Measurement Unit (IMU) and the radar altimeter within their designated mounting points. This step ensures physical integration prior to testing of the GNC system.

For ME 1.2, the requirement that the chassis must interface correctly with all other subsystems will be verified through interface analysis. CAD models and Interface Control Documents (ICDs) will be reviewed to confirm that power, data, structural, and thermal interfaces align in accordance with the system architecture. The verification will include digital simulations using NX or SolidWorks to visualize assembly tolerances and potential clashes before physical integration.

Structural performance in high-wind environments (ME 1.1) will be validated using Finite Element Analysis (FEA). The chassis will be modeled under simulated wind loads reflecting

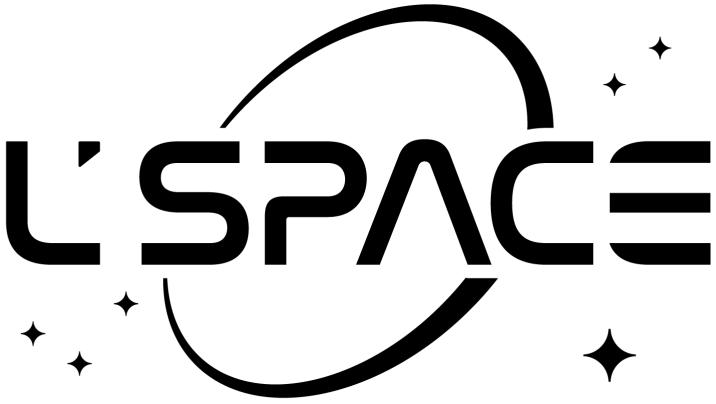


atmospheric pressures expected during descent and cruise phases over Venus. Static and dynamic loads will be tested to assess deflection, stress concentrations, and potential failure modes. These analyses ensure that the structure meets safety factors while minimizing mass.

Material suitability (ME 3.0) will be verified through environmental testing. Test samples of Ti-6Al-4V and other materials used in chassis fabrication will be exposed to thermal and chemical environments that simulate the upper atmosphere of Venus, including high temperatures and corrosive acidic compounds. After exposure, the samples will be examined for corrosion, surface degradation, and structural weakening. Data from this test will confirm whether the selected materials are suitable for sustained operation.

Precision of the navigation system (ME 2.1), specifically maintaining altitude estimation within 5% error, will be validated through analytical simulations and laboratory tests. Sensor data will be collected in a controlled test environment, and algorithms will be run to estimate position and altitude. Accuracy will be compared to ground truth data to evaluate performance. The system will be tuned as necessary to meet accuracy thresholds.

Throughout the verification process, all results will be recorded in formal verification reports and matched against the original requirements traceability matrix. In the event of any discrepancy or failure to meet the requirement, a corrective action plan will be developed and implemented. This structured verification approach ensures that the Mechanical Subsystem will perform reliably throughout the mission under extreme environmental conditions.



2.1.2. Power Subsystem Overview

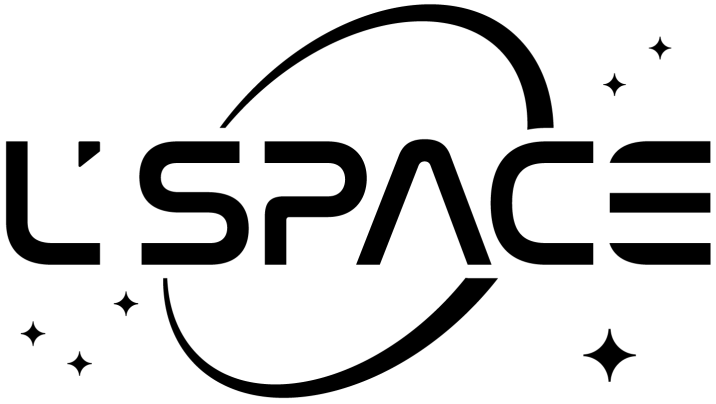
A reliable and efficient power subsystem is essential for the success of the Venus Aerobot Mission. Given the extreme atmospheric conditions of Venus and the mission's limited duration, the aerobot's power system must maximize energy efficiency while meeting the varying demands of scientific and operational subsystems. This section provides an overview of the power architecture, explains how each component works in concert, and details how power is scheduled over distinct mission phases to optimize performance.

The aerobot's power system is composed of three main elements: the primary battery source, the power regulation unit (PRU), and the wiring harness. Each component has been selected based on stringent criteria involving power density, thermal stability, mass efficiency, and aerospace reliability.

Primary Battery System:
The mission will utilize SAFT LSH20 lithium-thionyl chloride ($LiSOCl_2$) cells, which are known for their high energy density (around 500 Wh/kg), excellent shelf life, and wide operating temperature range. These batteries are configured in series-parallel arrays to meet the total voltage and capacity requirements for 48-hour operation. $LiSOCl_2$ cells are primary (non-rechargeable), making energy conservation and intelligent scheduling critical to extending mission functionality.

Power Regulation Unit (PRU):
The PRU, provided by Crane Aerospace & Electronics, ensures a stable and efficient conversion of raw battery output to regulated voltage rails used by each subsystem (e.g., 3.3V for logic, 5V and 12V for sensors and instruments). The PRU includes essential protections such as overcurrent shutoff, thermal cutoffs, and undervoltage lockout to prevent brownouts or subsystem failures.

Power Switching Unit (PSU):



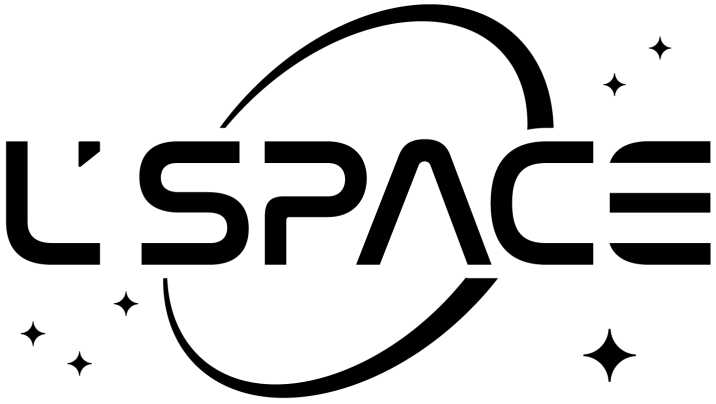
The Power Switching Unit (PSU) is a vital component of the aerobot's power subsystem, responsible for distributing electrical power from the primary lithium-thionyl chloride ($LiSOCl_2$) batteries to all operational subsystems. It enables selective power delivery based on mission phases—such as data collection, communication, or thermal regulation—allowing for efficient energy use throughout the 48-hour mission. The PSU also provides fault protection by isolating malfunctioning circuits and ensuring continuous operation through redundant power pathways.

Wiring and Distribution:
To transmit power and data reliably, the system employs Harbour Industries' PTFE-insulated wires for high-temperature durability, Radix Wire silicone-insulated conductors for flexibility, and Gore shielded twisted-pair cable for low-noise signal transmission. Wiring routes are optimized for minimal mass and interference, with Kapton-insulated Thermax wire selected for high-temperature areas.

To support the Venus aerobot's scientific goals while maintaining efficient energy use, the power system has been carefully designed around mission phase cycles, with a special emphasis on frequent data relay sessions to the orbiter. This approach reduces the risk of data loss and enables the mission team to adapt dynamically to the conditions of Venus' harsh environment. The aerobot's power management strategy accounts for each subsystem's operational demands across a continuous 48-hour mission, prioritizing periods of active science, data processing, and communication while leveraging low-power modes to conserve energy.

The mission begins with a 4-hour traverse phase, during which the aerobot stabilizes its altitude, engages basic navigation, and adjusts thermal balance as it prepares for data collection. During this phase, only essential subsystems such as guidance, Command and Data Handling (CDH), and thermal management are active, resulting in a moderate power draw of approximately 20 watts. This is followed by a 12-hour window dedicated to intensive data collection using the UV-Vis spectrometer and the Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES). These instruments require the highest power input, drawing about 35 watts while actively scanning the surface and atmosphere of Venus.

To enhance data integrity and reduce risk, the mission incorporates multiple short data relay sessions spaced at regular intervals. The first relay occurs at hour 8, midway through the first science block, and lasts one hour. During this time, only the communications and CDH systems



are active, consuming around 10 watts. A second relay window opens at hour 16, following the completion of data collection, ensuring the transmission of valuable data before the aerobot shifts into its data analysis phase.

Between hours 17 and 22, the aerobot processes collected data using onboard computing resources, with power demands decreasing to around 15 watts. This window allows for compression, prioritization, and packaging of data for future relay. To conserve energy, the system then enters a low-power mode from hour 22 to 24, drawing only 5 watts to maintain critical thermal and CDH functions. A third data relay is scheduled at hour 24, again lasting one hour, providing another opportunity to offload stored data. This is immediately followed by an extended low-power mode from hours 25 to 32, maximizing battery conservation in preparation for the second science block.

The second data collection session begins at hour 32 and runs through hour 40. Like the first collection phase, the scientific instruments demand a full 35 watts. During this period, another data relay occurs at hour 40. Subsequently, a second round of data analysis takes place from hours 41 to 44, using roughly 15 watts as onboard systems process the new findings. A final data relay session at hour 44 ensures all remaining data is transmitted before the mission enters its final low-power mode from hour 45 through 48.

This revised mission schedule enables more frequent and reliable communication with the orbiter while balancing energy consumption through carefully timed subsystem activation. By limiting the number of systems active at any given time and strategically interspersing high-energy operations with low-power intervals, the power subsystem maximizes the scientific return without exceeding the energy budget. This modular, cyclical approach to energy management not only supports scientific goals but also adds robustness to the overall mission design.

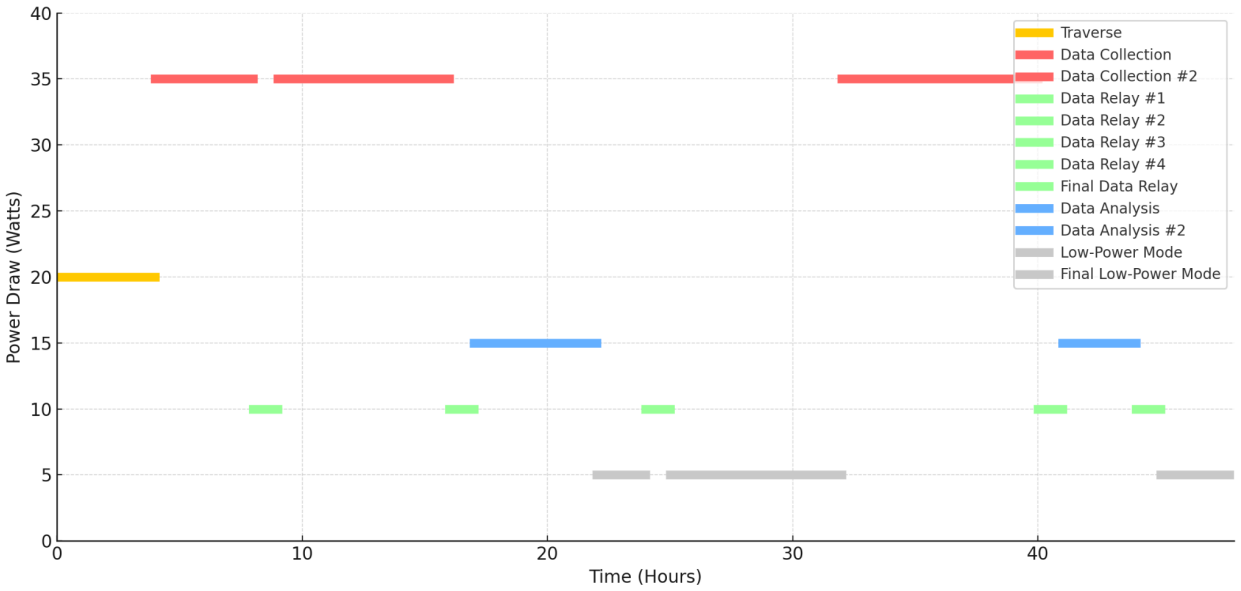

Traverse
Data Collection
Data Collection #2
Data Relay #1
Data Relay #2
Data Relay #3
Data Relay #4
Final Data Relay
Data Analysis
Data Analysis #2
Low-Power Mode
Final Low-Power Mode
Power Draw (Watts)
Time (Hours)
0
5
10
15
20
25
30
35
40

Figure 2.1.2 A

2.1.2.1. Power Subsystem Requirements

Requirement ID	Requirement	Rationale	Parent Requirement	Verification Method
PWR.1	The power subsystem shall provide a minimum of 200 W of continuous power to all operational subsystems, including payload and data handling.	To ensure sufficient power for all subsystems, especially during Venus' harsh environment.	SYS.1.0	Test, Demonstration
PWR.2	The power subsystem shall have low-mass, high-density energy storage to minimize mass and volume.	Reduces the overall mass of the aerobot to meet mass constraints while maintaining energy supply.	SYS.3	Test, Demonstration
PWR.3	The power subsystem shall operate effectively in extreme Venusian conditions, including temperatures up to 475°C.	Ensures reliable operation in Venus' harsh environmental conditions.	SYS.1, SYS.4	Test, Analysis
PWR.4	The power subsystem shall include redundant power paths to ensure continuous operation in the event of a failure.	Redundancy increases system reliability and provides a backup in case of component failure.	SYS.1	Test, Demonstration
PWR.5	The power subsystem	Ensures power stability	SYS.1	Test,

	shall incorporate power regulation electronics to distribute power safely to all aerobot subsystems.	and prevents damage to subsystems by providing correct voltage and current.		Demonstration
PWR.6	Battery capacity shall be optimized for a single-use, non-rechargeable system to minimize weight	Reduces mass by removing recharging hardware, simplifying design for a 48-hour mission.	CC.1	Inspection

Table 2.1.2.1 A

These requirements balance performance and practicality for a 48-hour Venus balloon mission. By opting for primary (non-rechargeable) batteries and eliminating unnecessary complexity, the design maintains low mass and high reliability. Redundancy and regulation ensure that, even in extreme conditions, all critical systems receive the power they need. Thermal survivability and smart energy distribution are crucial, especially when data relays and payload operation cannot afford sudden losses of power.

2.1.2.2. Power Sub-Assembly Overview

The Venus Aerobot Mission's power subsystem is critical to mission success, providing sustained and regulated power to all operational components within a highly challenging environment. The mission requires continuous operation over a 48-hour period, navigating through Venus' upper atmosphere where extreme temperatures and pressures, coupled with limited energy storage capacity, demand a well-engineered and optimized solution. This overview outlines the integration, design, management, and operational strategy of the power sub-assemblies used in the mission.

The power subsystem is comprised of three core sub-assemblies:

- Energy Storage Unit (ESU): Primary lithium-thionyl chloride (LiSOCl2) batteries sourced from SAFT (LS Series) and Tadiran (TLH Series) serve as the mission's energy

reservoir. These batteries are chosen for their high energy density, long shelf life, and performance reliability under wide temperature ranges.

- Power Regulation Unit (PRU): Supplied by Crane Aerospace & Electronics and SynQor, this unit ensures regulated voltage and current to prevent overloading or damaging subsystems. It also isolates power paths for redundancy.
- Power Distribution Unit (PDU): This component distributes power to each subsystem (mechanical, payload, thermal management, CDH). Wiring is sourced from Harbour Industries (MIL-Spec), Belden (data/signal), and Thermax (Kapton-insulated for extreme temperatures).

Load Distribution

Subsystem	Average Power Draw (W)	Peak Power Draw (W)
Payload	30	35
CDH	10	15
Thermal	15	20
Communications	5	10
Travel	10	15

Table 2.1.2.2 A

The power management strategy for the Venus Aerobot mission is designed to maximize the system's efficiency while ensuring uninterrupted operation throughout the 48-hour mission duration. The mission is divided into several distinct phases, each with varying power requirements:

- **Traverse Phase (0–4 hrs)**: This phase involves low to moderate power consumption (~20 W), mainly driven by mechanical operations such as steering and stabilization. The aerobot’s navigation system and sensors use minimal power during this time.

- **Data Collection Phase (4–8 hrs, 9–16 hrs, 32–40 hrs)**: This phase experiences the highest power draw, with peak usage of up to 35 W from the payload subsystems (UV-Vis Spectrometer and pHyTES). The payload is actively collecting atmospheric data, which requires substantial energy.

- **Data Relay Phase (multiple 1-hour windows)**: Data transmission to the orbiter and subsequent relay to Earth is a critical phase. Communications require minimal power (~10 W) but are necessary to ensure timely data delivery. These relay events are distributed throughout the mission to ensure efficient use of the power budget.

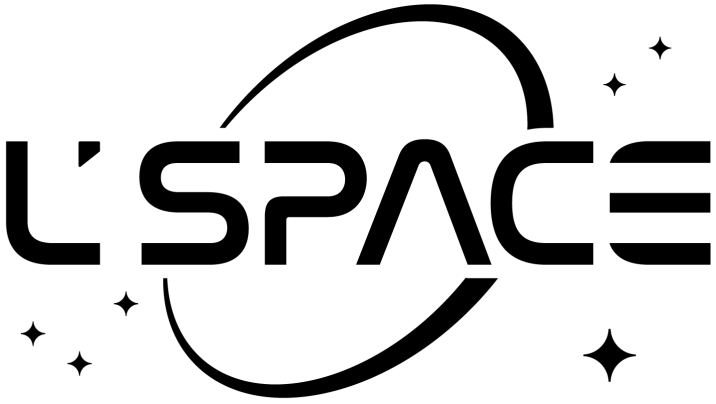
- **Data Analysis Phase (17–22 hrs, 41–44 hrs)**: This phase primarily involves the processing and analysis of collected data. The command and data handling subsystem, along with the thermal management system, draws approximately 15 W of continuous power. Thermal regulation ensures that systems remain within operational limits.

- **Low-Power Mode (various windows)**: To extend the operational lifetime of the mission, low-power modes are activated during less active periods. This phase consumes around 5 W, allowing the aerobot to rest and conserve energy for upcoming data collection and communication.

These phases interleave and repeat to create a smooth power schedule that prevents periods of excessive power consumption from overlapping. By adjusting the operational states of different subsystems, the power subsystem ensures that power is distributed efficiently throughout the mission's timeline. Redundant relay periods are included to increase the likelihood of successful data transmission and ensure that power is always available for essential operations.

The power consumption pattern throughout the mission is visualized in the Power vs. Time plot, which shows fluctuations in power demand over the 48-hour mission duration. The plot highlights key moments of high usage, such as during data collection, as well as periods of low power usage, such as when the aerobot is in low-power mode or during relay events.

The highest power demand occurs during data collection phases when the payload instruments are actively gathering information. The power profile also includes dips in consumption during



less active phases like data analysis or low-power modes, which allow the system to conserve energy for the more power-intensive operations. By designing the mission phases to avoid overlapping high-demand periods, the power subsystem ensures that the energy budget is maintained efficiently throughout the mission.

Operating on Venus presents unique challenges that influence the power subsystem design. Extreme temperatures—up to 475°C—require the subsystem components to be highly durable. Batteries, wiring, and electronics must be rated to withstand the harsh conditions, and special materials are used to protect sensitive components from the corrosive atmosphere of Venus.

The primary power storage solution is a non-rechargeable lithium-thionyl chloride ($LiSOCl_2$) battery, selected for its high energy density and long shelf life under extreme conditions. This choice simplifies the design, as it eliminates the need for recharging capabilities, which would increase system complexity and weight. However, this means the mission must be carefully planned to maximize the use of the available power during the 48-hour operational window.

The power subsystem also includes redundancy in the power distribution paths, ensuring that if one path fails, another can take over without disrupting critical operations. This redundancy is vital for mission success, as it mitigates the risk of power loss from system failures.

Furthermore, the power regulation electronics play a key role in distributing power safely to all subsystems. These units ensure that voltage and current are maintained within safe limits, protecting subsystems from potential damage due to power surges or voltage fluctuations.

The power subsystem of the Venus Aerobot Mission is a sophisticated and carefully engineered solution to meet the unique challenges posed by Venus' extreme environment. By balancing mass efficiency, power availability, and system redundancy, the design ensures that the mission's subsystems can function optimally throughout the 48-hour mission. The use of high-density energy storage, robust power regulation, and efficient load distribution allows the aerobot to complete its scientific objectives, including atmospheric sampling, surface thermal imaging, and data transmission to Earth.

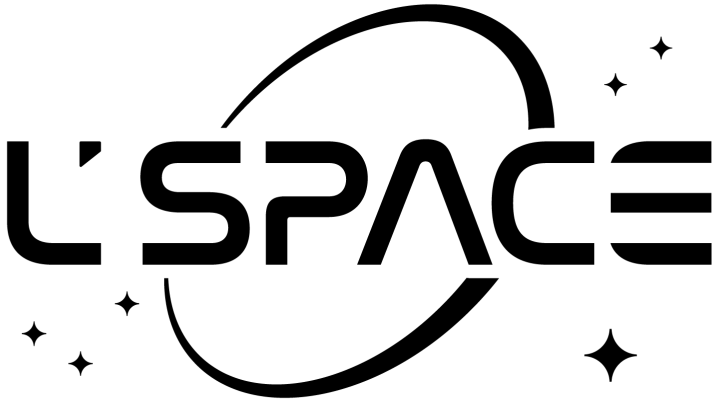
Through careful operational planning and strategic power management, the Venus Aerobot Mission will be able to collect valuable data on Venus' atmosphere and surface, providing

insights that will further our understanding of this enigmatic planet. The success of the power subsystem is fundamental to the success of the entire mission, as it ensures that all systems remain operational, even under the harshest conditions.

TRL Assessment

Subassembly	TRL	Rationale
Primary Battery Pack (LiSOCl2)	8	LiSOCl2 batteries have been used in space missions, providing reliable and long-lasting power solutions.
Power Distribution Unit (PDU)	7	Power distribution systems are mature and widely used in satellite and spacecraft missions.
Power Switching Unit (PSU)	8	Power switching units have extensive heritage in space applications, providing stable power output.

Power Primary Subassembly						
Criteria	**Explanation**	**Grade**	**Weight**	**Lithium-thionyl chloride battery**	**Alkaline battery**	**Lithium-ion battery**
Energy Density	The amount of energy stored per unit mass. Higher energy density allows for a lighter power system.	10 = Energy density > 500 Wh/kg. 1 = Energy density < 100 Wh/kg	25%	10	6	7
Temperature Tolerance	Ability to operate in the high-temperature Venus	10 = Can operate continuously in Venus-like conditions	20%	9	5	7



	environment.	(450°C+) 1 = Fails or degrades rapidly under Venus surface temperatures				
Mass	Total mass of the power system. Lower mass is preferable for reducing overall system weight.	10 = Total power subsystem mass < 5 kg 1 = Total power subsystem mass > 20 kg	15%	9	7	8
Longevity	Ability to sustain consistent power output over the 48-hour mission duration.	10 = Capable of sustaining consistent power output for > 48 hours 1 = Lasts < 12 hours before power loss	20%	10	6	8
Availability/Cost	Availability of components and cost to integrate the system into the aerobot.	10 = Readily available commercially 1 = Rare, custom-built, and high cost	10%	6	10	7
Complexity	Complexity of integration and management within the aerobot's power subsystem.	10 = Minimal or no need for thermal conditioning 1 = Requires complex thermal control	10%	8	10	9
		TOTALS:	**100%**	64.50%	45.50%	52.50%

The goal of this trade study is to evaluate and select the most suitable power source for the Venus Aerobot mission, focusing on performance, feasibility, and longevity in the extreme Venusian environment. The criteria, scoring, and weights were carefully selected to reflect the mission's unique challenges and priorities.

Criteria	Explanation	Grade	Weight	Battery Management System	Power Regulation Unit
Power Efficiency	How efficiently the unit handles and transfers power	10 = dynamically allocates power, handles multiple voltages 1 = cannot regulate voltage accurately	25%	8	10
Thermal Stability	Ability to operate at high temperatures (~475°C)	10 = no additional insulation required for survival 1 = would require heavy insulation	25%	7	9
Mass & Volume	How compact and lightweight the unit is	10 = integrates multiple functions in a small form factor 1 =requires extra housing	25%	6	9

Integration Complexity	Ease of integrating the unit with existing systems and wiring	10 = minimal development effort required for integration 1 = mismatched voltage levels or physical connectors	10%	7	9
Voltage Regulation	How well the component meets mission power requirements	10 = Highly efficient, with low loss and smart regulation 1 =Very inefficient, with large heat losses or conversion inefficiencies	15%	9	10
		TOTALS:	**100%**	66.00%	85.00%

This trade study evaluated the suitability of a Battery Management System (BMS) versus a Power Regulation Unit (PRU) for use in the aerobot's power subsystem. Using weighted criteria—such as functionality, reliability, thermal tolerance, and mass efficiency—each system was graded on a scale from 1 to 10. The PRU emerged as the more favorable option due to its superior performance in critical areas like power efficiency, voltage regulation, compactness, and reliability in Venusian conditions. While the BMS offers benefits in energy monitoring and safety, the PRU aligns more closely with the mission's power distribution and stability requirements, making it the preferred choice for the aerobot's operational architecture.

2.1.2.3. Power Subsystem Recovery and Redundancy Plans

The extreme environment of Venus necessitates a robust and strategically designed power subsystem that accounts for both recovery mechanisms and redundancy to ensure uninterrupted operations during the mission's limited 48-hour operational window. Given the lack of opportunity for repair or resupply, the aerobot's power architecture must prioritize fault tolerance, self-reliance, and intelligent fail-safe pathways. This section outlines the redundancy strategies and recovery protocols embedded in the aerobot's power subsystem to mitigate risks associated with component failure, environmental stress, and unforeseen power anomalies.

Redundancy Architecture

Redundancy in the aerobot's power subsystem is implemented at both the component and functional levels, ensuring that single-point failures do not compromise mission success. While full subsystem redundancy is limited by mass and volume constraints, a selective redundancy approach has been adopted to support critical functions such as power distribution and regulation.

Key redundant elements include:

- Dual power paths: The primary and secondary power buses are designed to feed mission-critical subsystems (Payload, CDH, Thermal Management) independently. In the event of a failure in the main power path (e.g., short circuit, regulator malfunction), the secondary path automatically engages via the Power Switching Unit (PSU) to maintain operation.

- Redundant Power Regulation Units (PRUs): Two PRUs sourced from Infineon Technologies are implemented in parallel, with one active and the other on standby. An automatic switching mechanism monitors output voltage levels and thermal performance; if the primary PRU degrades or shuts down, the backup unit is activated with minimal latency.

- Fault-tolerant wiring: Power cabling from Harbour Industries and Thermax includes shielding and redundant routing in high-stress areas. This mitigates risks from insulation failure or conductor breakage due to thermal cycling or structural flexing.

While battery redundancy was deemed infeasible due to strict mass constraints, the chosen lithium-thionyl chloride cells are highly reliable, with minimal self-discharge and proven aerospace heritage. The design instead relies on mission-length reliability supported by thermal insulation and load management protocols.

Recovery Strategies and Load Prioritization

In addition to hardware redundancy, the power subsystem features intelligent recovery strategies that respond dynamically to real-time power conditions. Recovery protocols focus on load prioritization, graceful degradation, and low-power operational modes to extend mission longevity in the event of partial system failure.

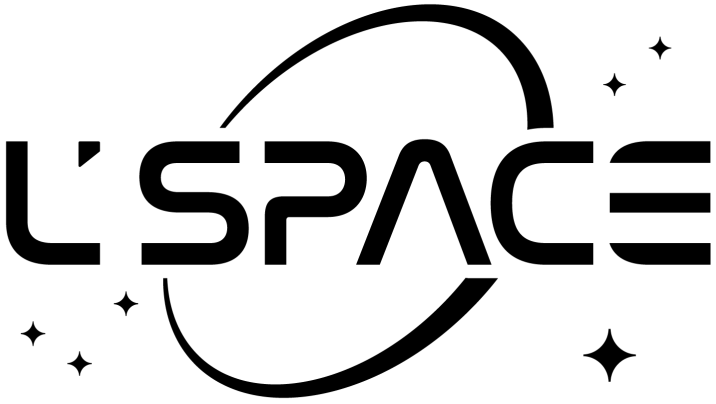
1. Load Shedding and Prioritization
Subsystems are classified into three priority tiers based on mission-criticality:

- Tier 1 – Essential: Power Regulation, CDH, Communications, and Thermal Management
- Tier 2 – Scientific Payloads: UV-Vis Spectrometer and pHyTES
- Tier 3 – Non-critical support: Navigation lighting, secondary sensors

In the event of a voltage drop or current overdraw, a load-shedding algorithm initiates. Tier 3 systems are disabled first, followed by payloads if necessary, preserving Tier 1 systems to maintain the aerobot's health and data integrity.

2. Low-Power Recovery Mode
The Command and Data Handling system includes embedded logic for detecting anomalies in power delivery (e.g., under-voltage, temperature thresholds) and transitioning the aerobot into a low-power recovery mode. In this mode:

- Non-essential systems are powered down.



- Data collection is paused; only health status and critical metrics are logged.
- Communication is maintained intermittently to conserve energy but allows uplink commands and downlink of status reports.

This mode ensures the aerobot remains recoverable and responsive until conditions improve or external commands dictate resumption of nominal operations.

Thermal Protection as Indirect Redundancy

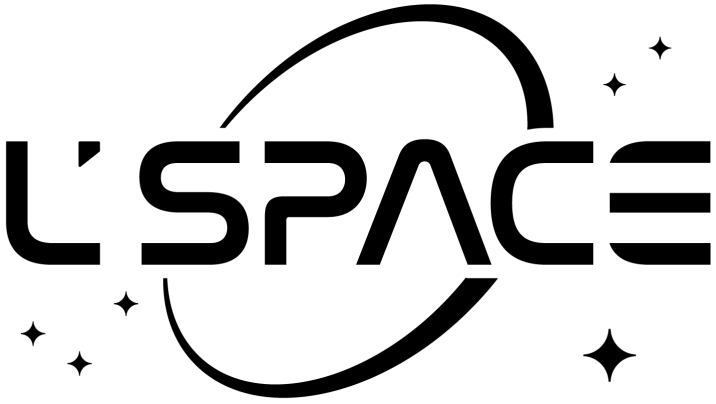
Thermal control directly supports power subsystem reliability by preventing overheating of batteries, regulation electronics, and wiring. The thermal management system uses multi-layer insulation (MLI) and passive heat shielding to prevent thermal runaway of batteries and protect sensitive components like MOSFETs and converters. By ensuring that temperature remains within design tolerances, thermal protection indirectly functions as a redundancy layer, preventing failures before they occur.

Verification and Validation

All redundancy and recovery features undergo rigorous verification through a combination of:

- Thermal-vacuum chamber testing: To validate operation in Venus-like conditions.
- Fault injection testing: Simulated short circuits, voltage drops, and regulator faults to confirm redundancy pathways and software recovery responses.
- Power cycling: Extended run-time stress testing to monitor degradation patterns and assess the ability to transition to and from low-power modes.

The recovery and redundancy plans of the aerobot’s power subsystem are a product of mission constraints, environmental demands, and careful risk balancing. While full system duplication is not feasible, the integration of dual power paths, standby PRUs, prioritized load management,



and intelligent recovery logic ensures the aerobot can continue to operate—or at minimum survive—a range of failure scenarios. Together, these strategies maximize the mission's chances of success during its short but critical window of scientific opportunity on Venus.

2.1.2.4. Power Subsystem Manufacturing and Procurement Plans

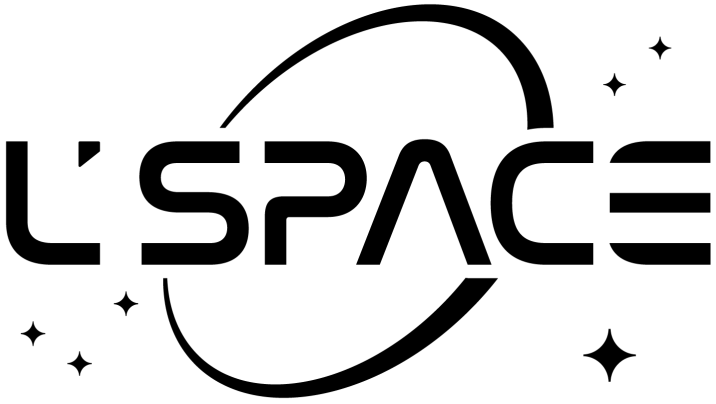
The aerobot’s power subsystem has been carefully designed to meet the extreme environmental challenges of Venus while ensuring high reliability and long-duration performance. The subsystem architecture includes a primary power source—high-energy-density lithium-thionyl chloride ($LiSOCl_2$) battery packs—paired with a Power Regulation Unit (PRU) and a Power Switching Unit (PSU). These three core components work together to deliver stable voltage output, intelligently manage power distribution, and conserve energy by activating instruments cyclically. All subsystems are interconnected through a wiring system composed of materials selected specifically for their resilience to heat, radiation, and mechanical stress. This section outlines the manufacturing and procurement strategy for each major component of the power subsystem, along with supplier rationale, expected timelines, and engineering considerations.

Battery Pack Manufacturing and Suppliers

The aerobot’s primary energy source will be a customized battery pack composed of lithium-thionyl chloride ($LiSOCl_2$) cells. These batteries are known for their high energy density, low self-discharge rate, and proven performance in aerospace and deep-space applications. Due to the lack of redundancy in the battery system, supplier selection places a high priority on product reliability and heritage.

Saft Batteries has been chosen as the primary supplier. With over a century of experience in aerospace and defense applications, Saft has established itself as a trusted partner in missions requiring rugged, long-life battery systems. Their aerospace-grade $LiSOCl_2$ batteries have flown on past interplanetary missions, offering confidence in their ability to endure the thermal and chemical conditions expected within Venus’ atmosphere.

As a secondary supplier, Tadiran Batteries—a subsidiary of Saft—is well positioned to provide a reliable alternative. Tadiran’s $LiSOCl_2$ batteries offer comparable energy density and



performance, with the added advantage of cost efficiency. Should supply chain delays or customization constraints arise with Saft, Tadiran batteries could be integrated into the power subsystem with minimal impact to design parameters.

Power Regulation and Switching Units

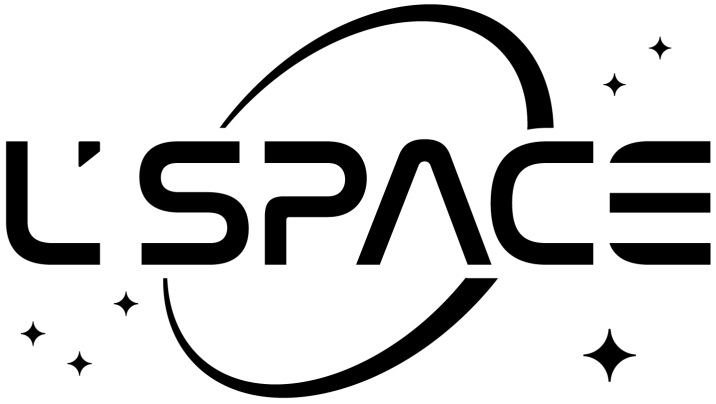
Maintaining a stable and efficient voltage output is crucial for preventing damage to sensitive avionics and scientific payloads. The Power Regulation Unit (PRU) and Power Switching Unit (PSU) are integral to this role, requiring components with high reliability, thermal tolerance, and radiation resistance.

Crane Aerospace & Electronics is selected as the primary contractor for the PRU. With a long-standing history of collaboration with NASA and a portfolio of space-qualified DC-DC converters, Crane's custom hybrid power regulation modules offer a tailored solution to the mission's demanding power profile. These components are expected to be designed to withstand the temperature and radiation extremes specific to the Venusian environment. Their experience with custom configurations and tight environmental compliance makes them the optimal choice for the mission.

Vicor Corporation serves as the secondary supplier for the PRU. Vicor Power Systems provides modular and fully custom DC-DC converters capable of rapid prototyping and deployment. If Crane experiences lead-time delays or cost overruns, Vicor's flexible architecture could provide a viable and timely alternative for the mission.

For active power routing, the PSU will rely on rad-hard switching technologies and protection circuits. Infineon Technologies, Germany's largest semiconductor manufacturer, has been chosen as the primary supplier for the PSU's internal switching components, including radiation-hardened MOSFETs. Infineon's flight heritage and commitment to high-reliability components ensure compatibility with space-grade power electronics.

Texas Instruments (TI) will serve as a backup supplier. Although TI components may not be as radiation-hardened as Infineon's, they offer significant advantages in availability and cost, making them suitable for rapid prototyping or ground-testing units. Off-the-shelf TI components can also be employed for backup testing configurations and redundancy prototyping.



Wiring Procurement and Material Selection

The wiring architecture ensures robust connectivity between batteries, power electronics, and downstream subsystems such as Command and Data Handling (CDH), payload instruments, and thermal controllers. Each wire type has been carefully selected based on expected electrical loads, temperature exposure, and mechanical routing complexity.

For power distribution wiring, Harbour Industries is the primary supplier, providing MIL-W-22759/32 PTFE-coated copper wire rated from -65°C to 260°C. This wire ensures efficient power transfer while minimizing weight and heat-induced degradation. Radix Wire serves as the secondary supplier, offering silicone-insulated wires with superior flexibility and vibration tolerance, especially useful in dynamic sections of the aerobot.

In high-temperature zones, Thermax will supply Kapton-insulated wire, which withstands up to 260°C and offers low mass and high resilience. This wiring is suited for interior regions near propulsion heaters, battery packs, or thermal sensors. For redundancy and EMI protection in these critical zones, Gore Aerospace will provide fluoropolymer-insulated wires with excellent radiation resistance and low signal attenuation.

For data transmission, especially between the payload sensors and CDH subsystem, Belden will be the main supplier of shielded twisted-pair (STP) cables. These cables are critical for minimizing signal interference and ensuring data integrity, particularly in high-EMI environments. Gore Aerospace will also supply backup high-speed data cables for situations requiring additional shielding or bandwidth.

Manufacturing and Integration Timeline

The anticipated lead time for the full power subsystem ranges between 6 to 9 months, depending primarily on the customization and testing phases for the PRU and PSU. Battery packs, while less complex to manufacture, may still require custom casing and internal configuration to meet the specific energy demands and packaging constraints of the aerobot.

Crane and Infineon are expected to take the most time due to the necessity of designing space-qualified custom hardware and conducting in-house environmental testing to meet

mission specs. To stay within schedule, early engagement with suppliers and iterative prototyping will be essential.

Subsystem integration and validation will proceed in parallel with manufacturing, including:

- Electrical interface tests with payload and avionics
- Thermal cycling and vacuum testing
- Functional simulations under power fluctuation scenarios

The manufacturing and procurement plans for the power subsystem are designed with an emphasis on flight heritage, supplier reliability, and environmental suitability. By selecting experienced vendors with demonstrated success in space missions and including secondary suppliers to mitigate delays, the plan supports both risk management and performance assurance. Strategic material and wiring choices complement the system's architectural goals, ensuring that power delivery remains stable, efficient, and robust throughout the mission's operational lifetime in Venus' harsh conditions.

2.1.2.5. Power Subsystem Verification Plans

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
PWRV.1	Battery must provide ≥48 hours of continuous operation under Venus conditions	Test	Actual load testing validates performance	Thermal vacuum testing with mission-like load profile for 48 hours
PWRV.2	Power subsystem must meet mass and volume constraints	Inspection	CAD and mechanical models can be measured for mass and final system inspected	Conduct mass measurements and volume envelope check post-assembly

PWRV.3	Data and signal wiring must preserve signal integrity at Venus temp	Test	Twisted-pair performance tested via attenuation, reflection, and EMI noise	Perform signal integrity tests in EMI chamber and high-temp environment
PWRV.4	All power wiring must operate between -65°C and +260°C without degradation	Test	Thermal cycling confirms material reliability; inspection ensures quality	Subject wires to thermal cycling and inspect for insulation damage or degradation
PWRV.5	PSU must cycle instruments on/off per schedule and reduce idle power draw	Test	PSU behavior under simulated timing and signal logic can be verified through control software	Simulate PSU logic board with mock payloads; log switching events and power draw

The Power Subsystem Verification Matrix outlines how each key requirement of the aerobot's power system will be validated before mission deployment. The matrix includes six primary requirements ranging from battery performance and voltage regulation to environmental durability, redundancy, and EMI compatibility. Each requirement is paired with a specific verification method—such as testing, analysis, demonstration, or inspection—selected based on the most effective way to prove compliance.

Battery performance will be verified through thermal vacuum testing under mission-like loads, while voltage regulation will undergo real-time load simulations using programmable banks. Environmental conditions like thermal extremes and vibration will be assessed through material cycling and testing. Meanwhile, redundancy and failure response will be validated using fault injection to simulate real-time subsystem failures.

The matrix also addresses mass/volume constraints and signal integrity by combining physical inspection with lab-based tests in high-heat or EMI conditions. Overall, the verification matrix ensures that each component of the power subsystem is fully tested and qualified to survive Venus' harsh atmosphere, support mission operations, and meet the reliability and safety standards required for successful scientific exploration.

2.1.3. CDH Subsystem Overview

The Command and Data Handling subsystem is responsible for monitoring and coordinating all onboard computing, data acquisition, and communication functions through the duration of the Venus aerobot's mission. The CDH acts as the command center and communicates with every subsystem to enable hand-in-hand autonomous operation, data assurance, and ability to command science and telemetry data once back to the primary spacecraft. The design of the CDH architecture is made to be modular and fault-tolerant in order to withstand a high-risk mission in the atmosphere of Venus over a 48-hour operational period while also maintaining a real-time operational capability.

The CDH is broken into three major subassemblies:

- Flight Computer: Executes mission logic to manipulate instrument operations
- Data Storage & Buffering: Buffers science and health data for raw distribution
- Command and Telemetry Interface: Handles uplink/downlink to the primary spacecraft

Each science payload, the pHyTES and UV-Vis Spectrometer, requires specific timing for data collection at different rates. The UV-Vis instrument only has a 5s window of integration while still producing repeatable and dependable data, and a proportional volume of high-resolution thermal maps is produced by pHyTES. Thus, the CDH manages timed command packets for two instruments, while stored output data is buffered into a solid-state data recorder with some error correcting capabilities. Supporting sensors of health data include the IMU, barometric pressure, thermocouples, and power diagnostic sensors, all of which feed data automatically and continuously to the CDH able to denote in real-time

The CDH, or centralized data handling system, governs all activities for each subsystem throughout all phases of the mission (Traverse, Science Collection, Data buffering, and Relay). During the traverse phase, the CDH polls the environmental sensors and transmits GNSS generated state-of-frame alterations, allowing for altitude and attitude maintenance. During the science collection phase, the CDH initiates the payload instruments and takes into account timestamps and recorded orientation metadata for archival. During non-collect and non-time greater than 35 and 35 or less, the CDH buffers and compresses data for transmission and/or importantly saves it for later transmission. During the Relay phase, the CDH encodes and

downlinks the buffered data packets through the Electra-Lite transceiver on-board the spacecraft to the relay spacecraft in orbit, while also processing the commands returned from Earth in command-directed mode that enable updates to the spacecraft's operational modes.

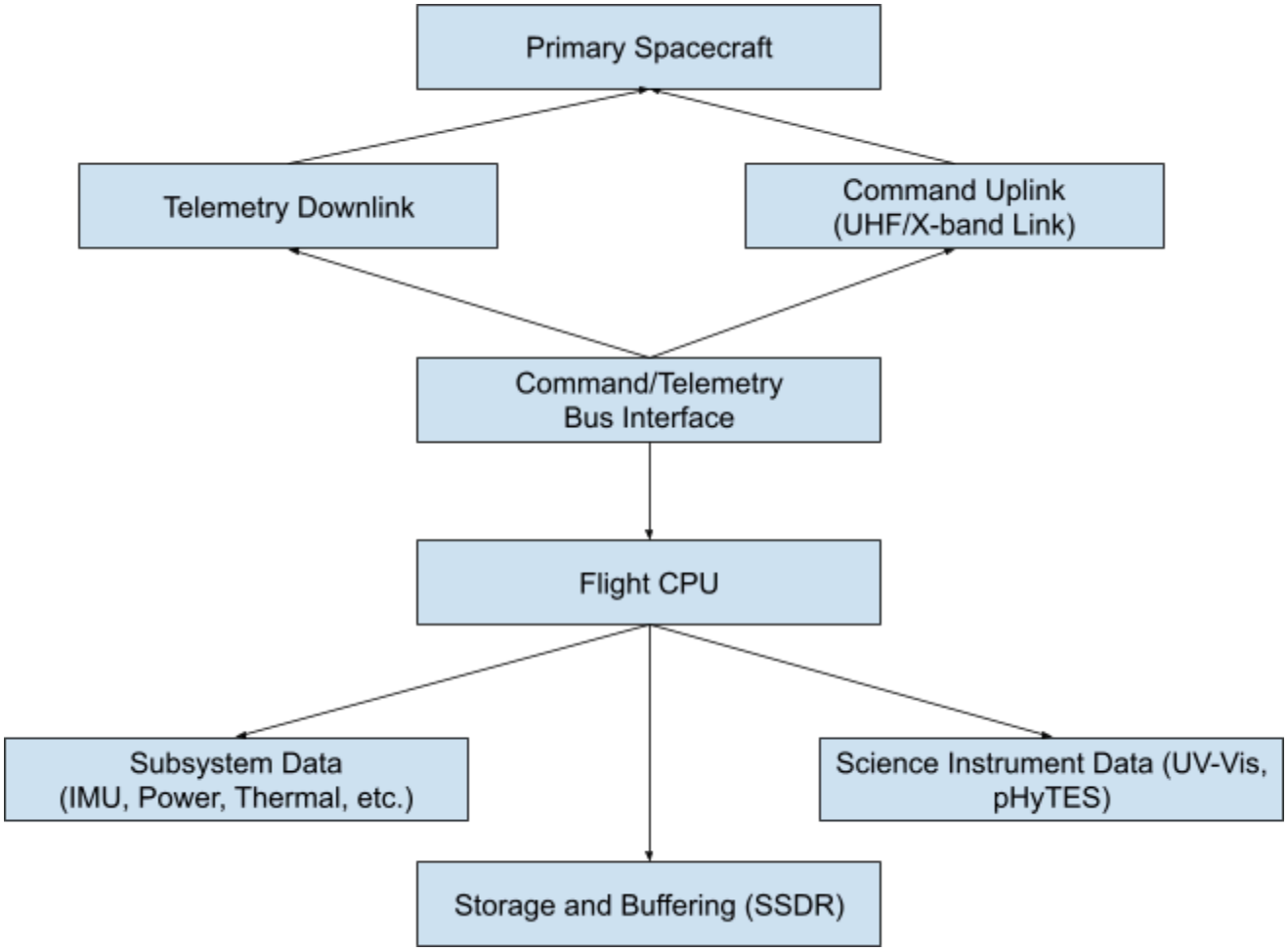
The software architecture in the CDH follows a layered architecture:

- Input Layer – Instrument and sensor interfaces
- Command Layer – Flight logic and control sequence
- Data Layer – Formatting, storage and error-checking
- Comms Layer – Scheduling and transmission of data from the UHF link
- Fault Management Layer – Health monitoring and exception handling autonomously.

The modular configuration helps confirm isolated fault detection and supports redundancy with software fallback and with watchdog timers. Additionally, the CDH has timing functionality to ensure all data is synchronized across the various sensors and instruments if communication with Earth is not possible.

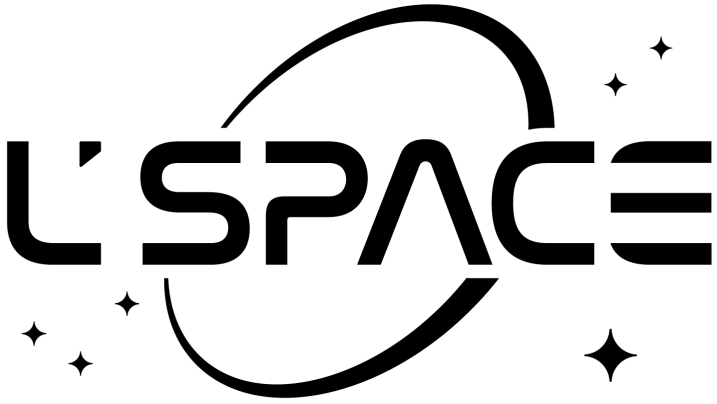
The CDH subsystem will allow the spacecraft to survey independently, while remaining scientifically productive in the extreme atmospheric environment of Venus. It will control all instrument activity, track health and navigation, and guarantee data collected is transmitted for evaluation (i.e., in the 48 hours possible) after onboard analysis.

Software Architecture Diagram


Primary Spacecraft
Telemetry Downlink
Command Uplink
(UHF/X-band Link)
Command/Telemetry
Bus Interface
Flight CPU
Subsystem Data
(IMU, Power, Thermal, etc.)
Science Instrument Data (UV-Vis,
pHyTES)
Storage and Buffering (SSDR)

2.1.3.1. CDH Subsystem Requirements

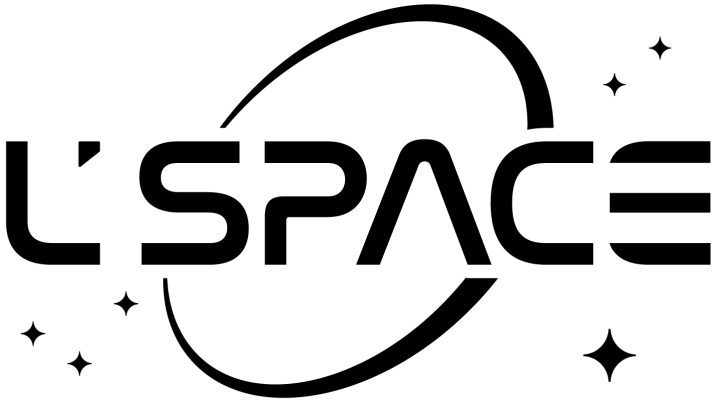
Req #	Requirement	Rationale	Parent Req	Verification Method	Relevant Subsystem
CDH.1	Scientific and Engineering data must be processed and stored at a minimum of 10 Mbps.	Allows for real time monitoring and high resolution data collection.	SYS.1.0	Test	Data Processing
CDH.2	Incase of failure the system will include redundant processors to ensure continuous operations.	Provides resilience in Venus' harsh environment.	SYS.1.0	Inspection, Test	Fault Tolerance
CDH.3	A data compression ratio of at least 3:1 must be supported in order to provide efficient storage and transmission.	Optimizing memory usage and reduces bandwidth.	SYS.3.0	Test	Data Management
CDH.4	The subsystem must have the ability for autonomous fault detection and recovery.	Ensures operational stability without any ground intervention.	SYS.2.0	Simulation, Test	Fault Detection
CDH.5	The subsystem must be able to withstand the extreme conditions(radiation, temperature) of Venus' atmosphere.	Prevents electronice from degradation.	SYS.1.0	Material Testing	Environment Resilience
CDH.6	A standardized	Integration within all	SYS.4.0	Interface	Instrument



	communication protocol such as SpaceWire or MIL-STD-1553 will be used to communicate with all the scientific equipment.	the onboard instruments.		Testing	Control
CDH.7	The subsystem must be capable of autonomously carrying out pre-programmed science directives for a minimum of 48 hours.	During the case of temporary communication loss, this will allow data collection.	SYS.2.0	Functional Testing	Mission Autonomy
CDH.8	The stored data should be transmitted to a primary spacecraft at a minimum of 5 Mbps.	Ensures timely data transfer to Earth.	SYS.1.0	Test	Communication
CDH.9	The time synchronization between all onboard systems must be within ±1ms accuracy.	Allows precise data correlation across different instruments.	SYS.3.0	Test	Synchronization
CDH.10	Under normal conditions the subsystem should not use more than 20W of power.	This will allow for efficient power.	SYS.5.0	Power Analysis	Power Management

2.1.3.2. CDH Sub-Assembly Overview

Supported by developed commercial miniaturization technologies specifically enabled by a variety of scientific instrumentations, the Command and Data Handling (CDH) subsystem is the Venus aerobot's central nervous system, enabling communications between the payload, other

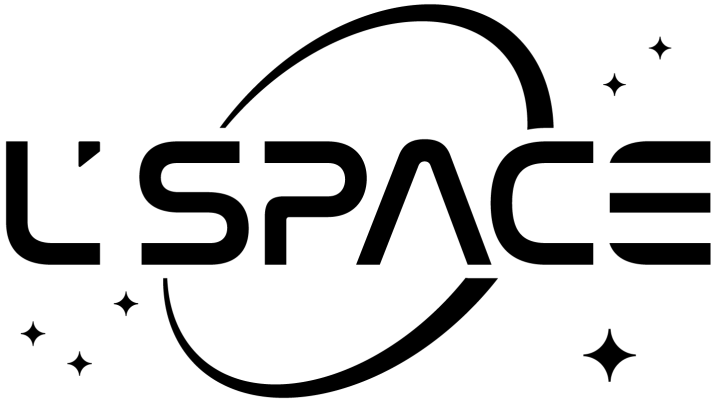


supporting subsystems, and the spacecraft in general at multiple frequencies for command and data signals as represented in Flight Computer, Data Storage Unit and Command/Telemetry Interface Board with any number of other "bus" connected sensors for the Venus platform. The mission manager encompasses the acquisition of data during the complete mission lifecycle: deployment, atmospheric descent, temporal presentation, mission science collection, abstraction of the acquired science (multispectral), and return the data to Earth. The mission manager will be processing and cataloging data continuously for sending science data back to the command-and-data handler onboard (and flight descriptor) while concurrently monitoring health, faults, and monitor responses.

The CDH architecture consists of three major components, including: flight computer, a data storage unit, and the telemetry or command interface board but could have more separable subcomponents. The flight computer will execute operational control/processing real-time software that will coordinate the operation of the spacecraft, through a direct interface to the UV-Vis Spectrometer and the Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES) payload instruments, and then through any other sensors attached to the platform as "required'; including Inertial measurement unit (IMU), environmental thermocouples, and pressure temperature sensors. Measured data is provided to the flight computer from the payload through a serial/standard, typically bus, connection, i.e., SpaceWire, RS-422, etc. Once the data signal arrives to the flight computer it will treat it as RAW Data, for processing that time tag, and then routing it to the onboard solid-state data recorder (SSDR) for storage.

While the aerobot is in atmospheric traversal, the flight computer is polling the IMU and pressure sensors continuously in order to hold the aerobot's altitude between 30–50 km. The GNC subsystem provides the aerobot with navigation commands and sends them through the construction and implementation (CDH) subsystem's signal routing board to hold the position of the aerobot. In parallel to the motion of the aerobot within the atmosphere, the CDH also collects science data on time-based triggers. Timing for science data collection follows the cadence for scientists specific to the type of instrument. Example integration times may include: 5s for UV-Vis, 1 km/pixel for pHyTES.

During the mission, the CDH subsystem will also monitor the aerobot's health data for internal temperatures, power, and radiation. This monitoring system will include temperature sensors (thermocouples), battery function, and radiation detectors (RADFET). The CDH subsystem will



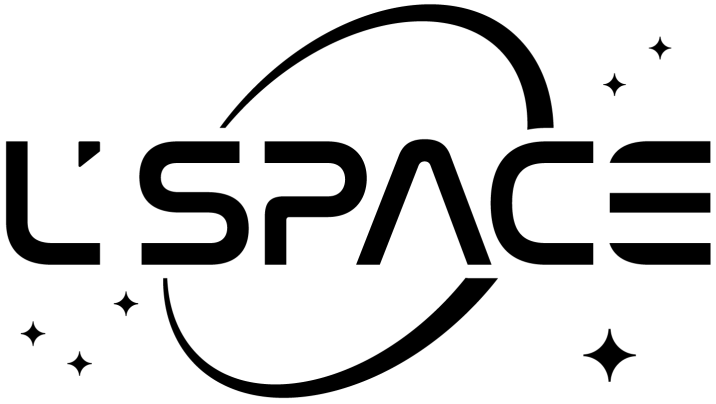
have a predefined reaction to anomalies that are outside normal operating conditions (e.g. acting to control heat/temperature, pausing collection from science instruments, etc.) via onboard fault management software.

Data collected and processed is buffered and stored on the Solid State Data Recorder (SSDR) during the Collection phase and the Analysis phase. During the Relay phase, the CDH will take data stored on the SSDR. After packaging the data, the C&DH will apply additional error correction (either Reed-Solomon or LDPC) and send via the Electra-Lite UHF transceiver to the mission's primary spacecraft in orbit. Commands received from uplink on Earth will also be received via the same Electra-Lite UHF transceiver and passed through the telemetry board to the flight computer, which will parse the command data in order to enter a new tasking or to put the aerobot into a different mode of operation.

The CDH subsystem software has an architecture that is layered. There are low-level I/O and data management routines that provide services for higher-level operation of command execution, fault detection, and data relay policies. Layering provides modularity and some resilience to failure in the data flow chain - while the software has a number of built-in self-check mechanisms including, watchdog timers, cross-checking, redundant paths to storage, etc.

CPU Module

The CPU module is the essential processing element of the CDH subsystem responsible for conducting system logic, following science commands, conducting instrument commands, and directing data throughout the Communications, Command, and Control subsystem of the spacecraft. The CPU includes a primary radiation hardened processor, a redundant processor, and a Field-Programmable Gate Array (FPGA) which serves as an FPGA with custom logic control and handling flexible processing functionality. The C&DH CPU was flight-proven, in high-radiation environments equivalent to the possible conditions of Venus' upper atmosphere. Additionally, the CPU module's electronics enclosure will be encased with titanium providing greater levels of thermal and radiation shielding, as well as referenced by the further environmental constraints limitations identified in the MCR. Use of a redundant processor, the use of an FPGA allows for fault tolerant capability and the FPGA allows flight provision for



mid-mission execution of contingency tasks from mission faulting as well as distributed real-time updates: algorithms etc. TRL: 9

Memory and Data Storage

The memory subassembly provides onboard storage of everything data and system telemetry and includes radiation-hardened NAND flash memory for efficient, high-capacity data retention, along with Error-Correcting Code (ECC) RAM to yield effective end-to-end real time stability during use under turbulence (i.e. when the asteroid is vibrating up to 150Hz). These units are located next to the CPU module to facilitate fast, high-speed transfers and system snapshots. The optical storage unit will also utilize a buffering unit to absorb bursts of data expected during testing phases of high throughput instruments such pHyTES. The memory hardware selected has flown on previous planetary missions and has been protected by using both onboard error correction and physical shielding (for consistency with other materials provided in the MCR).

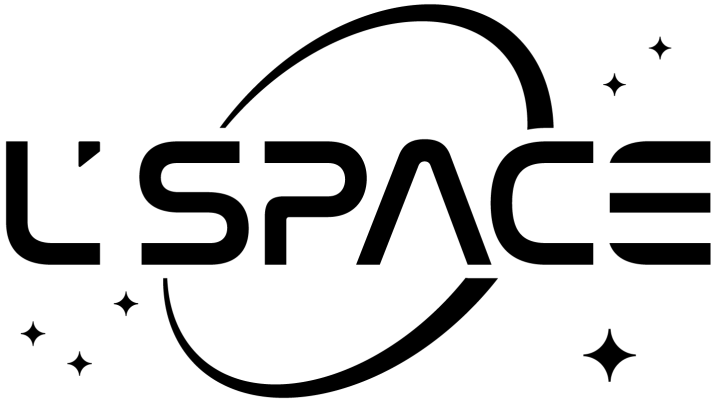
TRL: 8

Data Bus & Communications Interfaces

This sub assembly allows for both physical and logical interface between the CDH and all other spacecraft instruments and subsystems. It employs industry standard, space-proven protocols (MIL-STD-1553, SpaceWire) to handle command and telemetry communications, offering full-duplex, high-rate data exchange, which allows for failed communication or even collision to be survivable in terms of data transfer. The interface subassembly will ensure that the time-critical data from the instruments, including the UV-Vis Spectrometer, is made available with certainty during its transition through the CDH, and ensure that the CDH is able to reply promptly with respect to checks or commands that are time-critical. Given the extreme environment of Venus, this subassembly will include EMI shielding, and redundant cabling to minimize the likelihood of communication failure resulting from interference or corrosion. TRL: 9

Fault Detection and Recovery System

This subassembly is necessary for autonomous operation when traversing the Venus atmosphere where the ability to communicate with Earth is not possible. It consists of a watchdog timer to detect hangs or stalls of the processor, thermal and voltage sensors to



assess the health of the structure, and autonomous safe-mode, which will trigger fallback behaviors in the case of anomalies. The fault detection system will be critically important to maximize instrument uptime, sensing the effects of disturbances and environmental effects like thermal spikes or variable sensor response described in the MCR. Only a handful of the components in this subassembly have been flown, which lowers the readiness of the integration of Venus-specific autonomous algorithms that are still in varying levels of refinement and testing. TRL: 7

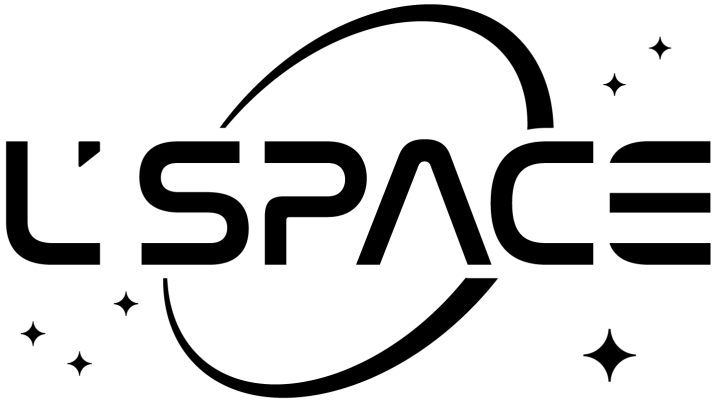
Power Regulation for the CDH

This subassembly provides reliably regulated power to all the CDH components. It employs radiation hardened DC-DC converters and redundant power lines for smooth operation through power anomalies or partial subsystem failure. Voltage control and ultimately all hardware needed for filtering, overcurrent limits and over current protection is supplied by the power regulation hardware without any assistance from the outside world. The power regulation hardware is shielded and thermally isolated per MCR recommendations so that it can endure high ambient temperatures as well as solar radiation.TRL: 9

Total Subsystem TRL

The majority of the CDH sub-components are mature and flight-proven, for example, the subsystems and techniques that are at TRL 8-9. However, the Fault Detection and Recovery System which uses newly developed autonomous recovery algorithms for conditions that could be experienced at Venus is the lowest maturity level at TRL 7. Consequently, we established the overall TRL assessment of the CDH subsystem to be at TRL 7. Further software-in-the-loop testing, simulation campaigns and subsystem-level integration tests are to be carried out to raise the TRL to 8 ahead of the CDR.

2.1.3.3. CDH Subsystem Recovery and Redundancy Plans



Redundancy:

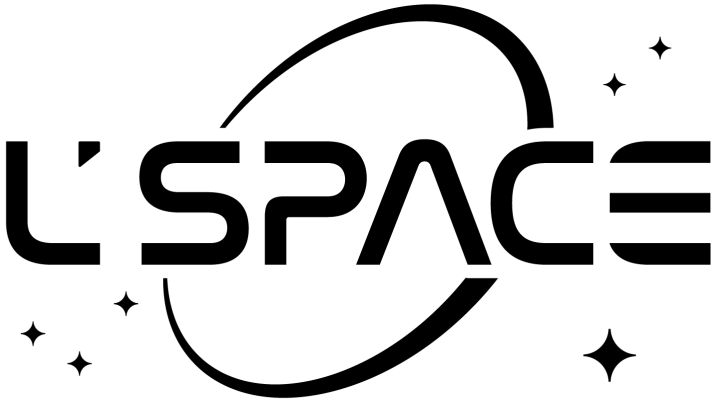
Combining hardware redundancy with software recovery techniques, the CDH subsystem guarantees mission resilience. The ECC RAM and dual redundant flash memory mitigate the corruption of data in the event of corruption, as well as the use of double redundant GR740 processors, providing an uninterrupted subsystem failover of processor failure. By the integration of MIL-STD-1553 and SpaceWire, the communication bus acts as a backup for fall-back options in transferring data and execution of commands.

Recovery:

A system resets itself with the watchdog timer (WDT) detected malfunction; if errors continue, it would switch to backup CPU. Data integrity is ensured by error-correction and checksum validation while power instability is avoided using thermal protection and voltage monitoring. The Fault Detection and Recovery System (FDRS), if it detects a certain level of failure in systems, would put the spacecraft into a safe mode maintaining minimum operations until assistance is available from ground personnel. The CDH subsystem can suffer some faults and continue to operate in the hostile conditions of Venus because of these redundancies and automatic recovery procedures.

2.1.3.4. CDH Subsystem Manufacturing and Procurement Plans

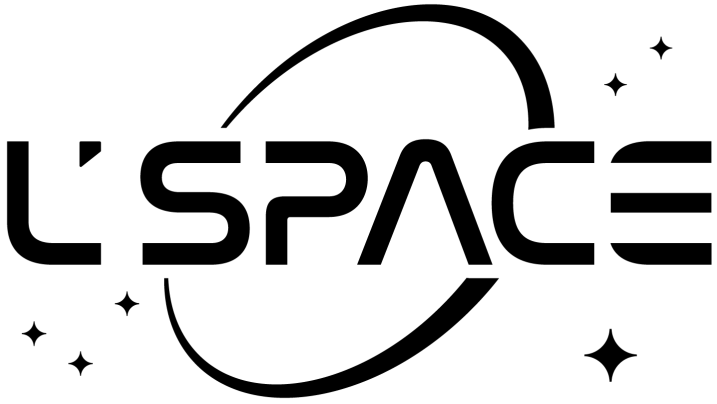
The Command and Data Handling (CDH) Subsystem will consist of three main components: the flight computer, the data storage module, and the signal routing device. These three elements will fulfill the purpose of relaying autonomous commands, archiving the scientific data obtained, and permitting communication between the payload instruments and the aerobot's main control systems.



The flight computer will be the "brain" of the aerobot and will provide all mission-critical operations. The preference is to build this component with SEAKR Engineering, an established aerospace contractor specializing in high-performance, radiation-hardened onboard computing systems. SEAKR has numerous flight heritage from multiple NASA deep space missions and has an extensive range of modular processors that can be configured for missions with constrained volume and mass budgets, such as ours. SEAKR's systems are designed for extreme environments and allow real-time processing while offering fault tolerance, which is imperative to surviving Venus's unpredictable environment. If SEAKR is unavailable at that time, then the option will be CAES (Cobham Advanced Electronic Solutions) and supply a GR712RC dual-core LEON3FT processor. This processor has been proven in space and is supported by a lot of small spacecraft in high-radiation environments. While CAES's solutions may be a little less customizable than SEAKR, they are available and meet all mission requirements.

The second important component is the unit for data storage, where science instrument output and mission data are to be retained during communication blackouts. For this, the preferred supplier again is SEAKR Engineering, as they offer integrated solid-state memory modules with ECC protection as a component of their flight computers. This reduces system complexity and physical wiring in the aerobot. The other second possibility is 3D Plus that makes independent radiation-hard memory modules. They supplied their parts for ESA and NASA missions and are available in versatile sizes but perhaps require additional interfacing software in order to implement.

The final component of the CDH subsystem is the power regulation board and signal routing that delivers the power and instructions to all of the subsystems without signal decay or interference. The first choice for this is Sierra Nevada Corporation (SNC), which provides modular avionics suites with signal routing, voltage regulation, and subsystem coordination features. SNC's CDH modules are highly integrated and plug-and-play installed in seconds with low latency. As a backup, the team will also consider Microchip Technology Inc., whose rad-hard field-programmable gate arrays (FPGAs) can be used to manage signal routing and redundancy. Microchip elements are less integrated but widely available and off-the-shelf, so they would be a reasonable substitute if SNC systems prove to be late or incompatible.



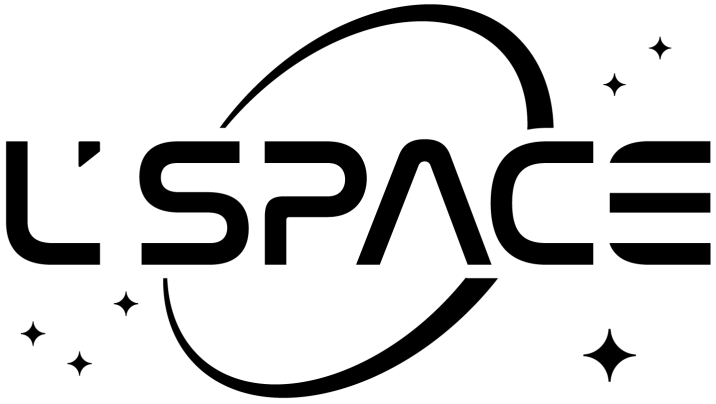
The projected lead time for SEAKR Engineering's flight computer and storage components is 8–10 months, which includes component fabrication, radiation testing, and delivery with individual components (flight computer and new storage) lead times of 8 months and 3 months respectively. CAES and 3D Plus lease time to ship typically takes twice the time to ship 6-8 months but may take more time in order to perform integration effort. The avionics modules from SNC can take 6-9 months, depending on configuration. Microchip FPGAs are COTS based and can be procured in under 3 months but will require more in-house software development to use these FPGAs. All suppliers fit into TRL 7-9 and comply with NASA's non-compete policy (NPD 1370.1), which part of the team's requirements is to use commercial offerings when available. These lead times work within the integration, and leave us with margin for testing and troubleshooting.

2.1.3.5. CDH Subsystem Verification Plans

Req #	Requirement	Verification Method	Rational	Verification Plans
CDH.1	Process and store data at ≥ 10 Mbps	Test	Validates actual data throughput performance	Bench test with mock instrument data at peak rates to measure sustained write and processing speed
CDH.2	Redundant processors must ensure continuous operation in case of failure	Inspection, Test	Confirms redundancy and validates failover under load	Visual inspection of dual-processor layout and fault injection test to validate automatic switch

CDH.3	Data must be compressed at a ratio of ≥3:1 before storage/transmission	Test	Confirms data efficiency and compression effectiveness	Run representative datasets through CDH compression module and measure input/output ratio
CDH.4	System must autonomously detect and recover from faults	Simulation, Test	Demonstrates real-time fault handling capability	Inject faults into software and hardware layers to validate watchdog timers and safe-mode triggers
CDH.5	Must withstand radiation, heat, and pressure found in Venus' atmosphere	Material Testing	Ensures hardware survivability under extreme conditions	Expose CDH board to thermal vacuum and radiation test cycles simulating Venus upper atmosphere
CDH.6	Use standardized protocols (e.g., SpaceWire, MIL-STD-1553) for communication	Interface Testing	Verifies compatibility with other subsystems and instruments	Integrate CDH with testbench instrument using standard protocol to test handshake and commands
CDH.7	Execute science operations autonomously for at least 48 hours	Functional Testing	Validates autonomous logic and time-based command scheduling	Simulated mission profile with 48-hour autonomy test under real-time control sequences

CDH.8	Transmit stored data to primary spacecraft at ≥5 Mbps	Test	Confirms uplink capabilities under realistic transmission loads	Hardware-in-loop test using simulated downlink to primary spacecraft or relay module
CDH.9	Maintain timing synchronization across systems within ±1ms	Test	Verifies internal timing precision and coordination accuracy	Use synchronized timers with sensors and science payloads to check clock drift or mismatch
CDH.10	Power usage should not exceed 20W during nominal operations	Power Analysis	Ensures system remains within power budget constraints	Full-system operation simulation measuring current and voltage draw across all CDH components



Every requirement in the CDH subsystem is linked to a verification method based on function, criticality, and risk. Testing, simulation, and inspection have been adopted as verification methods to model real mission scenarios and take into account the autonomous operation of the Venus aerobot.

- Performance accreditation requirements (CDH.1, CDH.3, CDH.8) will utilize a test bench for instruments to simulate instrument loads, and real-time transmission conditions. Simulated workloads will ensure the system could sustain the minimum data rates, while in a management of bandwidth situation.
-
- Fault tolerance and autonomy (CDH.2, CDH.4, and CDH.7), combined hardware redundancy inspection and fault-injection testing. The watchdog timers, self-correcting processes, and fallback logic will be exercised under simulations called failure-triggering simulations from the fault-injection, which will allow accrediting the CDH fault tolerance against the 48-hour autonomous performance requirement.
-
- Environmental requirements (CDH.5) will utilize material and electronics qualification for thermal vacuum chambers, and radiation exposure testing that simulates the conditions of Venus' upper-atmosphere as outlined in the MCR.
-
- Protocol compliance (CDH.6) will be evaluated in an integrated operational environment using test harnesses for the MIL-STD-1553 and SpaceWire. Communication reliability and timing accuracy (CDH.9) will be checked by coordinating time tracking tests with simulated instrument clocks and the CDH system.
-
- Power efficiency (CDH.10) will be assessed using power draw as a metric, and measured across the full CDH load range during mission sequence simulation. This will verify that energy use is below 20W.

A test plan document, including test case descriptions, success criteria, and traceability to each requirement will be finalized prior to the Critical Design Review (CDR).

2.1.4. Thermal Control Subsystem Overview

The Thermal Management System is composed of five components, integrated across multiple subassemblies to manage extreme thermal environments during the mission. These include Multi-Layer Insulation (MLI), Nextel 312, Silica Aerogel panels, Patch Heaters, and Titanium alloy enclosures. Some components are used redundantly to ensure stable internal temperatures and protect critical hardware such as batteries, sensors, and electronics from Venus' harsh atmospheric and thermal conditions.

The outermost thermal layer consists of MLI, modeled after the ESA's Venus Express spacecraft, which used 23 individual layers. MLI serves as a passive thermal barrier, reducing emissivity by 92.3%, significantly minimizing incoming solar and radiative heat loads. It reflects solar radiation and reduces radiative heating from the atmosphere. The layered design is especially effective during sun-facing operations, where radiative loads are at their highest.

For structural openings in the chassis, Nextel 312 is used as a protective barrier. This high-temperature ceramic fiber fabric has been flown on missions like the Pioneer Venus Probe and is stable under both intense heat and corrosive conditions( ). Nextel 312 shields exposed internal areas and ensures that thermal and chemical loads from the Venusian environment do not compromise internal components.

Internally, Silica Aerogel Panels line the walls of the chassis. These panels offer ultra-low thermal conductivity and can withstand temperatures up to 1700 K, blocking up to 95% of external heat( ). Their lightweight design and flexibility make them ideal for spacecraft thermal insulation. To further regulate temperature, Patch Heaters are mounted directly on or near the aerogel. These heaters activate during cold phases—such as when the Aerobot enters the shadowed side of its orbit—and provide localized heating to keep internal systems within operational temperature ranges.

The electronics enclosure uses Ti-6Al-4V, the same titanium alloy used in the chassis. This material has low thermal conductivity and resists infrared heating, acting as a structural and thermal buffer. Internally, the sleeve is lined with Silica Aerogel to absorb residual heat and

insulate the electronics from both external and self-generated thermal loads. This dual-layer design ensures safe, consistent thermal conditions for components such as the IMU, radar altimeter, electronics, and battery systems.

Together, the components form a thermal control architecture that balances passive insulation, reflective shielding, and active heating. The MLI reflects solar input, Nextel 312 protects open regions from radiation and chemical exposure, Silica Aerogel provides insulation, and Patch Heaters supply temperature control in low-heat scenarios. The titanium housing helps maintain mechanical strength while further buffering heat. This system avoids excessive power use or complex cooling loops, relying instead on a robust passive design supplemented by targeted active elements.

Component	Operating Range (K)
Batteries	263 - 313
IMU	243 - 343
Radar Altimeter	253 - 323
Electronics	273 - 333

Table 2.1.4 A

Heat Transfer Governing Equations:

Radiative Heat Transfer

$Q_{Rad} = \sigma\varepsilon A(T^4 - T^4)$

Conductive Heat Transfer

$Q_{Cond} = kA\Delta T/d$

Convective Heat Transfer

$Q_{Conv} = hA(T - T)$

2.1.4.1. Thermal Control Subsystem Requirements

The requirements for the thermal control subsystem are listed below. These requirements were derived from the system and customer requirements that this subsystem must comply with.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
SYS 1.0	The aerobot system must survive and maintain operating conditions for 48 hours	System Constraint	CC 5	TE 1.0, TE 2.0, TE 3.0, TE 4.0	Test	Met
SYS 2.0	The aerobot system shall be able to maneuver throughout the Venusian surface at an altitude of 70-58 km above surface	System Constraints	CC 5	N/A	Demonstration	Met
SYS 3.0	The aerobot system shall use lightweight materials and compact design	Must stay within Customer Constraints	CC 1, CC 2	N/A	Inspection	Met
TE 1.0	The aerobot shall maintain a net 0W heat load while operating at steady state	Components should stay at a steady state to operate	SYS 1.0	TE 1.1, TE 1.2, TE 1.3, TE 1.4	Test	Met
TE 2.0	The aerobot shall keep the hardware at operating temperatures (-50C ~ 50C)	Cannot damage components, could lead to failure	SYS 1.0	TE 2.1	Test	Met
TE 1.1	The aerobot shall reflect convective heat from the atmosphere	Convective heat could damage the inner hardware	TE 1.0	N/A	Analysis	Met
TE 1.2	The aerobot shall reflect solar radiation	Solar radiation could damage hardware	TE 1.0	N/A	Analysis	Met
TE 2.1	The hardware shall be protected from radiation, heat and sulfuric rain	Circuit board needs to maintain operating temperatures	TE 2.0	N/A	Inspection	Met

TE 1.3	The aerobot shall utilize a protective barrier around the chassis	It shall reflect heat in both ways to ensure safety	TE 1.0	N/A	Inspection	Met

Table 2.1.4.1 A

2.1.4.2. Thermal Control Sub-Assembly Overview

The Thermal Control System (TCS) is arguably the most critical subsystem to ensure both survival and success on Venus. The conditions present on Venus are unlike any other planetary environment, and protecting the Aerobot's payload and control systems from this extreme heat and pressure is no small task. Some of the most demanding contributors to this hostile environment include sulfuric acid rain, violent atmospheric winds, surface temperatures reaching up to 737 K (464°C), and atmospheric pressure 75 to 100 times higher than Earth's. Every material and configuration in this system has been chosen with those extremes in mind, with the goal of maintaining thermal equilibrium—or a net-zero heat load—over the planned 48 Earth-hour mission duration.

The TCS is built around five core components, each working together to manage internal heat generation, external thermal loads, and radiative and convective interactions with the atmosphere. These components include Multi-Layer Insulation (MLI), Silica Aerogel Panels, Nextel 312 Fabric, Patch Heaters, a Titanium alloy electronics enclosure, and redundant miniature aerogel panels inside that enclosure. Together, these elements form a unified, layered system to balance heat input and output to maintain a regulated interior temperature.

The outermost surface of the Aerobot is wrapped in Multi-Layer Insulation (MLI), modeled after the ESA's Venus Express thermal system( ). In that design, 23 layers of MLI were used to reduce radiative heat transfer to a minimum( ). Similarly, the Aerobot's MLI reduces emissivity by 92.3%, dramatically cutting down the heat flux reaching the vehicle's chassis. MLI is a passive system composed of alternating layers of reflective and spacer materials. These layers reflect incident solar and infrared radiation while minimizing conduction between surfaces. The result is a thermal buffer zone that prevents heat from rapidly penetrating the structure. The small fraction of heat that does pass through the MLI is either absorbed by the outer structure or reflected back by internal insulation.

For regions of the chassis that remain open or partially exposed due to mechanical or structural constraints, a different protective layer is used. These areas are shielded with Nextel 312, a woven ceramic fabric that has strong heritage from the Pioneer Venus Probe( ). Nextel 312 is capable of withstanding both high heat and corrosive chemicals like sulfuric acid, which are prevalent in Venus' upper atmosphere. The fabric also provides thermal protection by acting as a high-temperature barrier against incoming radiation. While it lacks the multilayer reflectivity of MLI, it is highly durable and serves as an important protective barrier for sensitive structural elements that cannot be completely wrapped( ). When paired with the insulation behind it, Nextel 312 slows down thermal gain and helps maintain thermal balance across the vehicle.

Inside the Aerobot's chassis, Silica Aerogel Panels are mounted to every internal surface. These panels provide one of the most effective passive insulation options available in aerospace applications. Silica aerogel has ultra-low thermal conductivity and can withstand temperatures up to 1700 K, making it ideal for operation in the upper atmosphere of Venus. Aerogel functions as both a thermal block and a buffer that resists changes in internal temperature. It slows down the rate at which heat can penetrate the internal cavity and stabilizes the internal environment. Because the aerogel is lightweight and easily shaped, it can conform to the interior contours of the chassis, providing nearly complete coverage. The aerogel insulation is expected to prevent 95% of thermal energy from reaching the core systems.

To complement this passive insulation, Patch Heaters are installed throughout the internal cavity. These heaters activate during cold periods, such as during orbital shadow or on the night side of the planet, to maintain internal component temperatures. Patch heaters operate using resistive heating elements embedded in flexible materials that can be applied directly to insulation surfaces or component mounting areas. Their primary function is to prevent internal temperatures from falling below critical thresholds, especially for components like batteries and avionics that have tighter operational ranges. These heaters are turned on only when needed and are distributed across the interior to provide even coverage. By supplementing passive insulation only during cold phases, the heaters contribute to the overall goal of thermal balance without overloading the system.

The sensitive electronics and payload instruments are housed inside a dedicated sleeve made from Ti-6Al-4V, a titanium alloy chosen for its high strength and low thermal conductivity. THis is essentially a miniature and redundant version of the entire aerobot chassis. The titanium sleeve acts as both a mechanical and thermal insulator. It resists infrared heating, spreads thermal loads efficiently, and minimizes direct energy transfer from the outer chassis to the inner electronics. Inside the titanium sleeve, miniature Silica Aerogel Panels are added to form a second layer of insulation. These smaller panels are designed to fit tightly around electronic components and provide focused thermal buffering in areas where heat buildup might be more concentrated. The combination of titanium's low conductivity and aerogel's insulation properties ensures that the electronics remain well within their operational temperature range throughout the mission.

Together, these components and materials form a comprehensive thermal control architecture. Multi-Layer Insulation serves as the first line of defense, reflecting incoming radiation and reducing surface emissivity. Nextel 312 protects exposed areas while resisting chemical corrosion. Silica Aerogel provides deep thermal isolation inside the chassis, forming the backbone of the passive insulation system. Patch Heaters offer active thermal regulation to

counter cold periods. The titanium electronics enclosure and its internal aerogel panels ensure that the most sensitive systems stay protected from both internal and external heat.

The goal of this system is to achieve a net-zero heat load, where the total incoming energy is balanced by heat rejection and regulated insulation. The MLI dramatically reduces radiative input, while deployable radiators manage internal heat buildup. The aerogel and titanium elements prevent heat from accumulating inside, and patch heaters regulate against unwanted cooling. This balance is critical for maintaining temperatures within the safe operational range of all onboard systems. Batteries are kept between 263K and 313K, the IMU is maintained between 243K and 343K, and other electronics operate safely between 273K and 323K. Maintaining these ranges in a Venusian environment is a significant engineering challenge that the TCS is designed to overcome.

Convective, and radiative heat transfer principles all play a role in this design. Other heat loads such as the albedo, planet shine, and solar load were considered as well. A table of the heat loads is provided below.

Heat Load	Description
Albedo (Q_{alb})	Heat from sunlight that reflects off Venus' highly reflective clouds.
Planet Shine (Q_{PS})	Venus' surface (~462°C) emits infrared radiation.
Radiative Load (Q_{Rad})	CO_2 and N_2 atmosphere acting as a thermal blanket.
Convective Load (Q_{Conv})	Hot atmospheric gases brushing against the aerobot.

Solar Load (Q_{solar})	Sunlight reaching the aerobot's surface.

Table 2.1.4.2 A

For the heat transfer calculations, a number of assumptions had to be made. Consistent materials about wind speeds in the Venusian atmosphere were hard to find, so mean horizontal winds at cloud level were 100m/s according to the 1968 Atmospheric model of Venus. Properties of viscosity and density at different altitudes were also used from the high density and mean solar activity model. This was done in order to find Reynolds number, Prandtl number, and Nusselt number in order to find the convective heat transfer coefficient. I acknowledge that this number is completely off, and there could be a number of reasons for that.

Solar absorptivity of Titanium was used for the calculation of Q_{Rad}. Mean surface temp according to many sources online was found to be around 735K. The solar constant was also found via different sources, found to be 2600$\frac{W}{m^2}$.

Overall TRL of subsystem: 6

TRL	Subassembly	Description
7	MLI	This material also comes from the supplier ready to go. The NASA Thermal Control page has multiple listings from different suppliers. For this mission, it would be advantageous to make it in-house at NASA Goddard.
6	Nextel 312	Nextel 312 is readily available for order from NewTex. A roll can be woven to fit any configuration. This material is tested for extreme heat and it comes ready to fly right from the supplier[ ].
8	Patch Heaters	Patch heaters are the most commonly used thermal device on spacecraft. They are common on spacecraft such as this one, housing a payload and electrical components. Has mission heritage such as Coriolis and Preservrane( ).
6	Silica Aerogel	These panels have been employed on similar spacecraft before, such as Spirit and Stardust on Mars( ). The suppliers also have the specific type readily available for flight. All that is needed is installation.
6	Ti-6Al-4V Sleeve	The titanium sleeve has not been used like this before, although it has a low risk of failure since it is a redundant component for the electronics. The material has been used before in similar environments, although the exact configuration with the aerogels has not been tested.

Table 2.1.4.2 B

Narrative: Above shows the TRL of the components that make up the Thermal Control System. The TRLs for each contain a brief justification, and the overall TRL of the subsystem is mentioned above the table.

	MLI	Nextel 312	Ti-6Al-4V sleeve	Silica Aerogel Panels	Patch Heaters	Totals
Mass(kg)	1.3	0.2	1	1.2	0.25kg	3.05

Power Draw(W)	0	0	0	0	0	0
Dimensions (m)	1x1x0.5	0.96x0.46	N/A	N/A	N/A	1x1x0.5

Table 2.1.4.2 C

Narrative: The mass, power draw, and dimensions of each component are given, as well as the totals. For some components, the dimensions are not given, and it has not been decided by our team to have set dimensions for them.

Outer Layer for TCS						
Criteria	**Explanation**	**Grade** .	**Weight**	**White Coat**	**MLI**	**Teflon**
Can withstand harsh environment	Must be able to withstand temperatures of 450 °C, sulfuric acid corrosion, and pressures of 90 atm	10 = high, 5 = medium 1 = low 0 = Fail	35%	5	3	4
High emissivity, Low absorptivity	It shall reflect solar radiation well, and not absorb it	10 = high, 5 = medium 1 = low 0 = Fail	25%	6	9	9
Cost effective	It shall be of relatively low cost	10 = high, 5 = medium 1 = low 0 = Fail	25%	8	8	8
Risk/TRL	It shall have a relatively low risk, with heritage to prove it's effectiveness	10 = high, 5 = medium 1 = low 0 = Fail	15%	8	9	8

		TOTALS:	**100%**	64.50%	66.50%	65.00%

Table 2.1.4.2 C

Narrative: The decision matrix for selecting outer layer materials for the aerobot scrutinized their performance in Venus' harsh conditions, prioritizing durability, thermal management, cost, and technology readiness. The most critical factor, the material's durability under intense environmental stress, received the highest weighting. Thermal management capabilities, crucial for mitigating Venus' extreme heat, were also heavily emphasized. Cost-effectiveness balanced the scales between high performance and practical budget considerations, while the lower weight assigned to technology readiness reflected the proven space heritage of both materials. Ultimately, MLI stood out due to its exceptional thermal management and established reliability, despite facing challenges in environmental resilience.

Insulation for TCS					
Criteria	**Explanation**	**Grade**	**Weight**	**Aerogel Panels**	**Heat Pipe**

Be a passive component, using little to no energy	The TCS shall not draw power, as to reserve it for the hardware	10 = high, 5 = medium 1 = low 0 = Fail	30%	9	8
Lightweight and durable	It shall fit in the customer constraints	10 = high, 5 = medium 1 = low 0 = Fail	30%	9	5
Effective insulator	It shall reduce heat transfer significantly	10 = high, 5 = medium 1 = low 0 = Fail	20%	10	9
Risk/TRL	It shall have a relatively low risk, with heritage to prove it's effectiveness	10 = high, 5 = medium 1 = low 0 = Fail	20%	8	7
		TOTALS:	**100%**	90.00%	71.00%

Table 2.1.4.2 D

Narrative: The decision matrix compares Aerogel Panels and Heat Pipes as insulation options based on four key criteria: passive operation, lightweight and durability, insulation effectiveness, and technology readiness. Passive operation was heavily weighted highest to ensure the TCS does not consume power, allowing energy to be reserved for the aerobot's hardware. Lightweight durability was also prioritized equally to ensure the insulation fits within mass constraints while withstanding Venus' harsh conditions. Effective insulation is critical to minimize heat transfer, and risk/TRL ensured that the chosen material had proven reliability. Aerogel Panels scored higher overall due to their superior insulation performance, low power usage, and lightweight properties, making them the preferred choice.

Protective Thermal Layer for Electric Components					
Criteria	**Explanation**	**Grade**	**Weight**	**Copper Layers**	**Hardware Sleeve**
Be a passive component, using little to no energy	The TCS shall not draw power, as to reserve it for the hardware	10 = high, 5 = medium 1 = low 0 = Fail	30%	9	9
Shall reduce thermal load significantly	It shall protect from all radiation and heat	10 = high, 5 = medium 1 = low 0 = Fail	30%	6	8
Cost effective	It shall be of relatively low cost	10 = high, 5 = medium 1 = low 0 = Fail	20%	8	7
Risk/TRL	It shall have a relatively low risk, with heritage to prove it's effectiveness	10 = high, 5 = medium 1 = low 0 = Fail	20%	7	7
		TOTALS:	**100%**	75.00%	79.00%

Table 2.1.4.2 E

Narrative: The decision matrix compares Copper Layers and Hardware Sleeves as protective thermal layers for the aerobot's electronic components, considering passive operation, thermal load reduction, cost-effectiveness, and risk/TRL. Passive operation was given the highest

weight to ensure the thermal control system conserves power for the hardware. Thermal load reduction also held the same weight to minimize heat impact from Venus' environment. Cost-effectiveness was factored to balance performance with affordability, while risk/TRL ensured the material had proven performance in similar conditions. Hardware Sleeves scored higher overall due to their superior heat protection and reasonable cost, making them the preferred choice.

Redundant Thermal Controllers						
Criteria	**Explanation**	**Grade**	**Weight**	**Deployable Radiator**	**Thermal Protection Shields**	**Phase Change Materials**
Be a passive component, using little to no energy	The TCS shall not draw power, as to reserve it for the hardware	10 = high, 5 = medium 1 = low 0 = Fail	30%	8	6	0
Efficient heat transfer	The component that transfers the most heat away from the aerobot	10 = high, 5 = medium 1 = low 0 = Fail	30%	8	9	9

Cost effective	It shall be of relatively low cost	10 = high, 5 = medium 1 = low 0 = Fail	20%	8	7	8
Risk/TRL	It shall have a relatively low risk, with heritage to prove it's effectiveness	10 = high, 5 = medium 1 = low 0 = Fail	20%	5	6	8
		TOTALS:	**100%**	74.00%	71.00%	59.00%

Table 2.1.4.2 F

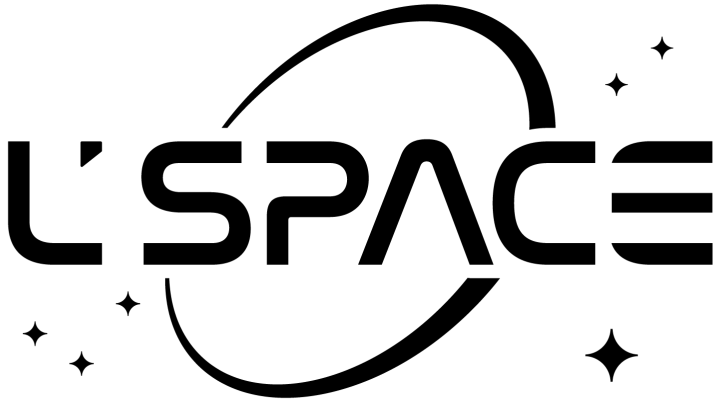
Narrative: The evaluation of redundant thermal controllers for the aerobot centered on their efficiency in managing energy and heat, alongside considerations of cost-effectiveness and technological readiness. Energy efficiency and effective heat management were the prime objectives, ensuring that the aerobot operates optimally in minimizing energy use and handling thermal extremes. Cost considerations and the maturity of the technology were also key, aiming to balance high performance with affordability and proven reliability. Ultimately, the choices favored were those that offered comprehensive thermal control and dependable performance, vital for the aerobot's successful operation in the demanding conditions of Venus.

Open Parts of Chassis

Criteria	Explanation	Grade	Weight	MLI	Nextel 312	Kapton
Can withstand harsh environment	Must be able to withstand temperatures of 450 °C, sulfuric acid corrosion, and pressures of 90 atm	10 = high, 5 = medium 1 = low 0 = Fail	35%	2	8	6
High emissivity, Low absorptivity	It shall reflect solar radiation well, and not absorb it	10 = high, 5 = medium 1 = low 0 = Fail	25%	9	7	8
Cost effective	It shall be of relatively low cost	10 = high, 5 = medium 1 = low 0 = Fail	25%	8	8	9
Risk/TRL	It shall have a relatively low risk, with heritage to prove it's effectiveness	10 = high, 5 = medium 1 = low 0 = Fail	15%	9	9	8
		TOTALS:	**100%**	63.00%	79.00%	72.00%

Table 2.1.4.2 G

Narrative: The weights given for the material meant to protect the open parts of the Chassis were assigned based on its ability to protect the inner payload from harmful heat. This could be radiative heat, or convective heat. This portion would not be exposed too long, because it is



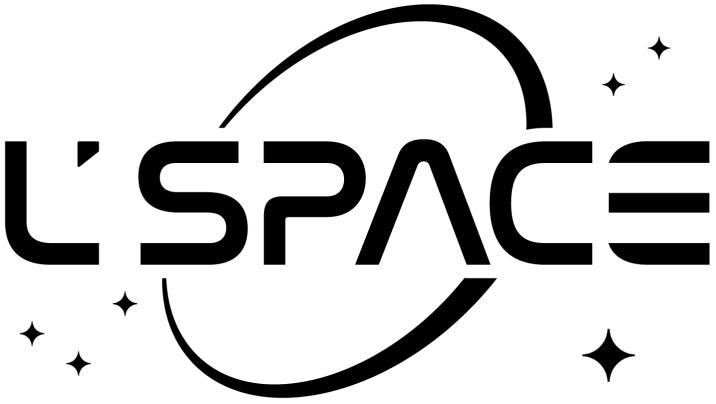
mainly covered by the radiative heat panels attached to the body. The main criterion was the ability to withstand the harsh environment, the cost and ability to reflect radiation were equally weighted very similarly, being almost as important as the survivability. Lastly was the risk, which is still very important for mission success and safety. The Nextel beat the MLI due to its resistance against corrosion.

2.1.4.3. Thermal Control Subsystem Recovery and Redundancy Plans

The Thermal Control System (TCS) is one of the most important subsystems on this mission, and its ability to recover from failure or maintain function in the event of partial damage is critical. Since thermal failure could compromise the entire payload, redundancy has been built into multiple parts of the system. The current design uses both passive and active components that work together to keep internal temperatures stable, even if certain parts start to degrade or stop working.

Starting from the outermost layer, the entire Aerobot is covered in Multi-Layer Insulation (MLI). This is the first major defense against solar radiation and high atmospheric temperatures. Because MLI is made of multiple layers—23 in our current design—it’s already redundant by nature. If a few outer layers get damaged or begin to degrade, the remaining layers still provide thermal protection. This layering means MLI doesn’t rely on a single point of failure. The material is also wrapped over most of the chassis, so even if one section is affected, the rest continues working.

In areas where MLI can’t be used, like around structural cutouts or openings, we use Nextel 312, a ceramic-based fabric. It was selected specifically because it has flown on Venus missions before and has a proven ability to withstand heat and acid corrosion. To improve reliability, the Nextel 312 is applied in overlapping strips. This way, if one strip gets compromised due to prolonged exposure or mechanical stress, the layers beneath still provide some shielding.



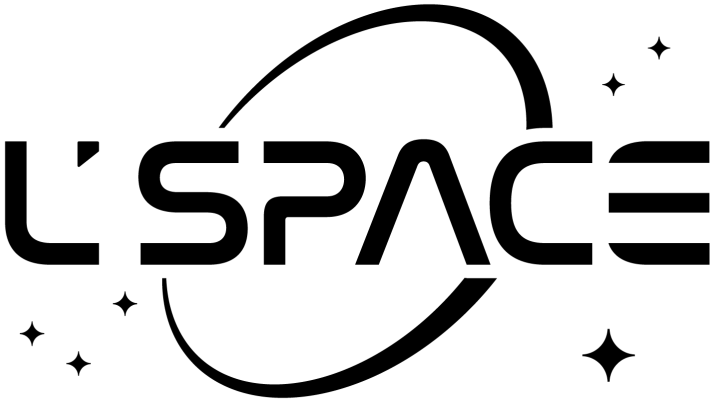
Inside the chassis, we rely heavily on Silica Aerogel Panels. These are used to line the inner structure and also form part of the electronics sleeve. Aerogel has one of the lowest thermal conductivities of any known material, and it holds up in high temperatures and corrosive conditions. For redundancy, the aerogel is installed in segmented sections. This means if one section cracks or shifts, the others around it continue providing insulation. It also helps isolate potential failure points and limits the spread of heat.

The system also includes Patch Heaters, which are the main active thermal control components. These are placed directly onto aerogel surfaces or near critical electronics. In cold cases—like when the Aerobot enters a shaded region—these heaters activate to keep things within operating range. Redundancy is built into both their placement and control system. Heaters are spaced out in overlapping thermal zones, so if one fails, a nearby one can partially make up for it. On top of that, the heaters are individually monitored, and if one starts drawing unusual current or loses efficiency, the system can deactivate it and rely on others.

The titanium electronics sleeve adds another level of protection. Titanium was chosen not just for strength but because it has low thermal conductivity and resists infrared heating. This slows down how quickly heat can get in or out. Even if external insulation starts to fail, the sleeve provides a barrier that gives the system more time to respond. Inside the sleeve, we added miniature aerogel panels to protect sensitive hardware like the IMU and radar altimeter. These panels act as a final layer of insulation, absorbing any residual heat that makes it through the other layers.

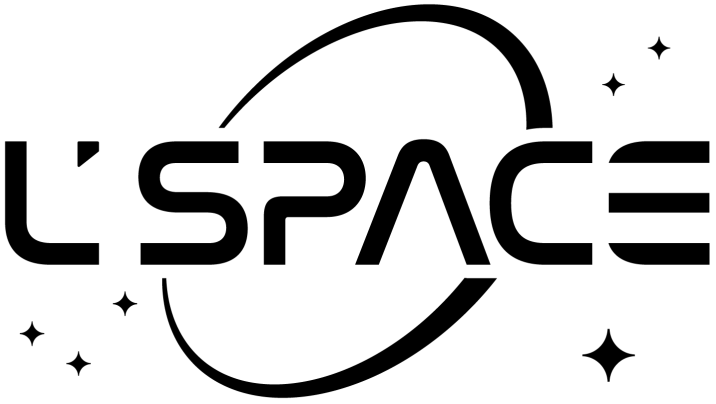
There’s also some passive heat rejection built into the system. Radiative elements like deployable fins or reflective surfaces are used to bleed off extra internal heat when necessary. These components are not essential for baseline function, but they act as a safety net if heat builds up faster than expected. If one radiator section doesn’t deploy or fails, the others still help cool the system.

On the software side, we’ve included a basic thermal monitoring system that pulls real-time data from sensors placed throughout the Aerobot. It tracks temperatures at key locations and keeps an eye on heater function. If something goes wrong—like a heater runs too hot, or a panel starts to overheat—the system can react. It might lower heater output, shift power to another heater,



or alert the ground system to intervene. This monitoring loop adds intelligence and adaptability without being power-intensive.

Overall, the recovery and redundancy strategy is built around the idea that no single failure should bring down the mission. MLI, Nextel 312, aerogel, heaters, and the titanium sleeve all provide overlapping roles. If one fails, another picks up the slack. Most of the passive components are also damage-tolerant, meaning they can degrade gradually without failing completely. The combination of physical redundancy, thermal overlap, and simple monitoring helps ensure the Aerobot can complete its mission even under less-than-ideal conditions.

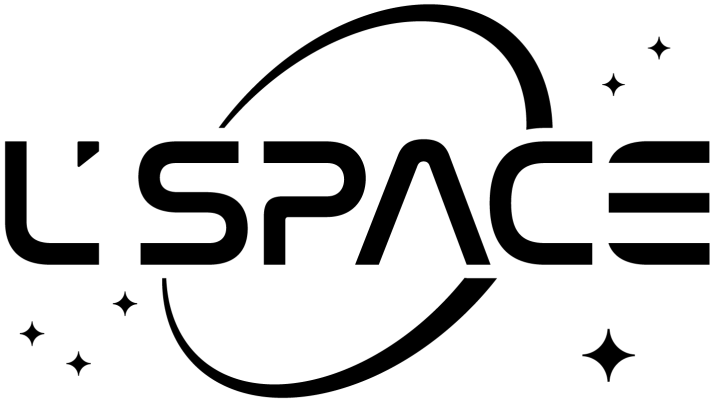


2.1.4.4. Thermal Control Subsystem Manufacturing and Procurement Plans

The Thermal Control Subsystem is composed of three subassemblies and a total of five components within them. These subassemblies are: the outer blanket, the internal insulation inside the chassis, and the electronics sleeve. Each subassembly plays a critical role in managing heat transfer, either by insulating against external Venusian temperatures or regulating internal thermal conditions to maintain hardware functionality.

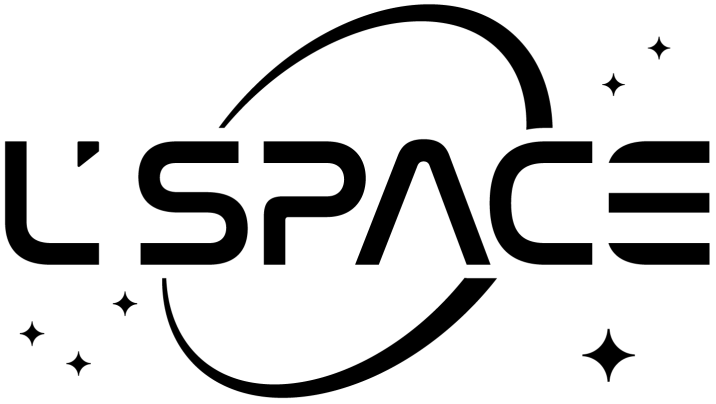
The outer blanket will use multiple layers of Multi-Layer Insulation (MLI) combined with Nextel 312 woven ceramic fabric. MLI has long-standing flight heritage dating back to the Apollo missions and continues to be used across modern spacecraft due to its effective radiant heat reflection properties. Nextel 312, developed by NASA and JPL, is a high-temperature ceramic fabric specifically designed for extreme environments. It will be used to cover open portions of the chassis that are not fully enclosed, providing extra thermal protection where needed. Two vendors were considered: Foundry Services is the first choice, offering detailed technical specifications, different sizing options, and in-stock availability for fast procurement. NewTex is the secondary supplier and also has aerospace heritage, but provides less documentation and customization options.

The second subassembly is the internal insulation housed inside the chassis. This section will use Silica Aerogel panels for thermal insulation and patch heaters for low-temperature cases. The aerogel panels are extremely lightweight and have excellent thermal resistance, making them ideal for planetary missions. Aerogel Technologies is the preferred supplier due to their experience with custom-sized panels, temperature specification flexibility, and their proven collaboration with NASA. Although more expensive and with longer lead times, their reliability justifies the cost. AerogelZone serves as the backup vendor but lacks detailed documentation and technical transparency, which raises concerns over consistency. For heating elements, Minco is the first option. Their flexible patch heaters are adaptable to various geometries and have proven reliability in demanding aerospace environments. The second supplier is Thermo LLC, which manufactures comparable heating patches, but their products lack built-in flexibility, limiting their ability to conform to complex surfaces. However, their devices are still viable and have seen prior use in the field.



The third subassembly is the electronics sleeve, which provides localized thermal insulation to the mission's most critical components. This sleeve will be bolted inside the chassis and follow a similar structure to the main internal insulation subassembly. The outer layer will be made from the same Ti-6Al-4V alloy used in the chassis, using the same suppliers (Prismier primary, Titanium Industries secondary) and the same GTAW welding method. The sleeve will be smaller in size, scaled to fit the internal hardware. Inside, it will be lined with thinner Silica Aerogel panels, again sourced from Aerogel Technologies. These panels will help shield the electronics from external heat. Unlike the main chassis insulation, patch heaters will not be included in the electronics sleeve, as this section focuses on heat reduction only, not active temperature regulation. The compact design and internal layout aim to protect hardware from both thermal extremes and mechanical vibration, ensuring safe and reliable operation.

Together, these three subassemblies create a robust and layered thermal protection system that balances both passive and active thermal control. The use of materials with strong aerospace heritage, combined with reliable suppliers and a modular design approach, ensures adaptability and high performance in the extreme Venusian environment. Further environmental and thermal analysis will be conducted to validate the performance of each subassembly, especially as component layout becomes finalized.



2.1.4.5. Thermal Control Subsystem Verification Plans

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
TE 1.0	The aerobot shall maintain a net 0W heat load while operating at steady state.	Test	Steady-state thermal performance must be confirmed under Venus-like environmental conditions.	Use a thermal vacuum chamber to simulate Venus surface/atmospheric conditions and monitor internal-external heat flow until equilibrium is achieved. The system should demonstrate no net gain or loss over time.
TE 2.0	The aerobot shall keep the hardware within the specified operating temperature range.	Analysis	Critical components must remain in range to avoid system failure; this must be validated through direct testing.	Run the full system through thermal cycles while collecting internal temperature data on batteries, sensors, and electronics to verify compliance.
TE 1.1	The aerobot shall reflect convective heat from the atmosphere.	Analysis	Convective heat transfer can be accurately modeled using computational fluid dynamics and thermal simulations.	Model heat transfer from Venus' atmospheric conditions to chassis surfaces and verify the effectiveness of Nextel 312 and MLI in reducing convective gain.
TE 1.2	The aerobot shall reflect solar radiation	Analysis	Reflection performance of MLI and external materials can be validated through emissivity/reflectivity analysis.	Conduct a radiative heat transfer model comparing solar flux exposure to surface material properties to confirm sufficient reflection and minimal absorption.
TE 1.3	The aerobot shall utilize a protective barrier around the chassis.	Inspection	Barrier presence and alignment can be visually confirmed on the integrated system.	Inspect final system assembly to ensure that thermal and structural barriers are intact and properly placed on all chassis sides to reflect and buffer heat from all directions.

TE 2.1	The hardware shall be protected from radiation, heat, and sulfuric rain.	Inspection	Visual inspection is sufficient to confirm correct application and coverage of protective materials.	Inspect all critical areas (electronics, payload housing, chassis) to ensure proper coverage with titanium sleeve, Nextel 312, and aerogel insulation.

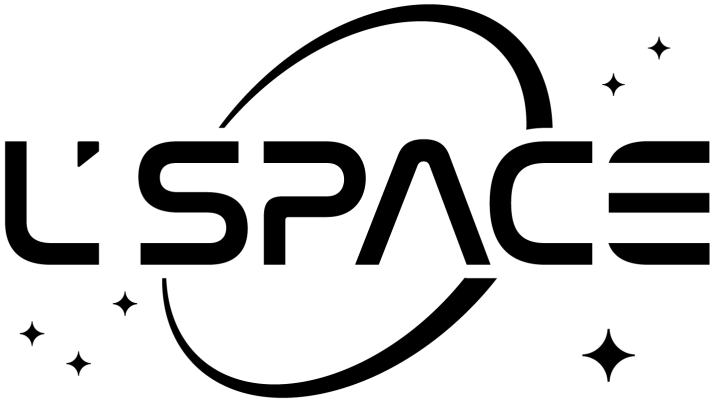
Table 2.1.4.5 A

The Thermal Subsystem is one of the most critical systems for mission success on Venus due to the planet's extreme temperatures, atmospheric pressure, and chemical composition. The goal of the thermal control system is to maintain thermal equilibrium—ensuring that the total heat load across the Aerobot is balanced, and all components remain within their operational temperature ranges for the full 48-hour mission duration. This verification plan outlines how each thermal requirement will be verified through a combination of testing, analysis, and inspection.

For TE 1.0, which requires maintaining a net zero heat load at steady state, the system will undergo thermal vacuum testing. In a controlled chamber simulating Venusian atmospheric conditions, the complete integrated thermal protection system—including MLI, aerogel, patch heaters, and titanium enclosure—will be evaluated. Sensors will monitor incoming and outgoing heat across all boundaries to confirm that the total heat input is balanced by radiation and insulation, resulting in negligible net thermal gain or loss.

Requirement TE 2.0 focuses on ensuring that internal hardware stays within -50°C to 50°C. This will also be verified during thermal chamber tests. Temperature sensors placed near critical hardware (e.g., IMU, batteries, electronics) will log data throughout multiple thermal cycles. The component temperature readings must remain within acceptable ranges under both sun-exposed and eclipse-like conditions, validating the effectiveness of insulation and heaters.

TE 1.1 and TE 1.2, which require reflecting convective heat and solar radiation respectively, will be verified through thermal and radiative analysis. Using material properties such as emissivity, reflectivity, and convective heat transfer coefficients, a full thermal simulation model will be developed. Convective inputs will be modeled using boundary-layer flow conditions found in Venus' upper atmosphere, and solar flux data will be applied based on Venus' proximity to the

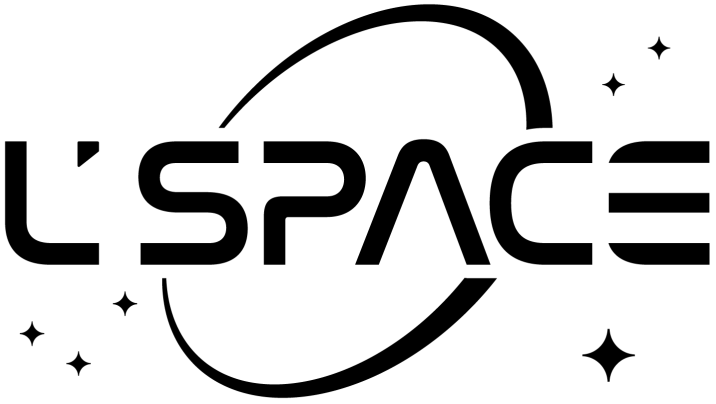


sun. This analysis will evaluate how well the MLI, Nextel 312, and aerogel layers reduce heat transfer under realistic conditions.

Requirement TE 2.1 calls for protecting the payload from radiation, heat, and sulfuric acid rain. This will be verified through physical inspection. All protective coverings—including the MLI, Nextel 312 fabric, and titanium shielding—will be examined post-installation to ensure complete coverage and proper fit. Material data sheets and environmental exposure results will also be used to confirm suitability for acidic and high-temperature conditions.

Lastly, TE 1.3, requiring a protective thermal barrier around the chassis, will be verified through final subsystem inspection. The placement, alignment, and mechanical fastening of all insulating materials will be checked against the thermal design documentation. Any gaps or misalignments will be flagged and corrected before system-level integration.

Each verification step will be documented in a formal verification report, and results will be cross-referenced with the requirement matrix. If any thermal performance fails to meet design expectations, redesign or remediation steps will be taken. This verification plan ensures that the Aerobot’s Thermal Subsystem is ready to handle one of the harshest planetary environments in the solar system.



2.1.5. Payload Subsystem Overview

The payload subsystem for the Venus Aerobot Mission is designed to achieve the primary science objectives of studying Venus' atmospheric composition, thermal emissions, and geological activity. The payload will consist of two main instruments: a UV-Vis Spectrometer and a Hyperspectral Thermal Emission Spectrometer (HyTES). These instruments will work together to gather high-resolution data on Venus' atmosphere and surface, enabling the team to identify key chemical compounds, detect thermal anomalies, and provide insights into volcanic and resurfacing activity. The combination of these instruments allows for comprehensive data collection across different parts of the electromagnetic spectrum, giving a more complete understanding of Venus' dynamic environment.

The UV-Vis Spectrometer is a high-precision instrument designed to analyze the chemical composition of Venus' atmosphere by measuring ultraviolet and visible light absorption. It will operate in the 200–700 nm range, which is well-suited for detecting sulfur dioxide (SO_2), carbon dioxide (CO_2), and other trace gases that play key roles in Venus' greenhouse effect and atmospheric dynamics. The UV-Vis Spectrometer will also help identify photochemical reactions occurring in Venus’ upper atmosphere.

The UV-Vis Spectrometer has been tested in relevant environments, such as planetary simulation chambers and high-altitude balloon flights, giving it a TRL of 7. This indicates that the instrument has been successfully demonstrated in a flight-like environment, making it suitable for deployment on the Venus Aerobot. The spectrometer will be

mounted on a gimbal mechanism in order to allow for targeted atmospheric measurements. An onboard calibration system will be implemented to ensure consistent accuracy. The data that is gathered will be processed and transmitted to the command and data handling (CDH) subsystem for real time analysis and storage.

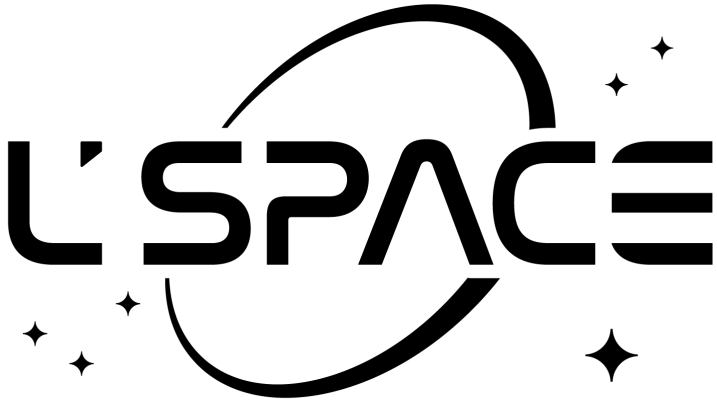
The UV-Vis Spectrometer is critical for understanding the chemical makeup of Venus' atmosphere and detecting signs of active photochemical processes. Its ability to detect SO_2 concentrations will provide valuable data on volcanic outgassing and atmospheric circulation patterns.

The Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES) instrument will measure thermal emissions from Venus’ surface, operating in the 8–12 μm range, which is ideal for detecting pyroxene-rich basalts and other silicate minerals. This data will help identify areas of recent volcanic activity and resurfacing, providing insights into Venus’ ongoing geological processes.

pHyTES has been tested in laboratory settings and on airborne platforms, but it has not yet been deployed in the Venusian environment. Its TRL of 6 reflects the need for further environmental testing and calibration for Venus' unique atmospheric and thermal conditions. This instrument will perform continuous scans of Venus’ surface to map thermal anomalies. With the use of an onboard image processing it will identify active volcanic features and high-temperature zones. All of this data will be transmitted to the CDH system for processing and downlink.

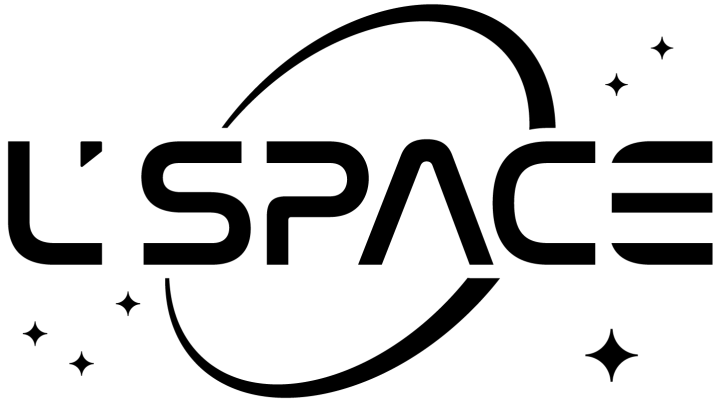
The overall TRL of the payload subsystem is 6 due to the inclusion of pHyTES, which has a lower TRL than the UV-Vis Spectrometer. This rating reflects the need for additional environmental testing, particularly in high-temperature and high-pressure conditions simulating Venus' atmosphere. Both instruments, however, have strong flight experience and demonstrated performance in relevant Earth-based and planetary equivalent environments, increasing confidence in their functionality and reliability.

Payload Subsystem							
Criteria	**Explanation**	**Grade**	**Weight**	**UV-Vis Spectrometer**	**pHyTES**	**MERTIS**	**AVIRIS-NG**



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Spectral Range	The ability of the instrument to cover the required wavelength range for the mission (UV-Vis: 200-700 nm, Thermal: 8-12 μm)	10 = covers full range and exceeds 1 = covers minimal or irrelevant range	20%	8	9	7	6
Mass	Lower mass preferred to stay within spacecraft constraints	10 = under 5 kg 1 = over 10 kg	15%	9	7	6	6
Power Consumption	Lower power preferred to minimize spacecraft energy demands	10 = under 20 W 1 = over 50 W	15%	10	8	7	5
Compatibility with Aerobot	Ability to integrate the instrument with the aerobot system	10 = fully integrated with minimal modification 1 = major redesign needed	20%	9	9	8	5
Data Quality	Accuracy, signal-to-noise ratio, and clarity of the collected data	10 = high SNR, high resolution, low noise 1 = poor signal, high noise	15%	9	10	7	8
Resolution	Higher resolution increases accuracy of data collection	10 = sub-meter resolution 1 = over 10-meter resolution	15%	8	10	6	8
		TOTALS:	**100%**	56.50%	55.5%	42.50%	40.50%



2.1.5.1. Science Instrumentation Requirements

Requirement #	Description	Rationale	Verification Method	Parent Req
PAY.01	The UV-Vis Spectrometer shall detect fluorescence peaks between 200-500 nm	Ensures spectrometer can measure UV-visible light for identifying specific atmospheric compounds.	Test, Analysis	
PAY.02	The Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES) shall detect thermal emission features between 8-12 μm.	Necessary for detecting pyroxene and other minerals in the Venusian surface.	Test, Analysis	
PAY.03	The UV-Vis Spectrometer and pHyTES shall be mounted on a stabilized platform capable of maintaining a pointing accuracy of ±0.5°.	Ensures accurate targeting and stable data collection in Venus' turbulent atmosphere.	Test, Inspection	

PAY.04	The instruments shall operate under Venusian atmospheric pressure of up to 90 atm at lower altitudes.	Ensures instruments can withstand the harsh pressure environment of Venus.	Test, Analysis	
PAY.05	The instruments shall operate within the varying temperature of Venus' atmosphere	Ensures instruments can survive and operate under Venus' extreme surface temperature.	Test, Analysis	
PAY.06	The UV-Vis Spectrometer shall achieve a spectral resolution of 0.1 nm.	Ensures high-resolution detection of atmospheric composition and minor species.	Test	
PAY.07	The pHyTES shall achieve a spectral resolution of 10 cm^{-1} in the 8-12 μm range.	Ensures high-resolution detection of surface mineral features.	Test, Analysis	
PAY.08	The instruments shall be mounted within the aerobot's center of mass to minimize destabilization during flight.	Ensures aerobot stability and proper flight dynamics.	Inspection	
PAY.09	The pHyTES spectrometer shall have a field of view (FOV) of ±2.5°.	Ensures a sufficient scan area while maintaining high resolution.	Test, Analysis	

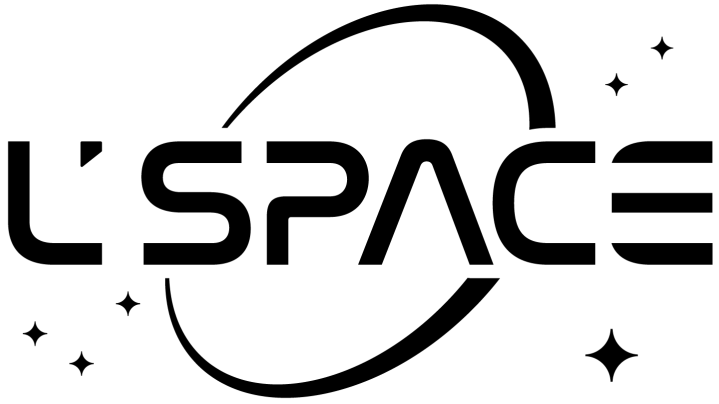
2.1.5.2. Payload Subsystem Recovery and Redundancy Plans

Redundancy:
The payload subsystem will not include redundant instruments due to mass and space limitations. The UV-Vis Spectrometer and pHyTES are highly specialized instruments that contribute significantly to the mission's overall mass and power requirements. Including backup instruments would exceed the mission’s mass and dimension constraints. The likelihood of failure is low due to the instruments’ high TRL (Technology Readiness Level) and heritage from prior planetary missions, which supports confidence in their performance without backups.

Although the payload subsystem lacks physical redundancy for the UV-Vis Spectrometer and pHyTES instruments, this decision is carefully justified by a combination of technical and logistical constraints. Both instruments possess a Technology Readiness Level (TRL) of 6–7, indicating that they have been flight-proven in relevant environments or thoroughly tested in simulated conditions. Their design is based on heritage instruments used in prior Earth-orbiting and planetary exploration missions, such as those on Mars orbiters and Earth science satellites. This heritage significantly lowers the risk of unexpected failures during the Venus mission.

From a systems engineering perspective, payload redundancy was deprioritized in favor of maximizing the scientific utility within strict mass, volume, and power envelopes. The aerobot platform requires a tightly balanced center of mass for controlled flight, and duplicating payload instruments would destabilize the system. Moreover, the thermal shielding and structural integration required to protect a second instrument would increase complexity and reduce space for other critical subsystems, such as communications, power, or thermal regulation hardware.

Finally, the scientific value of partial data from a single functioning instrument is considered high enough to justify operating without full redundancy. For instance, if only the UV-Vis



Spectrometer remains operational, it can still return valuable information about Venusian atmospheric composition. Likewise, if only pHyTES is functional, surface mineralogy and thermal profiles can still be characterized. This modular approach to mission success ensures that useful science can be achieved even under degraded conditions.

Recovery:
Recovery options for the payload subsystem will focus on software-level corrections. If the instruments encounter data acquisition errors, the command and data handling (CDH) system will be programmed to attempt a system reset or switch to a backup operational mode. If one instrument fails, the mission can still fulfill partial science objectives with data from the remaining instrument. Additionally, data prioritization algorithms will ensure that the most critical data is transmitted first in case of a payload malfunction.

The instruments will operate under a set of fault-tolerant software protocols coordinated through the CDH system. These protocols are designed to detect performance anomalies, trigger diagnostic routines, and implement automated recovery actions where appropriate.

For example, if the UV-Vis Spectrometer experiences a sensor communication failure or an unexpected drop in data quality, the CDH will attempt a soft reset of the instrument and reinitialize its operating state. If the failure persists, the system will fall back to a reduced-functionality mode that may limit spectral range or resolution but preserves basic data collection capabilities. Similarly, if pHyTES encounters thermal or signal anomalies, the CDH may deactivate certain sensor bands or reduce operational frequency to preserve core functionality while minimizing risk.

In addition to automated fault detection, ground-based operators will have the ability to issue override commands, upload updated control parameters, or adjust data collection priorities in response to evolving system behavior. For example, operators may choose to reduce pHyTES

activity and conserve power if it begins drawing unexpectedly high current or shows signs of sensor degradation. This flexibility allows the payload to adapt to changing conditions without requiring direct physical intervention.

A key part of the recovery plan also involves data management and prioritization. Given the limited bandwidth and short operational lifespan of the aerobot, the payload subsystem includes logic to classify data based on scientific value and compress or discard lower-priority data if necessary. If a malfunction is detected mid-flight, the system will prioritize the transmission of high-value data (e.g., full spectral sweeps, anomaly detections, peak fluorescence) while bypassing redundant or low-quality observations.

2.1.5.3. Payload Subsystem Manufacturing and Procurement Plans

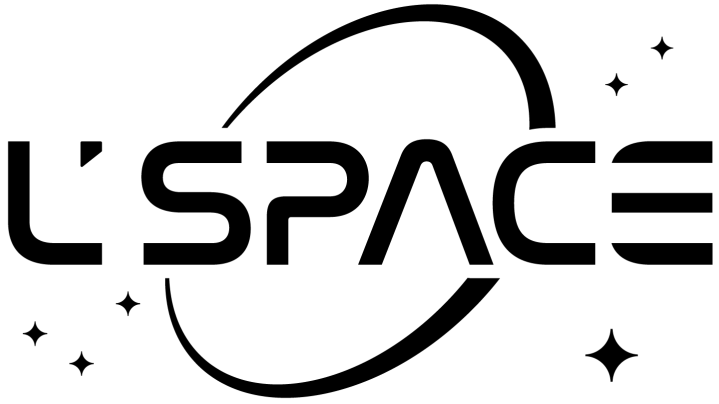
Ocean Insight was chosen as the primary supplier for the UV-Vis Spectrometer due to its extensive experience in miniaturized, high-sensitivity spectrometers for space and planetary science applications. Their HR Series and ST Series spectrometers offer a wavelength range of 200–700 nm, aligning with the aerobot's atmospheric composition analysis needs. Ocean Insight's spectrometers are also lightweight and compact, making them well-suited to fit within the mass constraints. Avantes was selected as the secondary supplier due to its AvaSpec-ULS Series spectrometers, which provide a similar 200–750 nm wavelength range with high-resolution capabilities. While their products are less space-proven than Ocean Insight's models, Avantes spectrometers have been used in harsh environmental conditions and feature radiation-resistant coatings and ruggedized optics, making them well-suited for Venus' extreme atmospheric conditions.

Due to the limited commercially available Hyperspectral Thermal Emission Spectrometers (HyTES), the best route to take is to have the manufacturing of the Hyperspectral Thermal Emission Spectrometer (HyTES) in-house. This decision will

offer significant advantages in terms of customization, mission-specific optimization, and risk mitigation. Unlike commercially available hyperspectral spectrometers, which are often designed for broader applications, a NASA-built HyTES can be precisely engineered to withstand Venus' extreme atmospheric conditions. The manufacturing can be done through NASA's Jet Propulsion Laboratory (JPL) as they have built and flight tested this instrument in the past. This development allows for smooth integration with the aerobot's thermal and CDH subsystems, ensuring optimized power efficiency and data transmission within the mission's constraints. Through NASA's expertise in planetary instrumentation, the spectrometer can be carefully tested in Venus-like conditions, reducing the risk of component failure in-flight. Furthermore, NASA's direct control over the design, fabrication, and calibration process ensures that the instrument meets the specific spectral resolution and sensitivity requirements needed for analyzing Venus' surface and atmospheric thermal emissions.

2.1.5.4. Payload Subsystem Verification Plans

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
PAY.01	UV-Vis Spectrometer shall detect fluorescence peaks between 200–500 nm	Test	Laboratory spectral testing ensures instrument sensitivity and wavelength calibration	Conduct optical bench testing with calibrated UV/visible sources in lab environment
PAY.02	pHyTES shall detect thermal emission features between 8–12 μm	Test	Validates IR spectral response and performance under expected conditions	Simulated blackbody tests in thermal vacuum chamber using known mineral samples



PAY.03	Instruments shall be mounted on a stabilized platform with ±0.5° pointing accuracy	Test	Verifies gimbal/stabilizer precision and mounting alignment	Perform ground-based pointing accuracy test using motion simulators and optical targets
PAY.04	Instruments shall operate under Venusian pressures (up to 90 atm)	Test	Confirms pressure survival of enclosure and electronicsy	Pressure chamber testing with instrument in Venus-simulated conditions
PAY.05	UV-Vis Spectrometer shall achieve 0.1 nm spectral resolution	Test	High-resolution test confirms instrument meets scientific needs	Use narrow-line UV calibration sources and measure instrument resolution performance
PAY.06	Instruments mounted at aerobot's center of mass	Inspection	Physical check ensures flight balance and integration	Perform CAD review and physical inspection of assembled aerobot and payload placement
PAY.07	pHyTES shall have a field of view of ±2.5°	Test	Validates optical design matches scanning and coverage requirements	Conduct field of view and resolution testing using optical targets and goniometers
PAY.08	Instruments shall operate within Venus' atmospheric temperature ranges	Test	Ensures thermal resilience and operational integrity	Perform thermal cycling in high-temp chamber replicating Venus atmospheric profiles

2.2. Interface Control

N^2 Chart

Payload	Instrument data					

Comand to begin/end collecting data	**Data Handling**	Instrument data	Navigation plan			
	Sends/receives comm signals	**Comms**	Receives nav updates			
	Telemantary	Send positioning	**Navigation**			
TCS	TCS	TCS	TCS	**Thermal**	Structural heat paths	
Mount support	Mount support	Mount support	Mount support	Mount support	**Mechanical**	
Power	Power	Power	Power	Power		**Power**

3. Science Mission Plan

3.1. Science Objectives

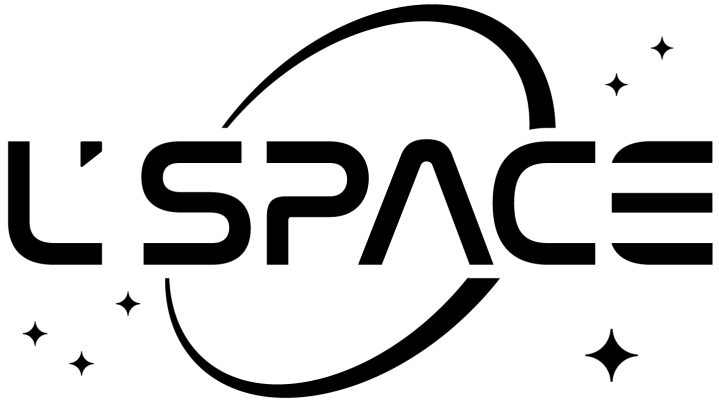
This mission aims to investigate the influence of Venus' volcanic activity on its atmosphere. The primary scientific goals are to measure the concentration of carbonyl sulfide (OCS) in the atmosphere and to detect surface minerals associated with volcanic

processes. These objectives were selected to enhance the understanding of Venus' geological history and current atmospheric dynamics.

OCS is a sulfur-containing gas that is unstable under the oxidizing conditions of Venus' atmosphere. Detecting OCS at measurable levels suggests a possible ongoing source, such as volcanic outgassing. Tracking its concentration helps to assess interactions between the surface and atmosphere, as well as the potential presence of active geological processes. This objective is considered the primary focus of the mission, as it directly supports the atmospheric science priority outlined in the Science Traceability Matrix and aligns with the mission's altitude-based design, which emphasizes measurements in the 30–50 km cloud layer where OCS is most likely to be detected before photolytic breakdown occurs. Additionally, understanding gas exchange between the surface and atmosphere has broader implications for planetary climate modeling and the study of greenhouse effects on terrestrial planets.

The second goal focuses on detecting pyroxene minerals commonly found in volcanic rocks. These minerals are indicators of past or recent volcanic activity and may provide information about resurfacing events and the thermal evolution of the planet's crust. Their presence can be used to analyze changes in surface composition and internal heat distribution over time.

To achieve these objectives, the mission will deploy two specialized instruments. A UV-Visible Spectrometer will be used to measure OCS concentrations within a wavelength range of 185–300 nm, specifically at altitudes between 30 and 50 km. Additionally, a Hyperspectral Thermal Emission Spectrometer (pHyTES) will examine the planet's surface in the 8–12 μm range to identify pyroxene mineral signatures. Both instruments have been selected for their ability to operate under the extreme environmental conditions of Venus and to meet strict limitations on mass, volume, and power consumption.



The data gathered from these instruments will contribute to a deeper understanding of Venus' atmosphere-surface interactions, provide insight into volcanic activity, and support broader planetary science research related to climate evolution on terrestrial planets.

3.2. Experimental Logic, Approach, and Method of Investigation

This mission focuses on studying the connection between Venus’ atmospheric composition and surface geology through a descending balloon platform equipped with scientific instruments. The main goals are to measure the concentration of carbonyl sulfide (OCS) between 30 and 50 km altitude and to detect volcanic minerals like pyroxene on the surface. These tasks are accomplished using two main instruments: a UV-Visible spectrometer for gas analysis and a thermal emission spectrometer (pHyTES) for surface mapping. These tools are supported by a network of onboard systems designed to function together under Venus’ extreme atmospheric conditions.

Subsystem Roles in Data Collection

The mission’s scientific instruments are supported by five integrated systems. The mechanical system provides the structure and secure placement for instruments. The power system uses high-density batteries with a management unit to ensure stable power for up to 48 hours. The thermal control system regulates the temperature of sensitive components using insulation and heat management materials. The command and data handling (CDH) system coordinates instrument timing, system health, and sends data to the orbiter. Lastly, the payload system houses and operates the sensors that gather science data.

As the balloon descends, the UV-Vis spectrometer scans Venus' atmosphere for OCS and other gases, using short sampling intervals and adjusting automatically to the light levels. At the same time, the pHyTES instrument focuses on detecting surface mineral signatures by observing infrared thermal radiation.

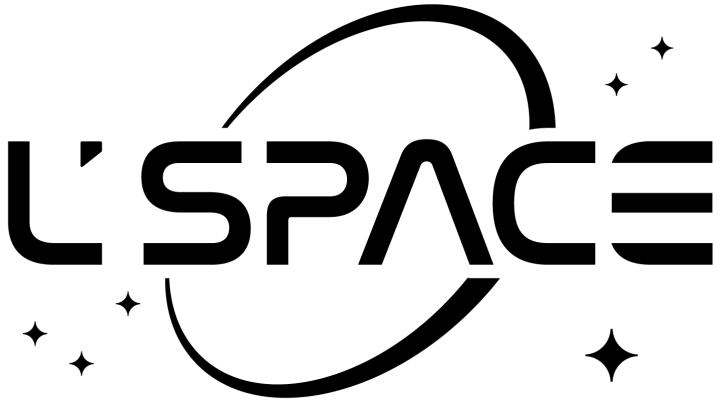
Planned Sequence for Data Collection

After deployment from the spacecraft into Venus’ atmosphere, the balloon inflates and starts a slow descent through various atmospheric layers. The following steps outline the experiment's timeline:

1. Initial stabilization (0–1 hour): The aerobot confirms proper orientation and altitude before collecting data.
2. Atmospheric gas scanning (1–12 hours): The UV-Vis spectrometer records gas absorption data in intervals, storing it locally.
3. Surface thermal imaging (3–20 hours): While passing over geologically interesting regions, the thermal spectrometer captures heat maps of the surface.
4. Data transmission: At regular intervals, information is relayed to an orbiter for delivery back to Earth.
5. Secondary pass (24–36 hours): Instruments re-scan to verify earlier results and detect changes over time.
6. Final phase (36–48 hours): As the balloon reaches lower, hotter layers, instruments shut down in sequence to preserve any last data.

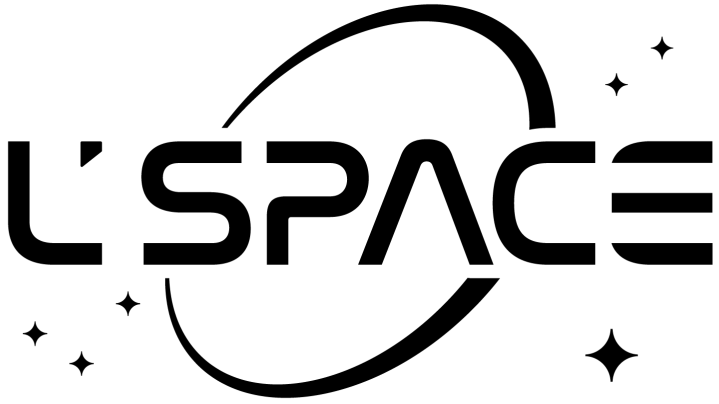
Planning and Site Selection

This procedure was shaped by mission limitations and refined using surface imagery and data from JMARS. Key regions like Idunn Mons, Maat Mons, and Alpha Regio were chosen for their likelihood of volcanic activity and history of high heat signals. These areas align with mission goals and are within reach of the aerobot’s altitude range. Their



geological features make them strong candidates for identifying both active gas release and ancient surface processes.

This experiment plan supports a short but focused mission lifetime, maximizing scientific return while managing environmental risks. The systems work in harmony to achieve key measurements within the mission's 48-hour operational window.

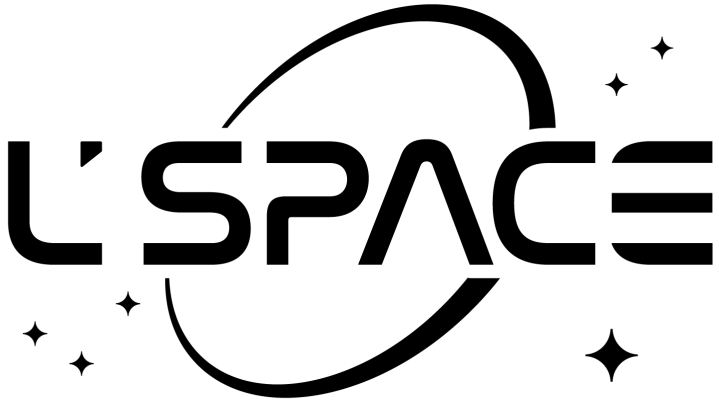


3.3. Payload Success Criteria

The scientific payload includes two instruments: the Ultraviolet-Visible (UV-Vis) Spectrometer and the Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES). Each instrument is designed to achieve one of the mission's primary objectives: atmospheric gas analysis and surface mineral detection. These instruments are responsible for supporting data collection throughout the 48-hour mission duration and enabling critical insights into Venus' geological activity and atmospheric interactions. The following paragraphs define the success thresholds and potential failure modes of each instrument, followed by a table summarizing all success criteria.

The UV-Vis Spectrometer will be tasked with detecting carbonyl sulfide (OCS), a volcanic gas that offers insight into Venus' potential active geology. The minimum success criterion for this instrument is its ability to detect UV absorption peaks of OCS within the 185–300 nm wavelength range while operating at altitudes between 30 and 50 kilometers. Optimum success includes achieving a spectral resolution of 0.5–1.0 nm, an integration time of approximately 5 seconds, and a detection sensitivity down to 10^{-7} mol fraction. A stretch goal includes performing continuous vertical profiling of OCS concentrations over time, altitude, and geographic location, particularly in volcanically significant regions such as Idunn Mons or Maat Mons. Failure modes include damage from sulfuric acid clouds or solar radiation, reduced signal-to-noise ratio from atmospheric interference, or instability in the aerobot's flight path affecting measurement quality. Detecting OCS with high resolution would provide indirect evidence of active outgassing and surface-atmosphere exchange, thereby supporting theories of ongoing geological processes on Venus.

The second instrument, pHyTES, will focus on detecting surface mineralogy through thermal infrared spectral data. The minimum success criterion is the detection of thermal emission signatures in the 8–12 μm range. Optimum performance includes



identification of pyroxene minerals, such as clinopyroxene and orthopyroxene, with a spectral resolution of ≤10 cm^{-1}, temperature sensitivity of ±1 K, and spatial resolution of ≤1 km per pixel. The stretch goal involves mapping mineralogical composition across selected regions, including Alpha Regio, Diana Chasma, and Fortuna Tessera, to assess recent geological resurfacing and crustal evolution. Failure modes include loss of visibility due to cloud opacity, limited operational power affecting measurement time, or thermal imbalance impairing sensor function. Successful pyroxene detection would validate hypotheses regarding recent lava flows and tectonic reshaping of the Venusian crust, expanding comparative knowledge of terrestrial planets.

	Category	Minimum Success	Optimum Success	Stretch Goal
	UV-Vis Spectrometer	Detect OCS absorption features in the 185–300 nm range	Quantify OCS concentration with high sensitivity (≤10^{-7} mol fraction), ≤1 nm resolution	Profile OCS over altitude, time, and location to associate with active volcanic regions
	pHyTES	Detect surface thermal emission in the 8–12 μm range	Identify pyroxene spectral features with ±1 K accuracy and ≤1 km/pixel resolution	Map mineralogical composition across key volcanic regions

3.4. Testing and Calibration Measurements

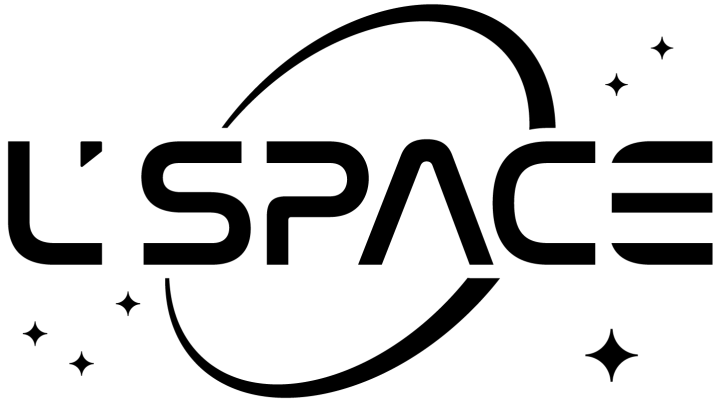
Upon arrival at the target altitude within Venus' atmosphere, in-situ testing procedures will be implemented to ensure that the scientific instrumentation operates as intended under environmental conditions specific to the planet. These tests are distinct from pre-launch verification and validation and are conducted to confirm the functionality and calibration accuracy of the instruments prior to full science operations. Each instrument will be individually assessed using reference conditions and known environmental baselines to verify performance and data integrity.

UV-Vis Spectrometer

To evaluate the performance of the UV-Vis spectrometer, initial measurements will be performed using direct solar irradiance as a spectral reference. This process involves capturing ultraviolet light within the 185–300 nm range while the instrument descends or floats at altitudes of 30 to 50 kilometers. The resulting spectral data will be compared to pre-calculated models and known irradiance curves to validate alignment and calibration. To establish a measurement baseline, observations will also be taken in regions expected to be free of carbonyl sulfide (OCS), enabling the differentiation of actual OCS signals from background atmospheric noise. Internal dark scans will be used to characterize instrument bias and sensor noise in the absence of light. Accuracy will be verified through repeated readings across consistent atmospheric layers, ensuring stability in spectral output. Control variables during these operations will include solar angle, altitude, and aerobot orientation, as each may influence the strength and shape of the recorded signal.

Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES)

For the pHyTES instrument, calibration will commence by observing atmospheric layers with well-documented thermal characteristics, prior to targeting surface emissions. Passive thermal scans of Venus' high-altitude clouds will be conducted to assess



baseline sensor output and detect any anomalies in thermal gain or calibration drift. An integrated blackbody reference source within the instrument will serve as a calibration standard, enabling real-time correction of thermal measurements. Once surface-directed scanning begins, initial observations will focus on geologically stable regions with minimal volcanic mineral presence to form a comparative baseline. These areas will provide a reference for later scans of regions with suspected pyroxene content. Instrument reliability will be verified by comparing successive passes over the same region, accounting for variations due to viewing geometry, local temperature gradients, and atmospheric opacity. Critical control variables include instrument pointing stability, ambient radiative background, and signal-to-noise ratio across thermal channels.

Summary

The outlined testing and calibration strategy ensures that each instrument is properly aligned and responsive to its measurement targets. By conducting controlled observations and internal diagnostics, the system will be capable of distinguishing between true scientific phenomena and sensor-related anomalies. These calibration procedures are critical for ensuring the credibility of all science data transmitted to Earth throughout the duration of the mission.

3.5. Precision and Accuracy of Instrumentation

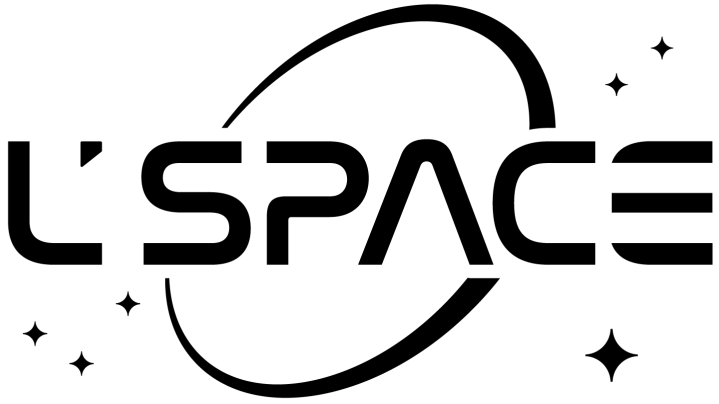
The precision and accuracy of scientific instruments are critical for producing reliable data and achieving mission objectives. For this mission, both selected payload instruments—the Ultraviolet-Visible (UV-Vis) Spectrometer and the Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES)—must provide consistent, high-fidelity measurements under the extreme environmental conditions of Venus. Each instrument's performance parameters directly influence the credibility of data related to volcanic activity and atmospheric composition. The following outlines the expected accuracy levels of each instrument and their implications for data quality. In cases where explicit specifications are unavailable, assumptions are made based on comparable instruments used in prior planetary science missions and are clearly stated.

UV-Vis Spectrometer

The UV-Vis Spectrometer is designed to detect carbonyl sulfide (OCS) by measuring ultraviolet light within the 185–300 nm wavelength range. Based on specifications of similar spaceflight instruments, the spectrometer is expected to maintain a wavelength accuracy of ±0.5 nanometers and detect trace gas concentrations with a precision of approximately ±10% at sensitivity levels down to 10^{-7} mol fraction. These tolerances are essential for resolving narrow spectral absorption features associated with trace gases in Venus' atmosphere. A deviation beyond the stated spectral accuracy could result in incorrect identification of gas species, while concentration errors may affect the estimation of volcanic gas flux and its spatial distribution. To reduce the impact of these uncertainties, the system will perform multiple scans across various altitudes and locations to increase confidence through data redundancy and statistical analysis.

Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES)

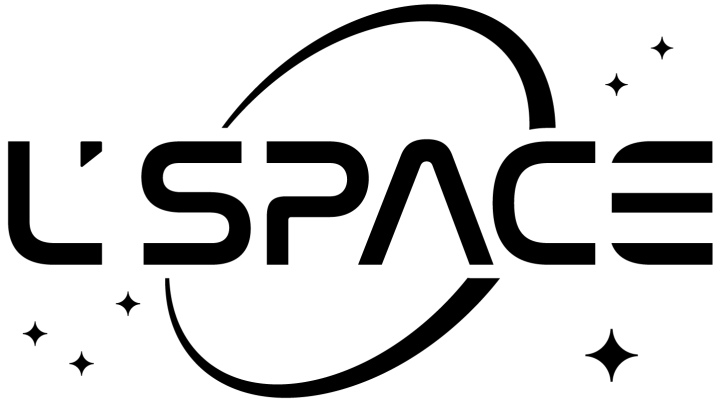
The pHyTES instrument is responsible for identifying surface minerals—specifically pyroxenes—through thermal infrared (TIR) spectroscopy across the 8–12 μm range.



Accuracy assumptions, based on similar hyperspectral systems such as those flown on Mars and Earth-observing missions, include a thermal sensitivity of ±1 Kelvin and a spectral resolution accuracy of ±0.1 μm. These tolerances allow for clear identification of thermal emission peaks and mineral-specific signatures. Errors beyond these thresholds could reduce the ability to differentiate pyroxene types or detect temperature anomalies, which are indicators of geologically recent activity. To ensure accuracy, the instrument will rely on internal calibration mechanisms such as onboard blackbody references and repeated measurements of geologically stable regions, which will serve as baseline comparisons for identifying anomalies.

Summary

Understanding the accuracy and precision of the selected instruments is essential to validate scientific findings. The performance tolerances of both the UV-Vis Spectrometer and pHyTES are appropriate for the mission’s scientific goals and support the reliable identification of atmospheric and geological phenomena on Venus. These specifications will guide post-processing methods and influence the interpretation of collected data in support of the mission’s core objectives.



3.6. Expected Data & Analysis

The scientific output of the Aeris mission will primarily rely on data collected from two key instruments: a UV-Visible (UV-Vis) Spectrometer and the Planetary Hyperspectral Thermal Emission Spectrometer (pHyTES). These instruments will work together to provide detailed insights into the interactions between Venus' atmosphere and its surface geology, focusing on the detection of carbonyl sulfide (OCS) gas and the identification of volcanic minerals such as pyroxenes. Data analysis will employ established remote sensing techniques tailored to the harsh environmental conditions of Venus, with interpretation based on comparative planetology and geophysical modeling.

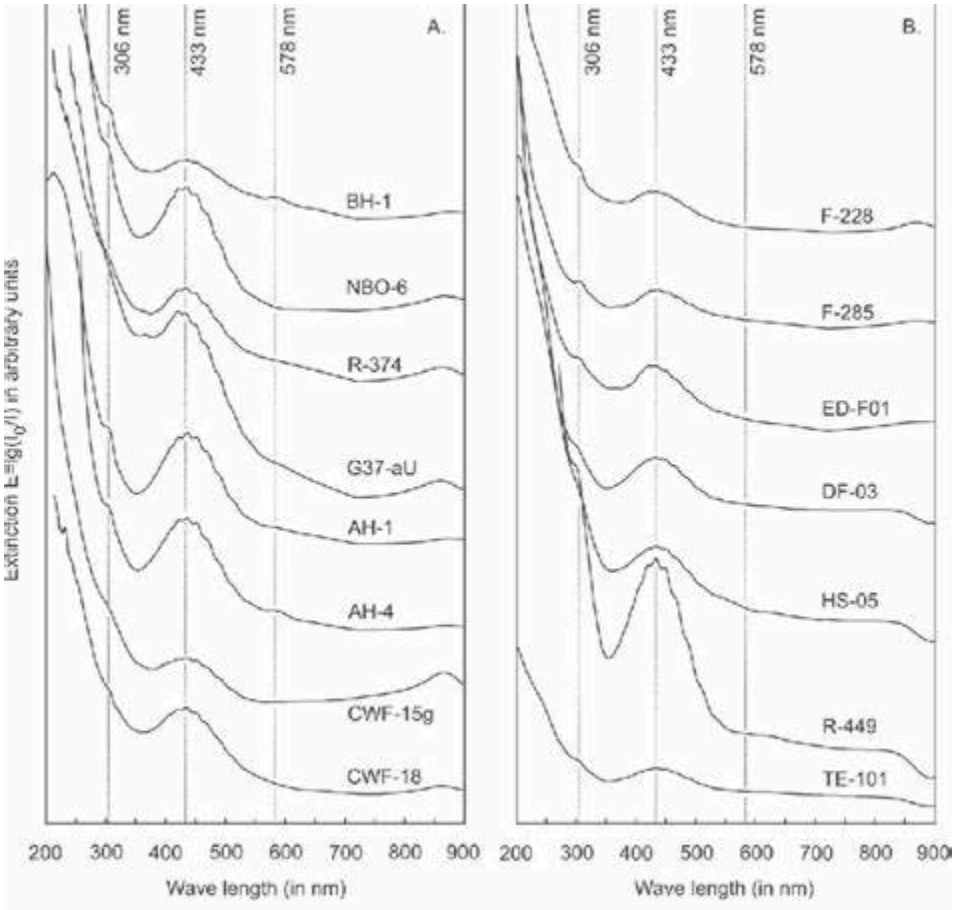


Figure 1: UV-Vis Spectral Data Showing OCS Absorption Features

The UV-Vis Spectrometer will produce high-resolution ultraviolet absorption spectra in the 185–300 nm range, targeting trace gases present in Venus' middle atmosphere. One gas of particular interest is carbonyl sulfide (OCS), a sulfur-bearing compound that serves as a possible indicator of active volcanism. OCS spectral features, notably those around 195–200 nm, will be identified using differential absorption spectroscopy (DAS). To confirm these signals, the measurements will be compared against solar irradiance reference spectra collected during the balloon's descent, as well as baseline readings from areas expected to be devoid of volcanic emissions. This method enables the separation of gas-specific absorption peaks from background atmospheric noise. Data from the SPICAV instrument aboard Venus Express demonstrate that it is feasible to resolve narrow OCS features under favorable conditions, and the mission's instrument is designed to match or exceed this spectral resolution of 0.5–1.0 nm.

The data analysis process will begin with on site calibration using solar irradiance and internal dark scans to establish measurement baselines and eliminate instrument bias. Once OCS absorption features are detected, the instrument will perform repeated scans at varying altitudes to create vertical concentration profiles. These profiles will be compared across different locations, especially near known volcanic regions like Idunn Mons or Maat Mons, to detect spatial variability and potential gas plumes. The concentration sensitivity goal is set at 10^{-7} mol fraction, with a wavelength accuracy of ±0.5 nm, making the expected uncertainty in OCS detection approximately ±10%. This precision will be adequate to identify significant trends in gas presence, especially when complemented with redundant sampling across different times and locations. Detecting OCS in the cloud layer, where it remains stable before photolytic breakdown, would provide indirect evidence of active geological processes and contribute to ongoing discussions regarding volcanism on Venus.

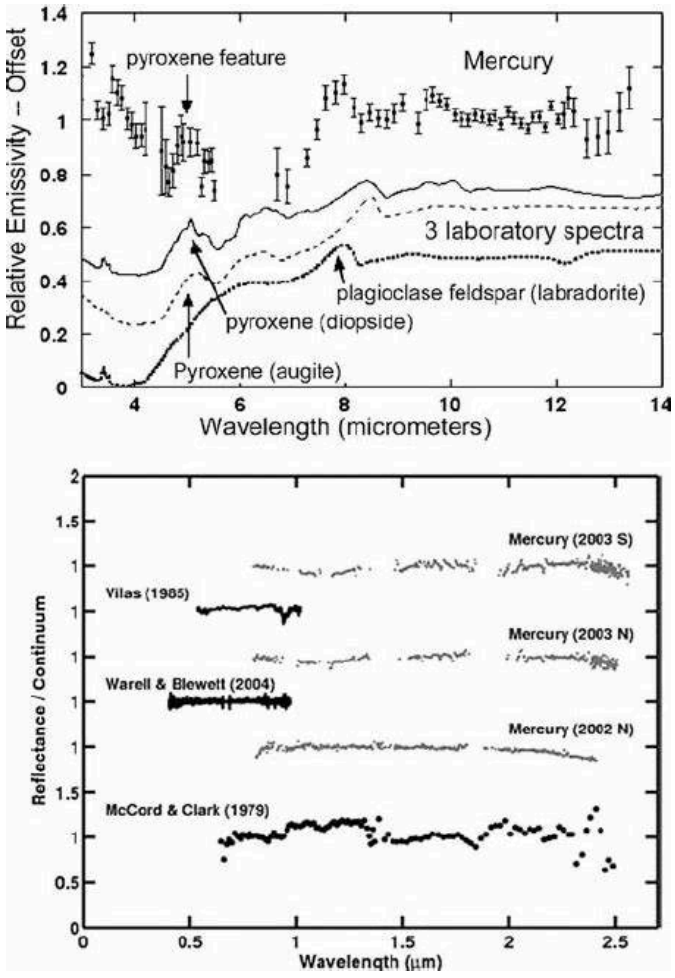
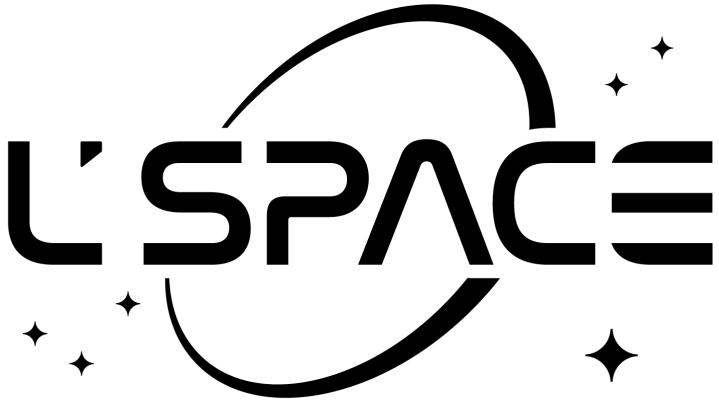


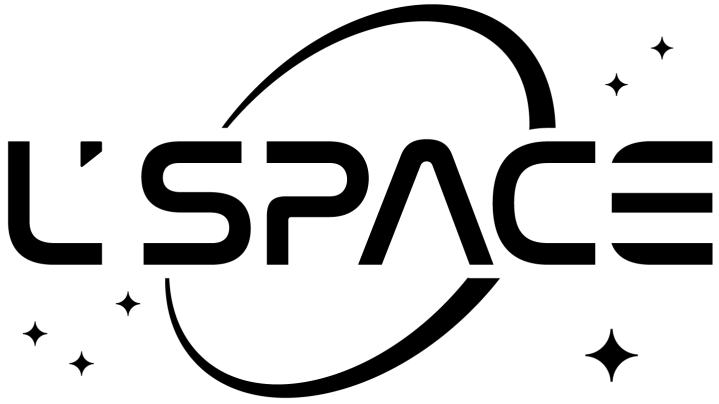
Figure 2: Laboratory thermal emission spectra of pyroxene minerals, indicating diagnostic absorption bands in the thermal infrared region.



The pHyTES instrument is designed to collect thermal infrared emission data within the 8–12 μm wavelength range to identify the mineral makeup of Venus' surface. This particular spectral window is well-suited for detecting silicate minerals such as pyroxenes, which are commonly associated with volcanic rocks. These minerals exhibit distinct absorption features—especially reststrahlen bands—that have been thoroughly documented in both Earth-based studies and Mars exploration. Instruments like the Thermal Emission Imaging System (THEMIS) on Mars, along with terrestrial hyperspectral systems, offer useful benchmarks for interpreting the data expected from this mission. Pyroxenes, for instance, generate recognizable spectral dips that shift slightly depending on their specific composition. With a targeted spectral resolution of 10 cm^{-1} or better and a thermal sensitivity of ±1 K, pHyTES is expected to resolve these variations effectively.

Surface mapping will commence once the balloon platform descends to altitudes where Venus' surface becomes observable through atmospheric windows. Initial scans will concentrate on geologically stable regions to establish baseline thermal signatures. These reference datasets will aid in distinguishing genuine mineralogical anomalies from sensor noise or cloud interference. High-priority targets such as Alpha Regio and Diana Chasma will then be scanned for localized temperature anomalies and spectral features associated with volcanic resurfacing. The instrument will utilize an onboard blackbody calibration source to correct thermal readings in real time, ensuring high fidelity under variable environmental conditions. The expected spatial resolution is ≤1 km/pixel, allowing for comprehensive yet detailed regional mineralogical mapping. Error estimation will be based on cross-comparison of repeated scans and anticipated thermal variability in selected control sites, with uncertainty margins constrained to about ±1 K.

Once the thermal emission data are collected, they will be analyzed to identify surface mineralogy, focusing on the presence and distribution of pyroxenes such as clinopyroxene and orthopyroxene. Detecting these minerals, particularly in areas with elevated thermal signatures, would offer evidence of relatively recent lava flows or



tectonic reshaping. The ability to map these minerals across time and space supports broader scientific goals aimed at understanding the crustal evolution and geological history of Venus. Furthermore, identifying variations in mineral composition across different regions will further enhance our understanding of the planet's geological processes.

4. Mission Risk Management

4.1. Safety and Hazard Overview

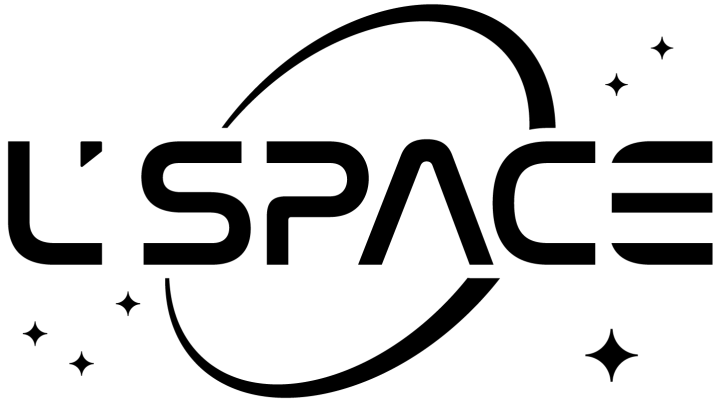
To ensure mission success and team safety throughout all phases of the Venus sample return mission, the project follows NASA's formal risk management guidelines using both **Risk-Informed Decision Making (RIDM)** and **Continuous Risk Management (CRM)** processes.

RIDM was used during early design to evaluate major trade-offs between scientific return and mission safety, with a particular focus on hazards posed by Venus's extreme temperature, pressure, and chemical environment. High-risk areas such as atmospheric entry, thermal insulation, and communication reliability were prioritized based on these analyses.

CRM is used continuously to track and manage risks throughout the mission lifecycle. Risks are logged, categorized, evaluated for severity, and assigned mitigation strategies. Each risk is monitored during weekly subsystem meetings, ensuring any increase in likelihood or consequence triggers a reevaluation or updated mitigation plan.

Key safety concerns addressed include:

- **Thermal and atmospheric hazards**: The spacecraft must withstand ~465°C temperatures and 90x Earth's atmospheric pressure.
- **Data and signal loss**: Harsh conditions may disrupt data relay from Venus back to Earth.
- **Subsystem malfunctions**: Particularly with power, structure, and communication systems.
- **Hazards to personnel**: While the mission is unmanned, all lab and test procedures follow NASA and OSHA safety protocols to protect team members during hardware



integration, battery handling, and simulation testing.

Mitigation strategies, including system redundancy, pre-launch testing, and scheduled design reviews, are applied based on severity levels. Descope actions were also considered. For example, reducing non-essential instruments helped minimize overheating risks without compromising mission objectives. A 20% schedule margin was also added to absorb unforeseen delays related to safety upgrades.

A detailed risk matrix (see Section 4.1.1) supports this hazard analysis and helps prioritize high-impact concerns.

4.1.1. Risk Analysis

The team used a structured framework for identifying and analyzing mission risks. This included internal brainstorming, review of historical Venus mission failures (such as Venera and Pioneer Venus), and mentor feedback provided during role meetings and in the MDR task spreadsheet.

Risk Identification Process

Each subsystem team compiled a list of potential risks relevant to their area. These risks were categorized into:

- Mechanical and Structural
- Thermal and Environmental
- Electrical and Power
- Communications
- Command and Data Handling (CDH)

- Science Payload
- Personnel and Programmatic

Risk Evaluation and Matrix Scoring

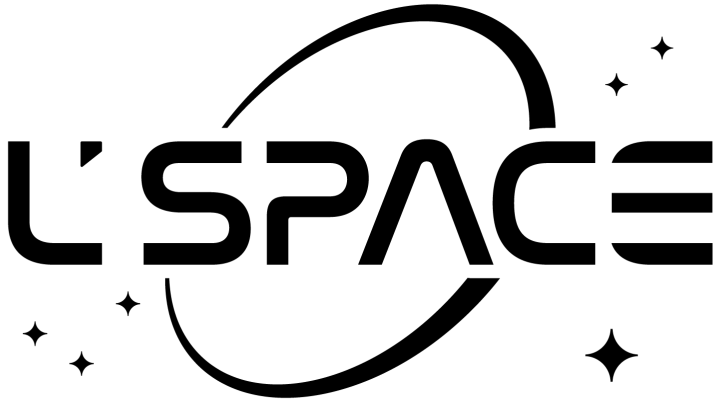
Risks were analyzed based on two main factors:

- **Likelihood (L)**: Probability the risk will occur (1 = Rare, 5 = Almost Certain)
- **Consequence (C)**: Severity of impact on the mission if the risk occurs (1 = Negligible, 5 = Critical)

These values were multiplied to determine a **Risk Score (R = L × C)** and classified as:

- **Low** (Green): 1–5
- **Moderate** (Yellow): 6–10
- **High** (Red): 11–25

A custom risk matrix was used to visualize these risk scores:



Rank	Risk ID #	Risk Owner	Risk Title	Risk Statement	L	C	Rating	Approach	Trend
Prioritization order of risks to addressed	assigned to each risk, may have a precursor code to designate what level the risk is tracked at	Which team member is assigned responsibility over seeing this risks closure	A short descriptive title of the risk	A more thorough description of the risk following the appropriate format of "Given that [CONDITION], there is a possibility of [DEPARTURE] adversely impacting [ASSET], which can result in [CONSEQUENCE]."	Likelihood of the risk to occur, see the Resources sheet for more information	Consequences if the risk occurs, see the Resources sheet for more information	High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LxC over its life cycle
1	THERM-001	Thermal Subsystem Lead	Venus Surface Thermal Overload	Given that Venuss surface temperature is ~465C, there is a possibility of thermal insulation failure adversely impacting lander systems, which can result in total system shutdown.	5	5	25 (High)	Mitigate (M)	→ Neutral
2	COMM-002	Comm Team	Signal Interruption	Given that the thick atmosphere may distort signals, there is a possibility of communication disruption adversely impacting Earth-bound data transmission, resulting in partial or total data loss.	4	4	16 (High)	Mitigate (M)	↑ Increase
3	STRUC-003	Structures Lead	Entry Shield Damage	Given the pressure and speed during entry, there is a possibility of shield damage adversely impacting core instruments, which can result in mission failure.	4	5	20 (High)	Mitigate (M)	↑ Increase
4	PWR-004	Power Systems	Battery Thermal Runaway	Given battery sensitivity to high temperatures, there is a possibility of thermal runaway adversely impacting onboard power, which can result in fire or shutdown.	3	5	15 (High)	Mitigate (M)	↓ Decrease
5	TEST-005	Test Operations Lead	Personnel Burn Risk	Given the handling of high-temp hardware in labs, there is a possibility of burns adversely impacting personnel, which can result in injury.	2	4	8 (Moderate)	Accept (A)	↓ Decrease
6	CDH-006	CDH Team	Software Crash During Entry	Given the fast phase changes and sensor overload, there is a possibility of software failure adversely impacting control logic, which can result in lost lander control.	3	4	12 (High)	Mitigate (M)	→ Neutral
7	SCI-007	Payload Scientist	Instrument Overload	Given the unknowns in Venus's chemical mix, there is a possibility of sensor failure adversely impacting data quality, which can result in partial mission value loss.	3	3	9 (Moderate)	Research (R)	↑ Increase

Mitigation & Tracking

All moderate and high risks have documented mitigation plans in the **Advanced Risk Log**, including:

- Use of heat-resistant materials
- Redundant communication paths
- Backup power supplies
- Fault-tolerant software
- Risk-aware descope decisions (e.g., trimming scientific payload)

Risk control is an ongoing process. Updates are shared with the systems engineering lead weekly, and the risk matrix is updated as conditions change. Trigger thresholds are defined for when risks escalate to a higher category, prompting action or review.

Mentor Feedback Implementation

Mentor comments from the MDR spreadsheet were integrated to:

- Reclassify under- or overestimated risks
- Include personnel safety as a separate category (addressed in Section 4.1)
- Ensure the risk matrix is visible and used consistently
- Align CRM steps with NASA's documented procedure

The team continues to use this living framework to anticipate and respond to technical and programmatic challenges throughout the mission lifecycle.

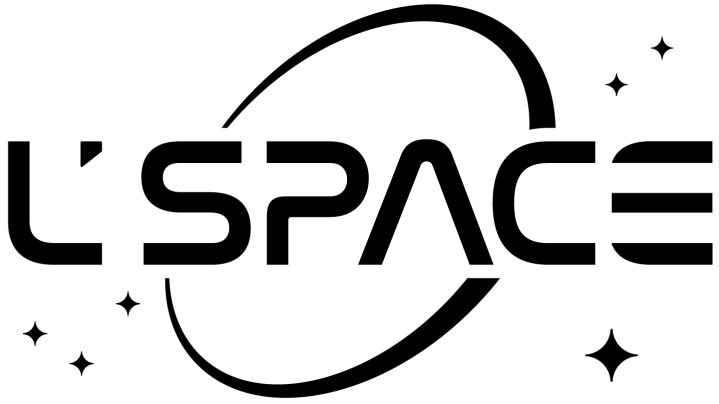
4.1.2. Failure Mode and Effect Analysis (FMEA)

As detailed by the stakeholders, the Failure Mode and Effect Analysis (FMEA) is a design tool used for a detailed analysis of how a system might fail and how this event carries into other systems. The FMEA works closely with the risks identified in the risk analysis and risk matrix. Where FMEA differs is, those risks that have been identified, are now looked at in a larger interconnected context. The FMEA Chart contains the severity of the consequences, if they were to occur, likelihood of occurrence, a proposed prevention method, scale of difficulty in detection, ranking for the prioritization of risks, and proposed mitigations (actions) if such risks continue to occur. The FMEA Chart featured below documents the most critical risks identified in the risk matrix and

analyzes its hypothetical continued impacts.The team will use this chart to actively track risks that pose any complications to individual subsystems.

[FMEA Chart]

Function	Failure Mode	Effects	Sev	Cause	Occ	Prevention	Det	RPN	Actions
Thermal Control	Insulation Failure	Overheating of internal system	9	Underrated insulation materials	4	Use high-temp rated insulation	3	108	Increase insulation layers
	Batter overheat	Power system shutdown	8	Prolonged Venus exposure	3	Add thermal shielding	4	96	Add temp sensors + emergency shutdown
Communications	Signal degradation	Loss of data transmission	8	Thick atmosphere interference	5	Redundant transmission paths	3	120	Switch to backup channel
	Antenna misalignment	No signal from lander	7	Mechanical shift post-landing	3	Autonomous realignment routine	5	105	Include auto-correcting gimbal
Structures	Sheilf crack during entry	Instrument exposure to entry h	10	Material failure at high temp	2	Stress test in extreme conditions	2	40	Redesign entry shield
	Fastener failure	Structural instability	7	Vibration during descent	3	Secure all joints with torque check	4	84	Use vibration-tolerant fasteners
Power Systems	Thermal runaway	Fire or explosion	9	Poor thermal design	4	Advanced cooling system	3	108	Include flame arrestor
	Battery disconnect	Loss of power to instruments	8	Faulty connector	3	Double-check connections during	4	96	Enable power routing redundancy



4.1.3. Personnel Hazards and Mitigations

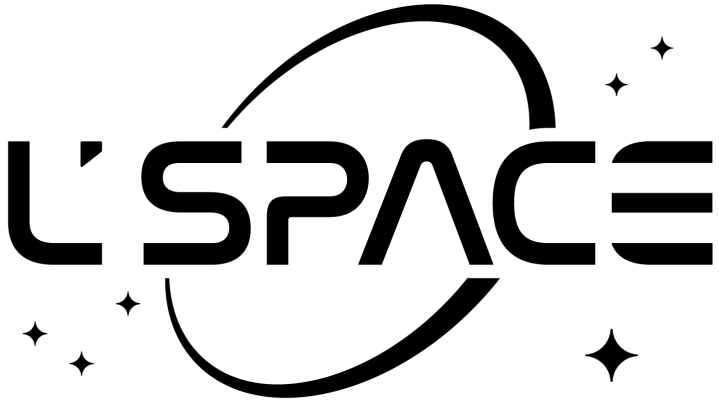
Understanding the importance of personnel hazards is crucial in order to carry out a safe and successful mission. During mission development such as testing within laboratories, manufacturing, and system integration phase, can pose serious threats to personnels' safety. "Per NASA NPR 8715.3, Section 3.8, "NASA hazardous operations involve materials or equipment that, if misused or mishandled, have a high potential to result in loss of life, serious injury or illness to personnel, or damage to systems, equipment, or facilities" (NASA Glenn Research Center 2019). Personnel hazards can include physical, chemical, biological, ergonomic, and psychosocial. Because of this, the Glenn Research Center's *Glenn Safety Manual* documents facility, aviation, vehicle & pedestrian safety, explosives, and toxic gas safety rules and regulations, and various other topics regarding personnel hazards. The identification of each categorical hazard is defined by the team and mitigation processes are defined by the Occupational Safety and Health Administration (OSHA) and the Centers for Disease Control and Prevention (CDC), as the team prioritizes expert judgement when assessing personnel hazards.

Physical:

Definition: Any hazard that threatens a person's physical safety due to external force or actions is deemed as a physical hazard.
Examples: Personnel working within laboratories or operating with heavy machinery may face hazards such as tripping, slipping, extreme temperature conditions, radiation, and etc.
Mitigation: Recommended mitigation strategy includes to "identify any exposures to excessive noise (areas where you must raise your voice to be heard by others), elevated heat (indoor and outdoor), or sources of radiation (radioactive materials, X-rays, or radiofrequency radiation)" (OSHA 2016).

Chemical:

Definition: Chemical hazards are hazards characterized by its ability to be airborne, toxic, or flammable.



Example: Manufacturing laboratories may utilize material modification techniques involving combustible materials, reactive or corrosive (acidic) chemicals, or contaminants such as dust, vapors, or fumes.
Mitigation: Established in 1983, the Hazard Communication Standard (HCS) provides a standardized approach to communicating workplace hazards associated with exposure to dangerous chemicals. It is highly suggested for personnel “to review product labels to identify chemicals at your job site that have low exposure limits, are highly volatile, or are used in large quantities or in unventilated spaces. Identify activities that may result in skin exposure to chemicals” (OSHA 2016).

Biological:

Definition: Biological Hazards occur when a personnel is exposed to bacteria, viruses, or fungi when handling high-contact, biological materials.

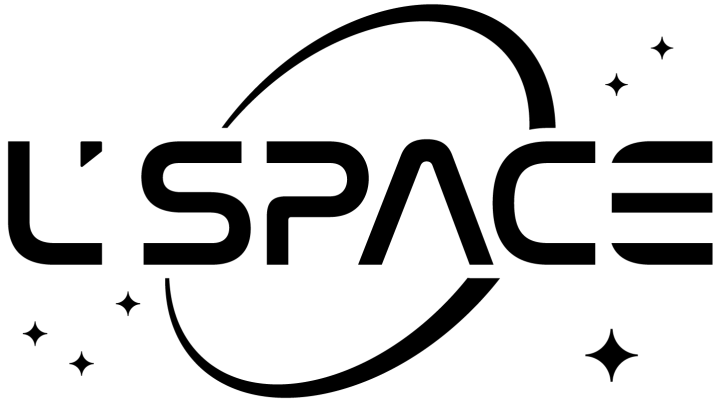
Example: Without safety rules or regulations, a laboratory may practice improper handling of hazardous materials, which may include cross-contamination, the growth of mildew or mold, and etc. If an employee suffers from an open wound or cut, the risk of biological hazard increases in the facility.

Mitigation: According to OSHA, all employees should make use of appropriate personal protective equipment (PPE such as gloves, gowns, or face masks/shields) when performing procedures involving blood (OSHA 2016).

Ergonomic:

Definition: Ergonomic Hazards are defined as hazards impacting personnel’s musculoskeletal system.

Example: Injuries suffered internally such as muscle strains, back injuries, tendons, and joint pain may result from poor workplace conditions and improper



facility use such as heavy lifting, lack of physical support, or disruptive vibration or repetitive movements.

Mitigation: Osha recommends examining “work activities that require heavy lifting, work above shoulder height, repetitive motions, or tasks with significant vibration” (OSHA 2016).

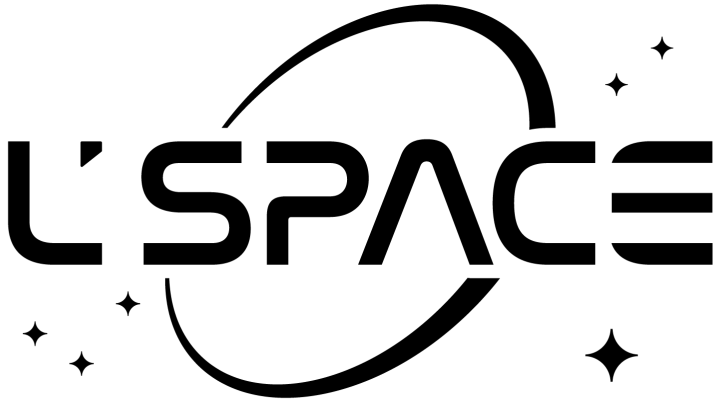
Psychosocial:

Definition: According to the CDC, psychosocial hazards refers to “factors in the work environment that can cause stress, strain, or interpersonal problems for the worker” (CDC 2022).

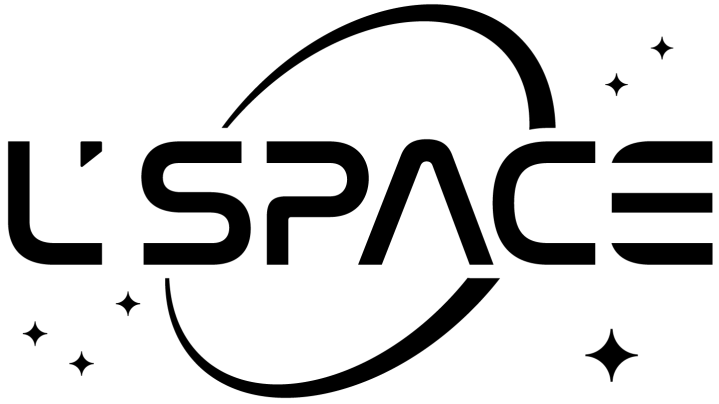
Example: Low quality work performance & productivity, a weaker sense of physical capabilities and daily functioning are characteristics of psychosocial hazards present in a workplace.

Mitigation: OSHA has done due diligence in recognizing that the workplace environment of an employee can cause immense psychological issues affecting not only the quality of their work but also their personal lives. To mitigate this, OSHA advises in their “Workplace Stress” guidance webpage suggesting tips and advice for employers such as being aware of individuals behaviors in social settings, showing empathy, and providing access to paid leave (OSHA 2024). OSHA also provides various outreach programs and resources for employees currently experiencing these hazards.

In general, quantitative exposure assessments & incident investigations will be performed as a preventative cautionary action. This could be done by conducting air



sampling or direct reading instrumentation. A clear plan is devised that entails those hypotheticals of who's involved, lines of communication, materials, and report templates. The plan offered by OSHA requests that it should be conducted with members from the management team and workers to investigate "close-calls", an event that almost resulted in an incident, and identify the root causes of it to address why the possibility of the incident had a probability of occuring. Aeris intends to have mission assurance specialists active throughout the mission planning process to identify and mitigate potential risks and personnel hazards.



5. Activity Plan

5.1. Project Management Approach

Accomplishing the Venus aerobot will be a tremendous feat. As recommended by the stakeholders, the team should be composed of at least 30-50 employees to ensure quality, smooth progression, and efficiency. Having more members will require more management positions as well. The size of the mission team will vary at various points along the mission lifecycle. Team Aeris' mission requires a diverse team of at least 30 personnel, including engineers, technicians, scientists, managers, administration, and potentially and occasionally various other specialty analysts. The engineering team, led by a Lead Engineer, will be responsible for system integration. The science team, led by the Chief Scientist, will oversee research objectives, with instrument scientists managing specific payloads and data analysts interpreting mission findings. Additionally, a project management team, including a Project Manager and Deputy Project Manager of Resources, will ensure smooth coordination, risk management, and budgeting, while a dedicated finance team will control costs and allocate funds. Outreach officers will engage with the public and stakeholders. If needed, reserve funds will be set aside for any contingencies, ensuring flexibility throughout the mission. This structured team approach will balance autonomy and collaboration, optimizing mission success.

The science team's "specific responsibilities include defining the science or demonstration requirements; ensuring the integration of these requirements with the payloads, spacecraft, ground systems, and mission operations; providing the algorithms for data processing and analyses; and performing data analysis and archiving" (NASA 2010).

The engineering and technician general responsibilities include the act of defining "the project space flight vehicle(s) and ground system and conduct trade studies. It includes the integrated planning and control of the technical program efforts of design

engineering, software engineering, specialty engineering, system architecture development and integrated test planning, system requirements writing, configuration control, technical oversight, control and monitoring of the technical program, and risk management activities" (NASA 2010). The engineering team will also be required to write detailed documentation on the development process such as the Interface Control Document and the Verification & Validation Plan.

The business team, or rather the management team, handles the "administrative planning, organizing, directing, coordinating, analyzing, controlling, and approval processes used to accomplish overall project objectives such as project reviews and documentation" (NASA 2010).

The level of autonomy possessed by each division will depend on the identified risks, complexity, or the phase in the mission life cycle. As suggested by the NASA Space Flight Program and Schedule Management Handbook, the business team will be handling the broader aspects of the finances of the mission development, while the engineering team will have full purchasing capabilities while operating under the allocated budget. If any use of the manufacturing margin, or reserve, needs to be used, the engineering team will consult with the business team for a collective decision. The same will be true for the other subteams as well.

Also, by recommendation of the customer and their historical mission development processes, the team has drafted a subteam organizational chart covering approximately the full mission lifecycle and for reference this chart will be displayed below.

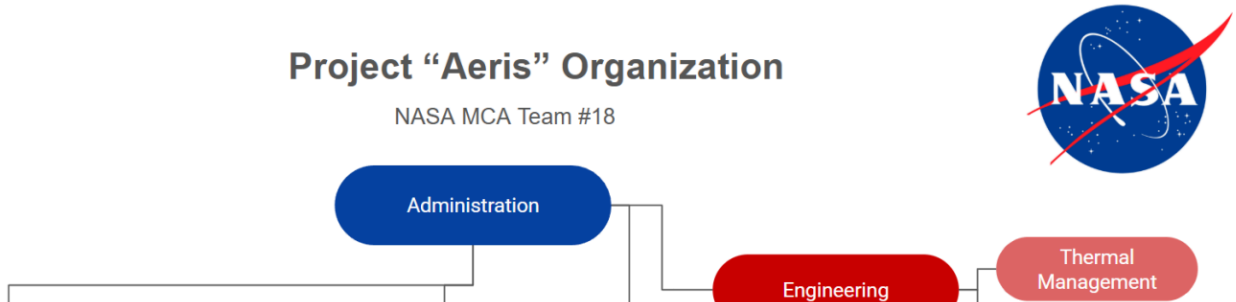
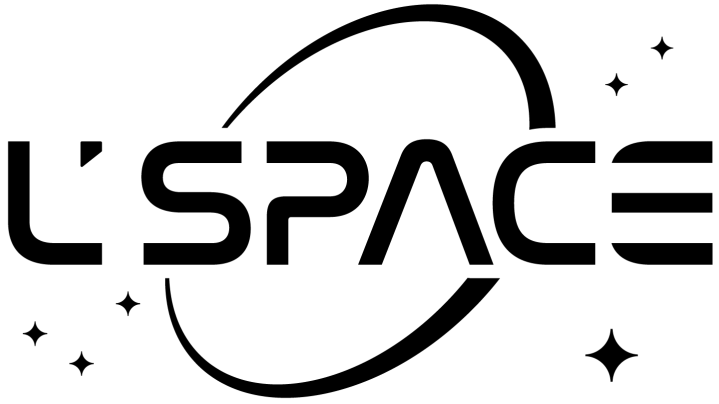


Figure: Full Mission Team Organizational Chart



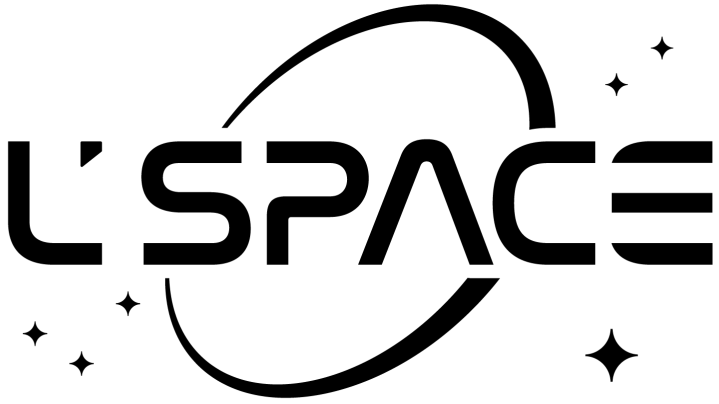
5.2. Mission Schedule

5.2.1. Schedule Basis of Estimate

This section documents the drivers, ground rules, and assumptions used in developing the schedule estimate which includes applicable model inputs, the team's justification for the analogies performed and displayed, and details supporting cost and schedule estimates (NASA 2010).

Science Drivers:
Science Drivers highlights and emphasizes the motive of the team's mission. The main drive of the Venus Aerobot Mission shares quite the same drive as current Venus explorations, such as the DAVINCI mission, to acquire new knowledge and investigate the surface of Venus to better understand its geological history. As the DAVINCI team explains, before the Mariner 2 was launched, Venus was thought to have similar conditions as Earth, as its surface suggests that some areas have mountains containing granite which is commonly thought of as the building blocks for Earth's surface. However, it is now known that this was not the case. By investigating Venus, scientists can study the similarities in various chemical processes that parallels recent extreme changes in climate and a looming future for Earth (NASA Goddard Space Flight Center 2025, "Science").

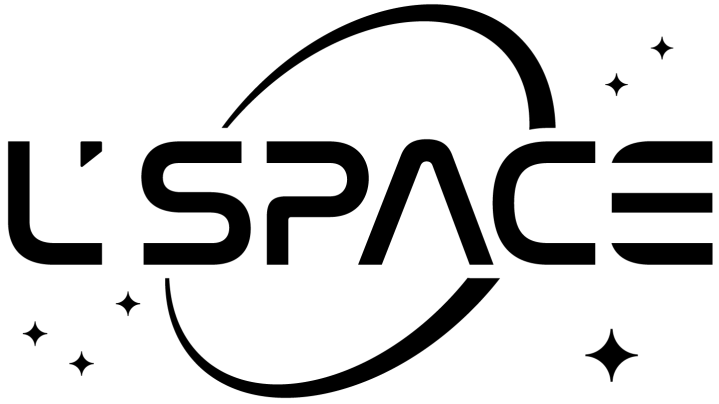
Ground Rules:
The ground rules are as follows: the schedule of basis estimate will only be accounting for Phases C - F. Pre-Phase A, and Phases A and B are regarded as "complete" at the start of the mission timeline. Based on NASA's Venus Express, it can be assumed that the duration for the aerobot to reach Venus will take around 4-5 months (NASA NSSDC 2025). As set by the customer constraints, the team's aerobot will need to be ready for integration with other subsystems by October 1st, 2028 with the launch readiness date



designated to be March 1st, 2029. These dates will be regarded as non-negotiable. The schedule developed by the team will need to incorporate 3 Key Decision Points (KDPs) between Phases C-F. NASA's Space Flight Program and Project Management Handbook states "KDPs for projects are designated with capital letters, e.g., KDP A. The letter corresponds to the project phase that will be entered after successfully passing through the gate". Completion of KDP C is crucial to the team and the stakeholders as this provides approval for implementation of the mission plan and system integration which will occur in Phase D. The customer and stakeholders state that for most planetary exploration missions, the schedule is determined by an optimal launch window where Earth is best aligned with the planetary body to be explored. Because of this, Launch Readiness Dates (LRDs) determined at the beginning of the project are firm-fixed dates.

Assumptions:

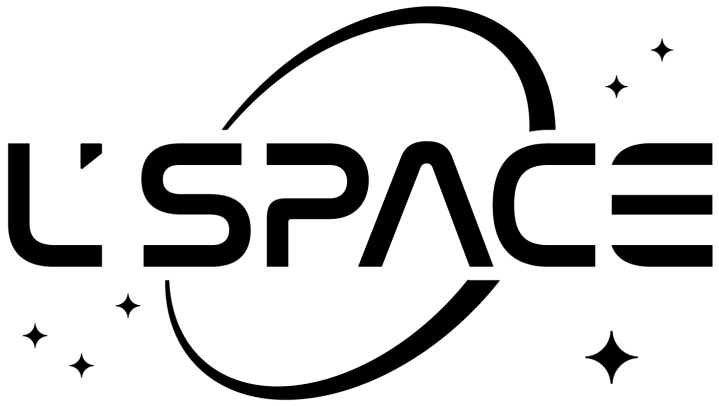
The assumptions addressed here will be formalized on grounds of recommendations received from the stakeholders and the team's best judgement. To estimate the team's schedule, an analogous approach will be taken. Analyzing previous missions allows a realistic estimate of the planned mission to be made. Since the team's target destination is Venus, the estimate will be mirroring similar previous or concurrent missions that also analyzed Venus. The Venus Aerobot System Mission schedule largely follows the typical life cycle within NASA standards based on similar historical and present interplanetary missions such as NASA's Venus Express. Throughout the mission life cycle, there will be required for about at least 30 personnel, consisting of scientists, engineers, technicians, administration, and managers, to travel the designated site locations at the respective dates. Phase C, the fabrication and final design, will begin in May 2025 and is expected to conclude by June 2028. Phase D includes the launch window so it is estimated that the phase will begin by July 2028 and end in March of 2029 following the launch date. This is the ideal date to give the team time to check all details of the system and be ready by the system integration date in October 2028.



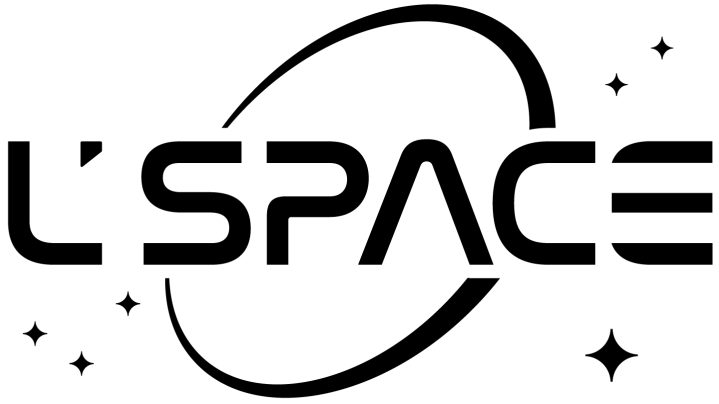
Based on NASA's Venus Express, it can be assumed that the duration for the aerobot to reach Venus will take around 4-5 months (NASA NSSDC 2025).

Life-cycle reviews are essential elements of conducting, managing, evaluating, and approving space flight programs and projects and are an important part of NASA's system of checks and balances.The schedule developed by the team will need to incorporate 3 Key Decision Points (KDPs) between Phases C-F. A portion of the team will travel to the 3 KDP meetings and the system integration site. For all travel costs, the fiscal years of 2024 and 2025 will be used to calculate all financial tasks. It is assumed that there will be one KDP date held in Washington D.C, which will last for about 5 days in total. The mission launch at Cape Canaveral, Florida also requires 5 days of stay for all team members traveling which includes an additional 2 days each of prior and after official meeting days. The successful completion of the KDPs allows for the team to progress forward to the next phase in the mission cycle. An unsuccessful completion would require more time in the schedule that would demand a need for a schedule margin.

A margin is "the allowances carried in budget, projected schedules, and technical performance parameters (e.g., weight, power, or memory) to account for uncertainties and risks. Margins are allocated in the formulation process based on assessments of risks and are typically consumed as the program/project proceeds through the life cycle" (NASA 2007). The expected margin is designated to be about 5-15% of a given task. This estimation was recommended by the stakeholders and was deduced by taking into account the individual task's lifecycle. It is also important to note that the length of the margin will vary as it depends on the task itself. Judgements from those with schedule expertise or contractors will have priority consideration in the recommended margin. Factors that could lengthen the margin include estimated lag time from manufacturers, shortage of personnel, or customer requests. As the phase progresses, long tasks are highly recommended to be broken down into smaller, defined tasks that should take no longer than one month to complete (NASA 2020).



The mission schedule will be efficiently charted and tracked using a Gantt Chart. For a better understanding of the projected schedule in process of mission development, the Gantt Chart featured in the Appendix will be provided.



5.2.2. Mission Schedule

To develop the mission schedule, the team analyzed task complexity, resource availability, historical data from similar missions, and NASA's standard timelines for each life-cycle phase. Duration estimates were refined through team discussions and expert consultation, particularly for hardware development and integration tasks.

Dependencies between tasks were mapped out to identify potential schedule-slippage and ensure that time-sensitive activities were prioritized. Schedule margins have been built in for non-critical tasks to absorb potential delays without impacting the launch schedule.

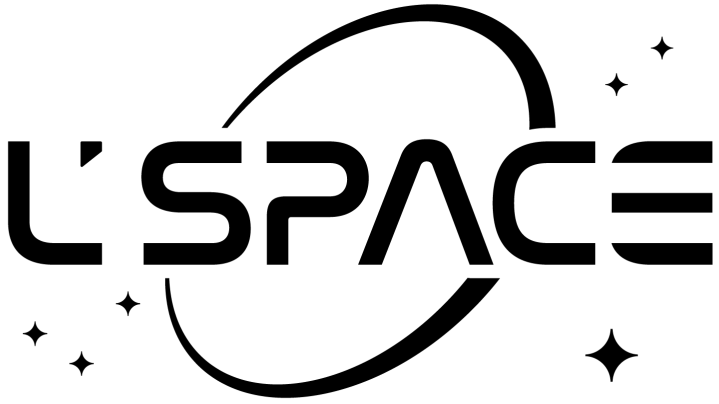
Key constraint dates, such as the mandated System Integration by **October 1, 2028**, and Launch by **March 1, 2029** were used to anchor the schedule and reverse-plan earlier milestones accordingly.

To map the team's plans, a Gantt chart was developed to highlight major task activities and milestones. The Gantt Chart, located in Appendix, visually represents the project timeline, helping the team track progress and identify overlaps or gaps in the schedule. It also highlights task dependencies and durations, which aids in recognizing the critical path and managing risk proactively.

As stated in the Schedule Basis of Estimate (BOE), the mission schedule should only focus on Phases C - F. Each phase includes the following planned activities:

- **Phase C:** Design finalization, subsystem development, CDR, PRR, SIR
- **Phase D:** Assembly, integration, testing, Launch, SAR, ORR, MRR
- **Phase E:** Mission Operations Initiation & Sustainment, CERR, DR

- **Phase F:** Closeout



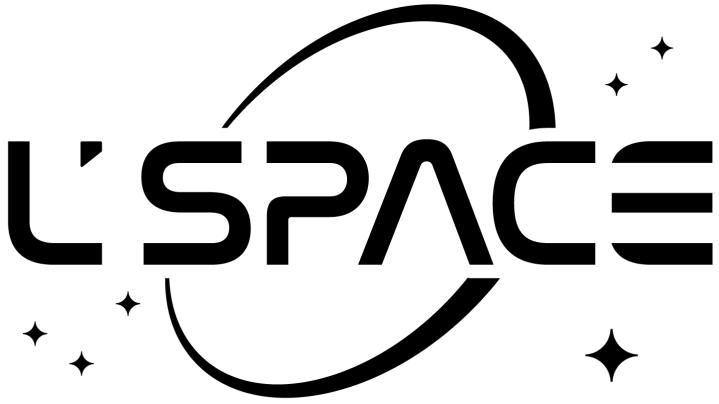
5.3. Budget

5.3.1. Budget Basis of Estimate

This section documents the drivers, ground rules, and assumptions used in developing the schedule estimate which includes applicable model inputs, the team's justification for the analogies performed and displayed, and details supporting cost and schedule estimates (NASA 2014).

Drivers:
As a Discovery Program mission conducting scientific research in support of future missions and advancements toward better understanding our solar system, there are many challenges that the mission team will need to consider. These challenges will ultimately have an adverse impact on the budget and its allocated funds for each section. Committing to the team's science goals is the main driver for the cost basis of estimate (BOE). Typically, Discovery Program missions have a cap of $500M, excluding the launch vehicle and mission operations.

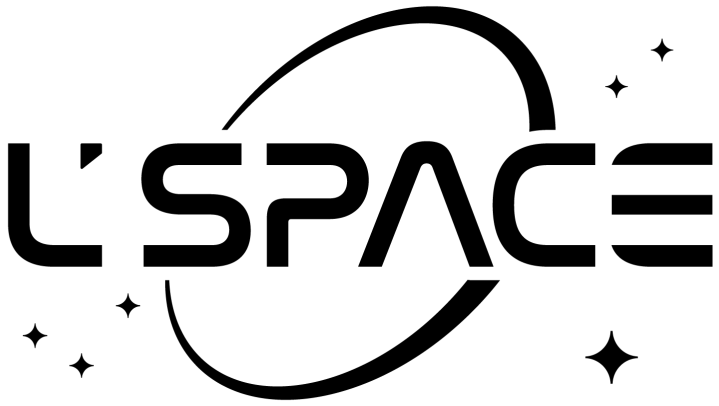
Ground Rules:
The ground rules for the cost BOE is determined and defined by the customers and stakeholders of the mission. As defined by the customer, the cost of all processes done by the team within Phases C - F shall not exceed $200M, as this is the budget limit. The customer explicitly states as well that "this cost is only associated with the Aerobot System you are responsible for. Other elements of this mission receive separate budgets that they manage themselves." All costs associated with launch and external subsystems such as the primary spacecraft and descent subsystem will be covered by various other contractors.



Personnel Costs includes the salaries for scientists, engineers, technicians, administrators, and managers. As an additional ground rule, It is mandatory that the budget accounts for all personnel salaries throughout the entire duration of the mission.

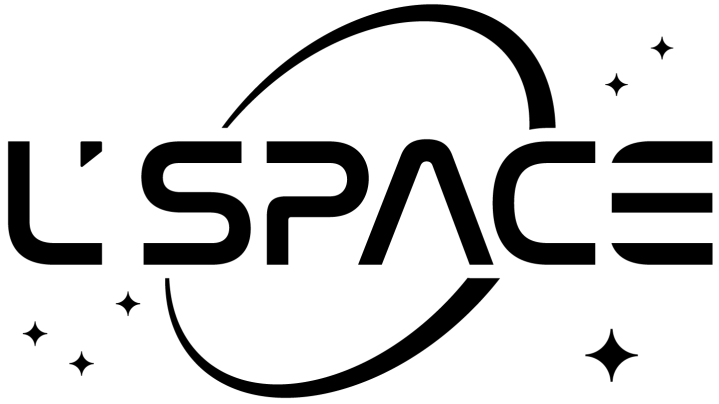
Assumptions:

Much like the Schedule Basis of Estimate (BOE), the cost BOE will also be estimated using the analogous method, a method comparing historical evidence and data, to NASA's DAVINCI Mission, as the Venus aerobot mission and the DAVINCI mission are within the Discovery class. To perform and display a brief calculation of the payload subsystems, parametric estimation will be used. Parametric method of estimating used statistical formulations involving drivers such as the mass, power, and design life of a given subsystem. Analogous relies on the research of historical budget costs from previous similar missions. The general Cost Basis of Estimate (BOE) will be categorized into 4 separate categories such as Personnel, Travel, Outreach, and Direct Costs (Total Spacecraft & Payload) using historical budget costs from previous missions. As previously mentioned, the salaries of all personnel will be calculated for a total of 30, initially. As the mission phases progress and the team expands, the cost BOE will need to be updated once further. Another main budget category is Outreach. Allocating funds for Outreach is also a non-negotiable hidden essential expense. NASA, the customer, is publicly funded by taxpayer funds from United States citizens. So to ensure transparency and education, NASA hosts a series of outreach programs, events, and various other opportunities to let the public know what their newest upcoming missions will be about and how to get involved. The outreach plans developed by the Outreach Officer will primarily determine how much funds will be needed to cover the expenses, although a general estimate of this was calculated based on the NASA’s Budget Requests and displayed in 1.10.5 Outreach Costs. The main expenses of a discovery mission will be the Direct Costs, which includes the payload, vehicle design, testing, and any additional expenses related, excluding the launch costs as that will be covered by the customer. Having sufficient funds for the appropriate instruments is crucial as a failed mission, while it can be a learning opportunity, is a costly consequence.



Cost margins for the Facilities & Administration (F&A), Manufacturing, Employee Related Expenses (ERE), and Total Costs are based on industry standards & stakeholder recommendations. The rate of inflation will remain constant over the course of the mission, based on…

Testing Facilities cost for Thermal Vacuum, Electromagnetic Interference, Vibration Testing, and Ambient Testing also needs to be factored into budget. As defined by the stakeholders, all previous tests mentioned combined accounts for ~30% of the overall instrumentation costs.

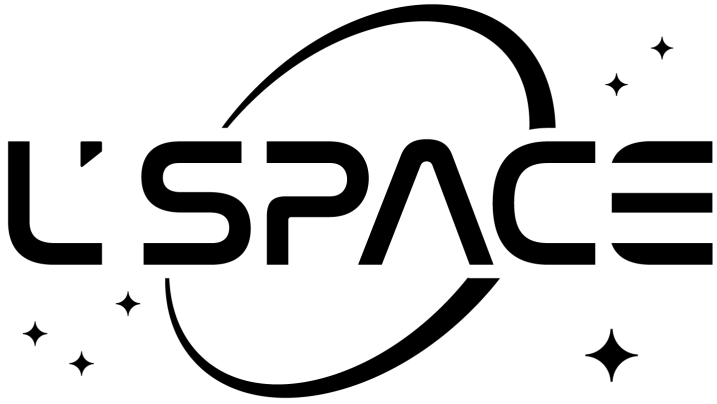


5.3.2. Total Mission Cost

The team's total mission budget of $198,151,226 falls within stakeholder constraints of $200 million. As previously mentioned, the total estimated cost of the mission has been developed based on projected expenses across all relevant phases of the NASA Mission Life Cycle, beginning at Phase C. Cost categories include Personnel (Salaries), Travel, Outreach, and Direct Costs.

To chart the team’s budget, a Budget Template was developed and refined to provide not only a visual representation of budget allocation of each phase but also provides support in calculating general expenses.

The Full Budget Template is displayed on the next page. The template shows the exact specifications of each phase and accounts for margins set by industry standards.



Additional Information

	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F
# People on Team	FY 1	FY 2	FY 3	FY 4	FY 5	FY 6
Science Personnel:	5	5	5	10	15	5
Engineering Personnel:	10	10	15	10	10	10
Technicians:	10	10	10	10	0	0
Administration Personnel:	5	5	5	7	5	5
Management Personnel:	4	4	4	5	4	4

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Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
PERSONNEL							
Science Personnel	$ 400,000	$ 410,400	$ 420,800	$ 862,400	$ 1,324,800	$ 452,000	$ 3,870,400
Engineering Personnel	$ 800,000	$ 820,800	$ 1,262,400	$ 862,400	$ 883,200	$ 904,000	$ 5,532,800
Technicians	$ 600,000	$ 615,600	$ 631,200	$ 646,800	$ -	$ -	$ 2,493,600
Administration Personnel	$ 300,000	$ 307,800	$ 315,600	$ 452,760	$ 331,200	$ 339,000	$ 2,046,360
Project Management	$ 480,000	$ 492,480	$ 504,960	$ 646,800	$ 529,920	$ 542,400	$ 3,196,560
Total Salaries	$ 2,580,000	$ 2,647,080	$ 3,134,960	$ 3,471,160	$ 3,069,120	$ 2,237,400	$ 17,139,720
Total ERE	$ 720,078	$ 738,800	$ 874,967	$ 968,801	$ 856,591	$ 624,458	$ 4,783,696
Personnel Margin	$ 15,000	$ 15,000	$ 15,000	$ 15,000	$ 15,000	$ 15,000	$ 90,000
TOTAL PERSONNEL	$ 3,315,078	$ 3,489,303	$ 4,234,224	$ 4,802,448	$ 4,350,545	$ 3,250,850	$ 23,442,447
TRAVEL							
Total Flights Cost	$ 16,120	$ 16,120	$ 16,120	$ 23,080	$ 16,120	$ 16,120	$ 103,680
Total Hotel Cost	$ 11,040	$ 11,040	$ 11,040	$ 17,800	$ 11,040	$ 11,040	$ 73,000
Total Transportation Cost	$ 350	$ 350	$ 350	$ 700	$ 350	$ 350	$ 2,450
Total Per Diem Cost	$ 8,500	$ 8,500	$ 8,500	$ 8,500	$ 8,500	$ 8,500	$ 51,000
Travel Margin	$ 15,000	$ 15,000	$ 15,000	$ 15,000	$ 15,000	$ 15,000	$ 90,000
Total Travel Costs	$ 51,010	$ 52,336	$ 53,663	$ 70,156	$ 56,315	$ 57,641	$ 341,121
OUTREACH							
Total Outreach Materials	$ 20,000	$ 20,000	$ 20,000	$ 20,000	$ 20,000	$ 20,000	$ 120,000
Total Outreach Venue Costs	$ 50,000	$ 50,000	$ 50,000	$ 50,000	$ 50,000	$ 50,000	$ 300,000
Total Outreach Travel Costs	$ 30,000	$ 30,000	$ 30,000	$ 30,000	$ 30,000	$ 30,000	$ 180,000
Total Outreach Services Costs	$ 60,000	$ 60,000	$ 60,000	$ 60,000	$ 60,000	$ 60,000	$ 360,000
Total Outreach Personnel Costs	$ 33,333	$ 33,333	$ 33,333	$ 33,333	$ 33,333	$ 33,333	$ 199,998
Outreach Margin	$ 100,000	$ 10,000	$ 10,000	$ 10,000	$ 10,000	$ 10,000	$ 150,000
Total Outreach Costs	$ 293,333	$ 208,620	$ 213,906	$ 219,193	$ 224,480	$ 229,766	$ 1,389,298
DIRECT COSTS							
Mechanical Subsystem	$ 156,993	$ 156,993	$ 156,993	$ 156,993	$ 156,993	$ 156,993	$ 941,958
Power Subsystem	$ 2,011,343	$ 2,011,343	$ 2,011,343	$ 2,011,343	$ 2,011,343	$ 2,011,343	$ 12,068,058
Thermal Control Subsystem	$ 1,281,743	$ 1,281,743	$ 1,281,743	$ 1,281,743	$ 1,281,743	$ 1,281,743	$ 7,690,458
Comms & Data Handling Subsystem	$ 1,165,667	$ 1,165,667	$ 1,165,667	$ 1,165,667	$ 1,165,667	$ 1,165,667	$ 6,994,002
Science Instrumentation	$ 8,678,333	$ 8,678,333	$ 8,678,333	$ 8,678,333	$ 8,678,333	$ 8,678,333	$ 52,069,998
Spacecraft Cost Margin	$ 2,603,500	$ 2,603,500	$ 2,603,500	$ 2,603,500	$ 2,603,500	$ 2,603,500	$ 15,620,999
Total Spacecraft Direct Costs	$ 15,897,579	$ 16,310,916	$ 16,724,253	$ 17,137,590	$ 17,550,927	$ 17,964,264	$ 101,585,529
Manufacturing Facility Cost	$ 80,000	$ 80,000	$ 80,000	$ 80,000	$ 80,000	$ 80,000	$ 480,000
Test Facility Cost	$ 120,000	$ 120,000	$ 120,000	$ 120,000	$ 120,000	$ 120,000	$ 720,000
Facility Cost Margin	$ 9,627,289	$ 9,627,289	$ 9,627,289	$ 9,627,289	$ 9,627,289	$ 9,627,289	$ 57,763,734
Total Facilities Costs	$ 9,827,289	$ 10,082,799	$ 10,338,308	$ 10,593,818	$ 10,849,327	$ 11,104,837	$ 62,796,377
Total Direct Costs	$ 25,724,868	$ 26,393,714	$ 27,062,561	$ 27,731,408	$ 28,400,254	$ 29,069,101	$ 164,381,906
Total MTDC	$ 15,897,579	$ 16,310,916	$ 16,724,253	$ 17,137,590	$ 17,550,927	$ 17,964,264	$ 101,585,529
FINAL COST CALCULATIONS							
Total F&A	$ 1,329,408	$ 1,370,742	$ 1,412,075	$ 1,453,409	$ 1,494,743	$ 1,536,076	$ 8,596,453
Total Projected Cost	$ 30,713,697	$ 31,514,715	$ 32,976,429	$ 34,276,614	$ 34,526,337	$ 34,143,435	$ 198,151,226

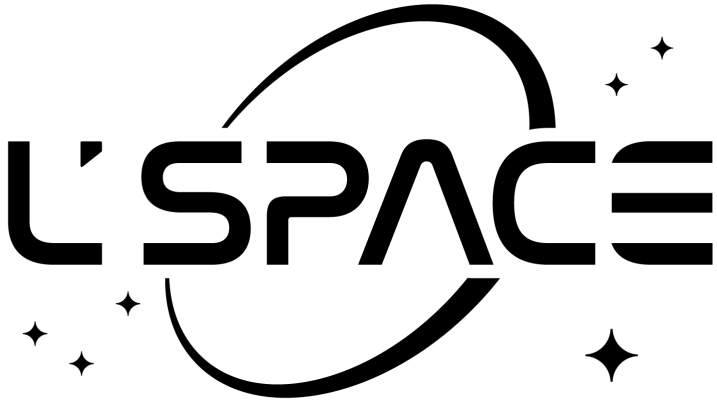
Figure: Budget Template

5.3.3. Personnel Budget

The personnel costs for Team Aeris will vary throughout the mission lifecycle, reflecting the evolving demands of each phase. At the initial planning and design phase, the team will consist of approximately 15 personnel. This includes a small engineering team, key scientists, and project managers, who will focus on mission feasibility, initial system design, and budgeting. This phase is crucial for establishing the mission's framework, including preliminary risk assessments and a clear definition of objectives.

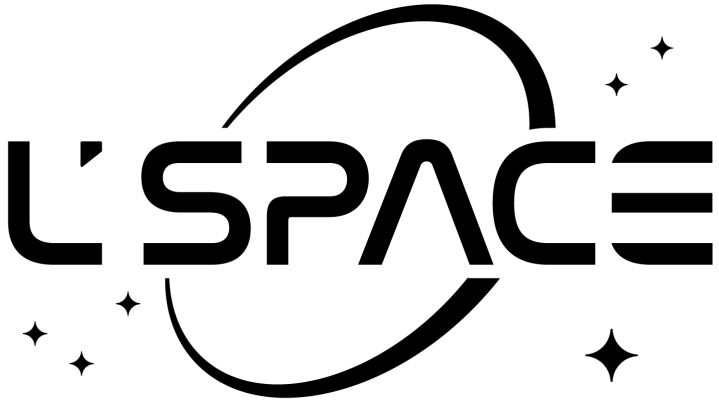
As the mission progresses into the development and testing phases, personnel requirements will increase to around 30 personnel. Additional engineers, technicians, data analysts, and management staff will be brought on board to handle system integration, prototype testing, and detailed risk management. The increased personnel count is essential for successfully moving from design to development, where collaboration across different expertise areas becomes necessary for ensuring the aerobot's systems and instruments function as intended. This period also involves extensive testing and troubleshooting, requiring additional resources for quality control and technical verification.

During the critical phases of final integration, launch, and initial operations, the staffing levels will remain at the peak of 30 people to ensure that all mission systems are functioning optimally. The heightened focus on system checks, testing under real-world conditions, and final adjustments requires maintaining a robust workforce to address unforeseen issues as they arise. After launch, the personnel requirements will begin to decrease, as the aerobot enters the operational phase of data collection.



Once the aerobot is deployed and data collection begins, the personnel requirements will shift. While the engineering team's role will diminish, the need for the science team, data analysts, and mission operators will remain. At this point, staffing levels will scale back to around 20 personnel. The focus will now be on monitoring the aerobot's performance, analyzing incoming data, and making operational adjustments as necessary. Although the team will be smaller, the tasks during this phase remain highly critical to the mission's success, as the team must ensure the aerobot continues to collect the necessary data on Venus' atmosphere and surface activity.

As the mission approaches its conclusion, staffing will be further reduced to about 10 personnel. At this stage, the team will primarily consist of data analysts, outreach officers, and mission managers, who will be responsible for the final analysis of collected data, documentation, and the archiving process. The reduction in personnel is due to the decrease in operational tasks, as the mission shifts from active data collection to final reporting and analysis.



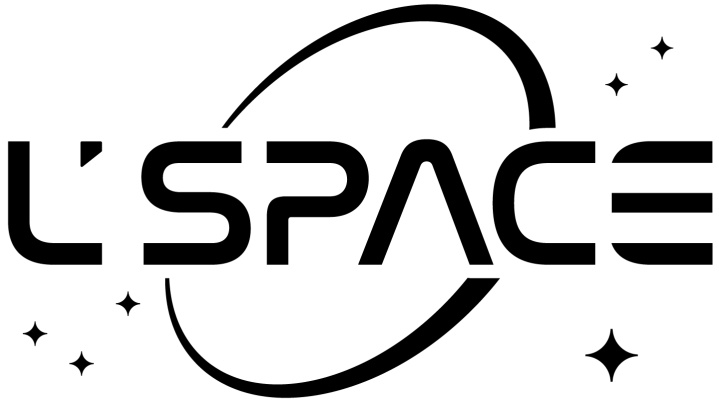
5.3.4. Travel Budget

In planning the travel costs for Team Aeris, several key considerations must be factored into the budget to ensure the mission progresses smoothly while maintaining fiscal responsibility within the $200 million lifecycle cost (LCC) allocation.

The team will plan for commercial coach flights using government rates from the City Pair Program (CPP). The cost of these flights will be determined based on the highest fare from various cities flying into the necessary locations, ensuring that the most cost-effective options are chosen for team members traveling to mission-related events. Given the nature of the mission, personnel will likely need to attend hardware development, system integration meetings, Standing Review Boards (SRB), and mission operations centers during the operational phase. Flight costs will be spread across multiple mission phases, especially during the development, testing, and launch phases.

Lodging costs will be calculated using Fedrooms, which partners with government-approved hotels. Each team member will require a room, and the team will plan for accommodations that provide suitable amenities at reasonable rates, ensuring that team members are comfortable during extended stays, particularly when they need to attend multiple review meetings or hardware testing events. The duration of stays will vary depending on mission milestones, but accommodations will generally be arranged for trips lasting several days.

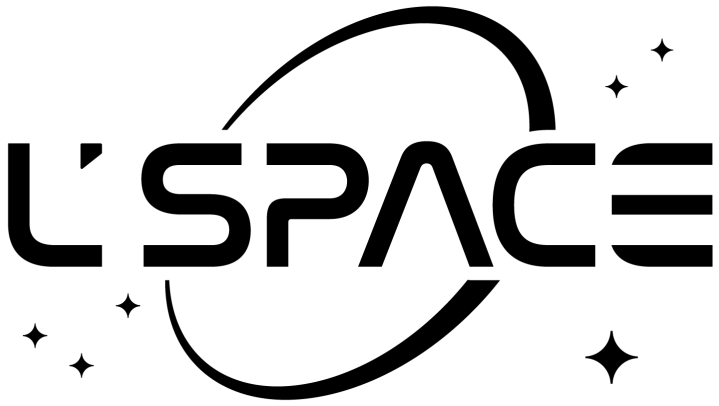
For ground travel, the team will budget for car rentals, considering varying schedules and travel needs. Car rentals are necessary to transport team members to and from airports, hotels, and mission-related venues. The team will assess each trip’s requirements, ensuring that vehicles are available to accommodate team members while balancing the need for cost-effectiveness.



Meals will be covered under the per diem rate allowances for food and lodging, as determined by the U.S. General Services Administration (GSA) Per Diem Website. Per diem will be provided for each team member for the duration of their travel, ensuring they are adequately compensated for meals while on business trips. These rates will be adhered to, and adjustments will be made depending on location and duration of travel.

Personnel costs will account for salaries across various roles, with scientists, engineers, technicians, and administrators all receiving specific salary rates. Scientists and engineers, each receiving an annual salary of $80,000, will be needed for the design and execution of the mission. Technicians, with an annual salary of $60,000, will assist with manufacturing, assembly, and testing, while administrators will manage the mission’s day-to-day administrative tasks. Managers, including the Project Manager and Deputy Project Manager, will be essential for overseeing personnel, schedules, and budgets, with an annual salary of $120,000. These salary figures will be multiplied by the duration of the mission phases to determine the total personnel cost.

The total travel and personnel costs will be planned and adjusted throughout the mission, ensuring that the team stays within the allocated LCC of $200 million for post-PDR phases (C-F), excluding sunk costs from Pre-Phase A, Phase A, and Phase B. Careful attention to planning, prioritization of travel, and resource optimization will help the team ensure that all critical activities are supported without exceeding budget constraints. The outlined approach will facilitate mission success while adhering to financial guidelines and ensuring cost-efficient operations.



5.3.5. Outreach Budget

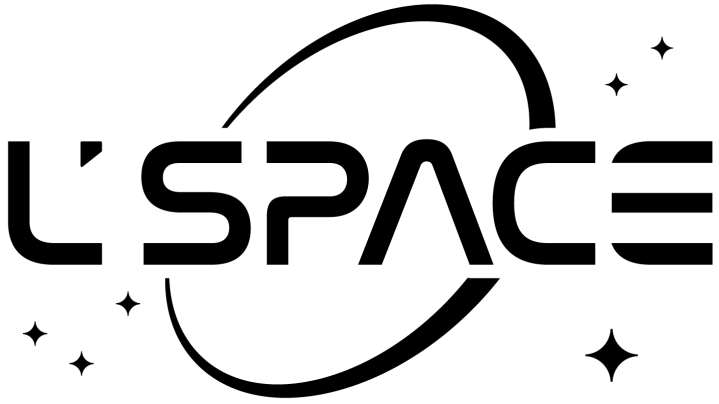
Team 18 has set aside $1.22 million for a public outreach plan in order to connect with a diverse audience. The goal is to make space exploration accessible and exciting for everyone from students and local communities to people following along on social media.

For K–12 schools, the team plans to spend $50,000 on virtual classroom sessions and $60,000 on in-person school visits. That covers everything from educational materials and teacher pay to travel costs and event setup. The idea is to bring space science straight into classrooms, making it fun, interactive, and easy for younger students to understand.

When it comes to colleges and universities, the plan includes $200,000 for student internships, $40,000 for guest lectures and workshops, and $30,000 for student-led events. These funds help cover things like speaker fees, travel, stipends, event materials, and more. It’s about giving students real opportunities to get involved, ask questions, and build skills that could lead to future careers in science and engineering.

Community outreach is a big part of the plan too. About $120,000 is going toward exhibits at museums and science centers, and another $90,000 is for running STEM workshops that focus on underrepresented communities. This includes costs like exhibit setup, travel, workshop materials, and staffing.

On the digital side, $200,000 will go into social media outreach to keep the public updated and engaged. Another $120,000 is set for building and maintaining a website and creating interactive tools. That budget covers things like content creation, paid ads, platform management, and website development.



Finally, the team is setting aside $120,000 just in case—this reserve will cover any surprise expenses or emergencies that might pop up along the way.

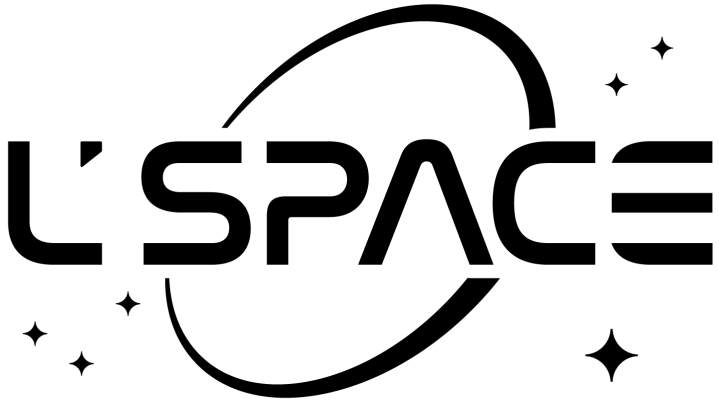
The $1.22 million budget is designed to support a well-rounded outreach plan that gets people excited about the mission, spreads awareness, and helps create a strong connection between the public and the Venus exploration mission.

5.3.6. Direct Costs

The direct costs of the mission covers the purchases of necessary hardware, contractor labor, and scientific instrumentation needed to complete the mission. It includes the vehicle itself and all systems associated with it. By the customers and stakeholders constraints, the Venus Aerobot vehicle must comply to their expectations, such as:

1. **Mass:** 50 kg
2. **Dimensions:** 1 x 1 x 1 m
3. **Prohibited Material:** Radioisotope Thermoelectric Generator (RTG)

Initially, the general, encompassing direct costs were estimated using analogous methods, such as comparing the historical cost of the payload and vehicle instruments used on the same class, previous NASA missions such as the Venus Express. Ideally, costs established in further deliverables will be estimated and finalized using a method of parametrics with the various NASA budget tools such as the Mission Concept Cost Estimate Tool (MCCET) to utilize the Cost Estimating Relationship equations. The MCCET takes into account system, subsystem, wrap costs, and Schedule Estimating



Relationship (SER), which determines the estimated time for the manufacturing, construction, and testing of an instrument. When using the MCCET, the inflation rate of the given year needs to be calculated to ensure accurate estimates. Because the subsystems have not been finalized at this time, the exact mass, power, and design life of each subsystem is unknown and thus will be estimated using both the parametric method and the analogous method with a similar class mission, the Venus Express. In the Venus Express mission, the following specifications were as followed:

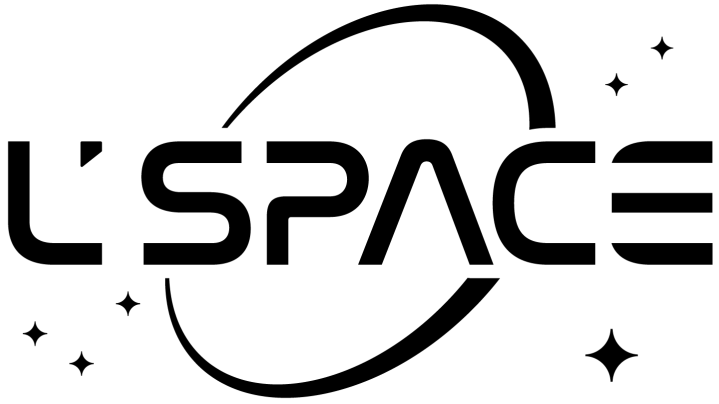
1. **Mass:** 1270 kg
2. **Dimensions:** 1.5 x 1.8 x 1.4 m
3. **Total Power Used:** 1.9 kW

Mechanical:

The mechanical subsystem includes the design and the material aspects of the mission vehicle. The spacecraft consisted of the main body chassis, thruster, antennas, radiators, and various other components. Majority of the scientific instruments onboard the Venus Express were mounted in the “bus”, body of the spacecraft. The vehicle was described as a “honeycomb aluminium box about one and a half metres wide” (ESA). Material Selection was also essential in maximizing the efficiency of its design.

For the Venus aerobot, any use of RTGs is prohibited. Any external radioactive materials can not exceed an accumulated mass of 5g. The team has not yet finalized material selection for the vehicle but the customer constraints will be regarded as ground rules for the design. Using the MCCET for the payload subsystem, the total properties of the mechanical subsystem are as follows:

1. Mass: 0.898 kg
2. Power: 18 W
3. Design Life: 16 years



Using the Mechanical Subsystem Model Cost Estimate Relationship (CER) in the Mission Concept Cost Estimate Tool (MCCET), which are formulated based on historical NASA missions, the total mechanical subsystem cost is about $941,952.

Power:

A vehicle's power subsystem refers to the supply with the ability to generate energy for the vehicle and payload. The power subsystem was found to have the following properties:

1. Mass: 16.5 kg
2. Power: 410 W
3. Design Life: 10 years

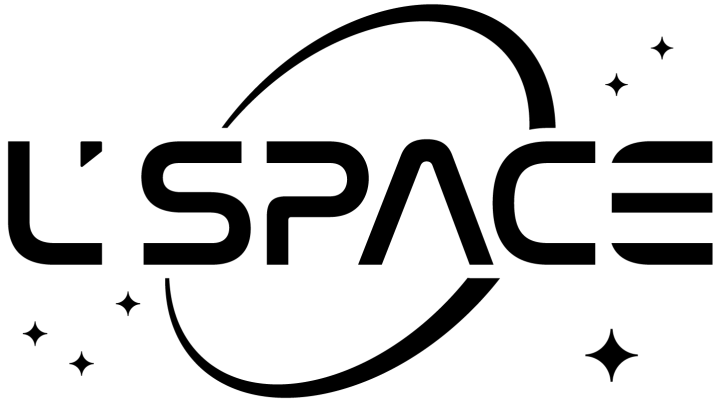
In total, the power subsystem estimated costs came out to be $12.07 million. This cost was determined by using the Electronics Subsystem Model Cost Estimate Relationship (CER) in the Mission Concept Cost Estimate Tool (MCCET).

Command and Data Handling (CDH):

The Command and Data Handling subsystem is critical in a spacecraft as it refers to the electronics or software responsible for all digital communications from the spacecraft to the home base. This subsystem controls all command executions, data processing and storage, and recovery. For the aerobot, the projected properties are as follows:

1. Mass: 18.2 kg
2. Power: 0.487 kW

The estimated cost of the electronics subsystem using the MCCET & Electronics Subsystem CER, was found to be $6.994 million.



Instrumentation:
“Instrumentation” refers to scientific instruments implemented into the vehicle’s payload to carry out the science objectives. Previously the team used analogous estimation by comparing, as provided by the NASA Space Science Data Coordinated Archive, the Venus Express designated approximately $89.2M for satellite construction and instrument integration. The team’s scientific instrumentation properties are as follows:

1. Power: 149 W
2. Design Life: 2 years (designed), 8 years (actual mission life)

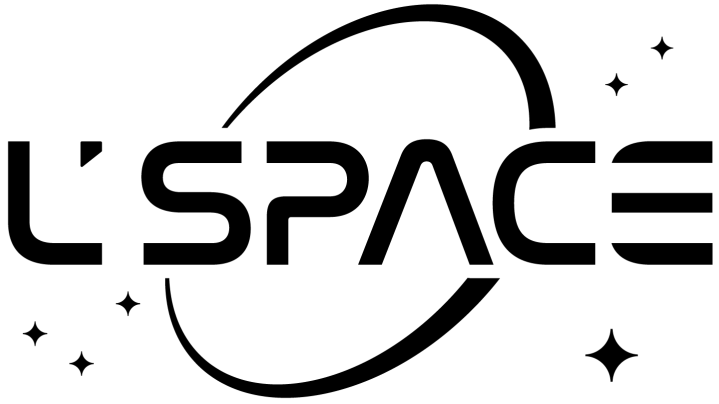
Using the appropriate CERs from the MCCET, the estimated cost of the scientific instrumentations is $52,069,998.

The total cost the team can expect for the system’s subsystems is $164,381,906, including manufacturing margin, facility cost margin, and spacecraft cost margin.

5.4. Scope Management

5.4.1. Change Control Management

The team follows a structure for change control in order to ensure that when changes are wanted, they are requested and evaluated through Request For Action (RFA) and Advisory Decision Validation (ADV) before they are implemented. This process is conducted so that there are minimal disruptions to the overall team mission and deliverable progress. Whether the changes are related to budget, technicalities, or schedule they must first be requested.

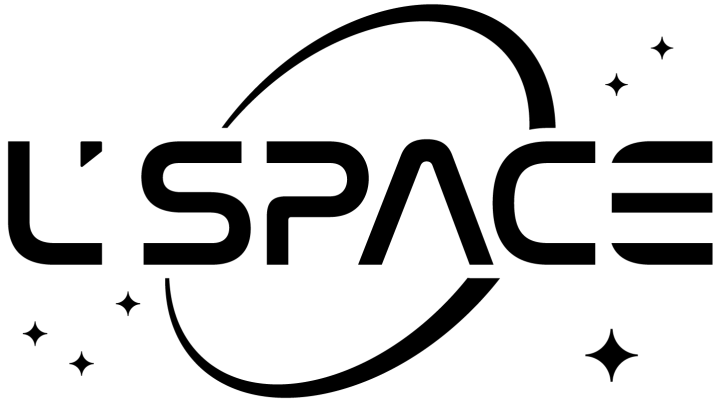


Team members must first identify a need for change based on risks or challenges that may arise. There must be a description of the change, reasons for the change, the potential impact on the mission, and supporting data. Once this is gathered, a member may bring the change request to their respective discipline leader.

The leaders of the Programmatics, Science, and Engineering teams will review the request and evaluate its impact on the mission. If the change is technical, the Engineering and Science team will lead the assessment. However, if the change is financial or logistical the Programmatics team will lead the assessment.

RFAs and ADVs are only provided by the customer/stakeholders, which are external changes given by a team or individual outside of the writers and contributors of the deliverable. The external feedback provided is prioritized over the internal feedback. This is done to ensure the team complies with the required standards and practices to ensure a successful mission. Once the evaluation of the deliverable is complete, a list of RFAs and ADVs along with feedback is sent to the team. These requests, RFAs, explain what is required in order to implement the change, including specific resources and time needed. After the Request For Action is completed, the Advisory Decision Validation (ADV) comes next. The ADV validates and confirms the decision in order to proceed with the change request. This document often provides justification and cost-benefit analysis of the proposed change to the team.

Internal changes originate from within the team or group working on the assignment. These changes could be in relation to the engineering, programmatics, or science aspects in the previous deliverable. If the team believes that a change is needed to correctly progress onto the next deliverable, then a Change Request Form can be made. Change requests within the CRF should be changes that are baselined, meaning that they have been previously established and approved by the stakeholders. For any internal changes, the team prioritizes any scientific change to be taken into



consideration first. The main goal of this mission is to carry out all the scientific goals detailed in the STM and maximize the usage of scientific instruments implemented on the vehicle.

Finally, a plan is created to implement the changes. This plan includes a timeline, assignments, and assigns responsibilities to Programmatics, Science, and Engineering teams. The team has decided that an effective way to measure and track progress would be to meet weekly and communicate on any internal feedback or requests for assistance. Weekly meeting notes are taken as well as attendance. From here, each team will collaborate to provide the necessary resources needed to implement the change. This may include adjusting the budget, schedule or systems. If issues arise during implementation, they will be monitored and changes will be made. In the event that a subteam needs to decide upon an urgent decision, they will hold a meeting within 72 hours to discuss ahead of the deadline.

Communication amongst the team will be solely done within Discord. Within Discord, the team discusses all questions, objectives, and other collaborative efforts. Additionally, the team shares all documentations, templates, meeting notes, and deliverables within a shared team Google Drive Folder. As a baseline of communication, the team prioritizes and emphasizes consistent communication and members who do not follow up with their subteam or project manager in a proper and timely manner will be communicated with to ensure their continuation in the mission development process.

For most decisions carried out amongst the team, it will be ultimately decided by a majority vote but the sub team leader can make the final decision if time does not permit the team to vote collectively. If further confusion occurs, the project manager will either make the decision or elicit guidance from the stakeholders.

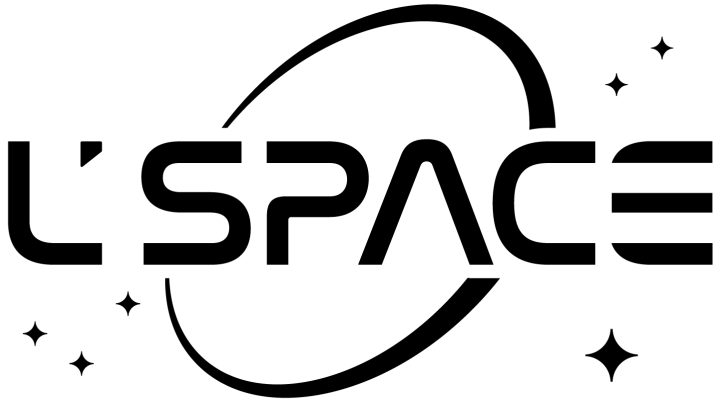
To track assignments or tasks delegated, the team creates a task list for each individual deliverable detailing the specifics of each section (1.1, 1.2, 1.3,...) in addition to

assigning a subteam or specific role with an internal deadline that is highly encouraged but not required. For reference, the figure below is a preview of the task list the team created for the System Requirements Review.

SRR Task List

Tasks	Assigned to	Due Date
1.1 Mission Statement		
Write Mission Statement	All	
1.2 Science Traceability Matrix		
TBD 1	Engineering	DONE
TBD 2	Engineering	DONE
TBD 3	Engineering	DONE
1.3 Summary of Mission Location		
Mission Location	Science	3/4
1.4 Mission Requirements		
Repeated from MCR	Science	3/4
1.5.1 Spacecraft Overview		
Top Level Subsystem Requirements	Engineering	3/6
Narrative Overview of Spacecraft/Subsystems	Engineering	3/6
CAD Drawings/Sketches of Spacecraft	Engineering	3/6

Figure: SRR Task List



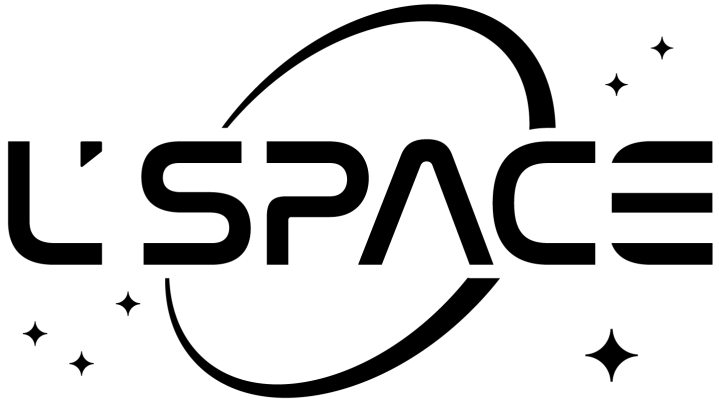
5.4.2. Scope Control Management

Descope – Cost

In the event that a scope change, or rather a descope, is required to proceed with the mission, the team will prioritize in addressing any concerns or doubts that the contractors may bring awareness to. Before planning alternative design concepts or changes to the science instrumentation, the budget is first analyzed to determine if any available option will cause a significant deviation from the allocated budget that was decided. “The development process for the architectural concepts should be recursive and iterative with feedback from stakeholders and external reviewers, as well as from subsystem designers and operators, provided as often as possible to increase the likelihood of effectively achieving the program’s desired ends while reducing the likelihood of cost and schedule overruns” (NASA Systems Engineering Handbook). Costs and schedule overruns typically refers to when the expenditures demanded exceeds the established budget and margin allocated. Because of the historical common occurrence of missions going over budget, a margin for manufacturing costs was placed to prevent any unpredicted expenses.

As time progresses within the mission planning process, the possibility of a descope increases significantly. Overtime, a contractor could locate an issue with one of the company’s instruments which will force the team to develop a descope plan. A clear example of how the team developed a descope plan for this specific instance is located in the Appendix under “Descope Slides”.

After addressing any cost or schedule concerns, the design of the team’s vehicle will be assessed for any potential changes to be done. The engineering team will examine for any redundant instrumentation that can be eliminated to either reduce mass or maximize volume. The team’s main goal for the creation of the aerobot is to adhere to our science objectives. Ideally, removing any instrumentation that might hinder the team’s ability to carry out all objectives will be considered the last option only if the team



faces a significant schedule or cost change. The science objectives the team set in place as shown in the Science Traceability Matrix (STM) are primarily regarded as non-negotiable, meaning that any scope changes will be planned around them.

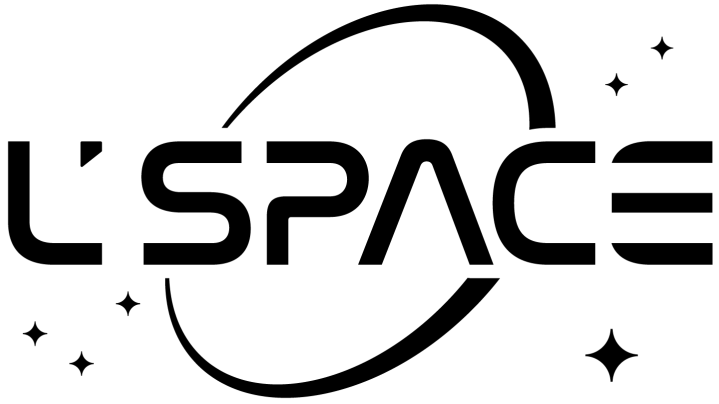
Descope – Schedule

“Project crashing” refers to the act of compressing the expected schedule timeline by reducing at least one task’s individual timeline. To counter the shorter time limit, additional resources are implemented (Simmons 2025). In turn, this will also add onto the existing costs and risks present. Meanwhile, “Fast-tracking” a project “refers to a method where tasks that would ordinarily be completed in sequential order are instead completed concurrently” (Gurnov 2024). In the event that any schedule change arises, the team has planned to develop a descope plan and implement schedule margins to serve not only as a buffer but also a placeholder for any additional time required to achieve full capability.

A descope plan assesses the potential negative impacts that will affect the project if the mission has a setback. Firstly, the team will review or develop the mitigation options and discuss their effects on all aspects of the project (Engineering, Science, & Programmatic). Fast-tracking or project crashing may be required if the team is closing in on an important milestone that is dictated by the customer (ground rule), such as the systems integration window or the launch date.

Increase of Scope – Cost

If an increase in scope occurs, the team will first conduct a comprehensive impact assessment to evaluate how the changes affect timelines, resources, and technical requirements. Key stakeholders will be consulted to ensure alignment on revised objectives and deliverables. The project timeline and budget will be adjusted accordingly, and responsibilities will be redistributed to maintain efficiency and accountability. Risk assessments will be updated to reflect the expanded scope, and



mitigation strategies will be revised as needed. Throughout the process, the team will maintain transparent communication and documentation to ensure all members are informed and aligned with the updated mission goals.

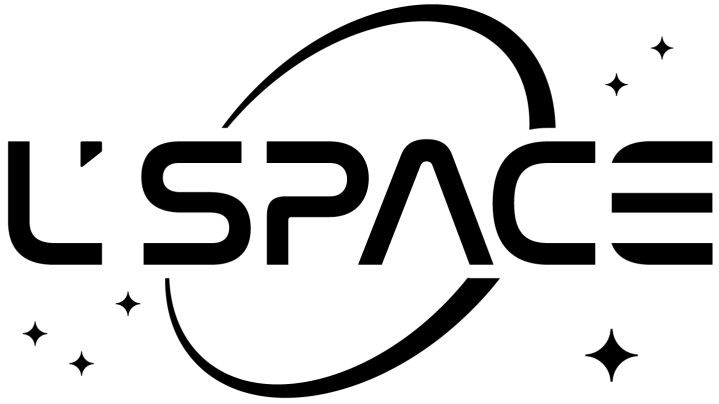
Any additional funds/time will be used to cover direct costs, expenses and testing dates in the schedule as this area tends to overrun the margin. This will be used to determine if all the subsystems are functional and compatible and if new scientific instruments could be included in the payload.

Increase of Scope – Schedule

If an increase in scope occurs, the team will first conduct a comprehensive impact assessment to evaluate how the changes affect timelines, resources, and technical requirements. Key stakeholders will be consulted to ensure alignment on revised objectives and deliverables. The project timeline and budget will be adjusted accordingly, and responsibilities will be redistributed to maintain efficiency and accountability. Risk assessments will be updated to reflect the expanded scope, and mitigation strategies will be revised as needed. Throughout the process, the team will maintain transparent communication and documentation to ensure all members are informed and aligned with the updated mission goals.

Any additional funds/time will be used to cover direct costs, expenses and testing dates in the schedule as this area tends to overrun the margin. This will be used to determine if all the subsystems are functional and compatible and if new scientific instruments could be included in the payload.

Throughout the mission planning process, it is important to remember that the engineering design, programmatics, and potentially the science objectives, will change ultimately after each deliverable.



5.5. Outreach Summary

Team 18’s outreach plan is about connecting with the public in a way that builds awareness of what is being done, and makes the mission feel accessible to everyone. The mission is focused on understanding how Venus' surface and atmosphere interact. Outreach efforts will target schools (K-12 and up), universities, local communities, and especially those who don’t always get a chance to be part of space science efforts. Through this, the goal is to publicize the mission, its findings, and future endeavors with the general public. It is extremely crucial to involve and educate individuals from all walks of life about the purposes of the mission.

For younger grades, fun and educational virtual classroom sessions will be offered, led by mission scientists and engineers. Free, teacher-friendly classroom resources like videos, worksheets, and even simulations will also be created. These tools will help explain topics like Venus’ extreme surface temperatures, its volcanoes, and how the aerobot will fly through its atmosphere. Using fun activities like these is a good way to make science engaging for the younger generation and create a growing interest in planetary science.

For middle and high school students, webinars will be hosted that dive deeper into science—focusing on topics like tectonic activity on Venus and how its extreme climate might have formed. The goal is to help students see how science is applied in real-life missions. Students at this level may also be more interested in watching informational videos of the mission.

College students are at a point where career paths are being shaped, so it's important to show that planetary science and space engineering are viable and exciting options. The outreach plan includes internships for undergraduate and graduate students in fields like planetary science, computer science, engineering, and data analysis.

These students will gain real hands-on experience by working alongside the mission team on tasks such as collecting and analyzing mission data. Guest lectures will also be hosted on university campuses, allowing students to hear directly from mission scientists.

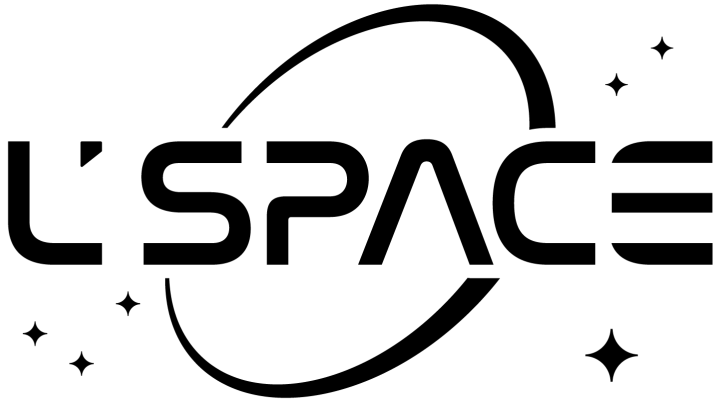
Student clubs and organizations will be invited to participate as well. Support will be given for activities like simple aerobot model-building projects and lab experiments simulating Venus' harsh conditions. The focus is on making space science more engaging and within reach for everyone.

A major aspect of the outreach strategy involves connecting with people from all walks of life, especially those in underrepresented and underserved communities. STEM workshops and mission materials will be brought directly to places that don't always have access to such resources.

Collaborations with local community centers, museums, and science centers will support public events that bring Venus science to life. These events will include hands-on exhibits that showcase what Venus is really like—such as extreme temperature demos, volcano simulations, and visualizations of the aerobot flying through Venus' clouds. These experiences are designed to be interactive, fun, and easy to understand for individuals with any level of science background.

Special STEM workshops will also be organized for low-income neighborhoods and for groups such as women in STEM and minority students. These workshops will feature hands-on activities like designing mini space missions or simulating volcanic eruptions and their impact on planetary climates. This ensures the mission is inclusive and accessible to all.

A strong online presence will also be maintained on platforms like TikTok and Instagram, which are widely used by the public to learn about science, news, and current events.

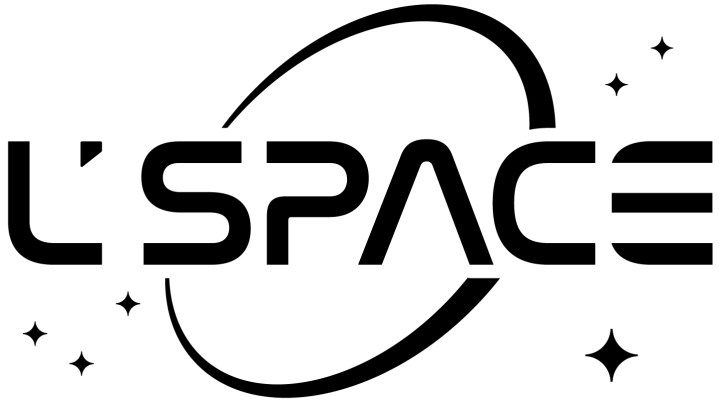


On TikTok, short videos will be shared featuring mission highlights, behind-the-scenes footage, fun Venus facts, and Q&A sessions where scientists answer public questions. This is a great way to keep the public engaged, even through casual browsing. Instagram will be used for visual storytelling, with posts including aerobot photos, infographics, mission updates, and interactive polls.

A dedicated website will be launched to serve as the main public hub for mission updates. The website will feature live mission tracking, interactive maps of Venus, educational resources for all ages, and simulation-based activities. Visitors will be able to explore the mission in real time and participate in interactive experiences—essentially a digital mission control for the public.

Team 18 will also explore partnerships with global outreach organizations, science influencers, and nonprofit organizations. By leveraging these connections, outreach efforts can expand beyond local and national boundaries to include international students and science enthusiasts. Translated content and subtitles will be created to make sure language is not a barrier to participation.

Science competitions and challenges tied to the Venus mission will also be launched. This will be an event organized for giving students the chance to submit creative ideas or science experiments inspired by the mission. Winners can receive virtual shout-outs, team merchandise, or even the opportunity to see the team in person as an internship experience.



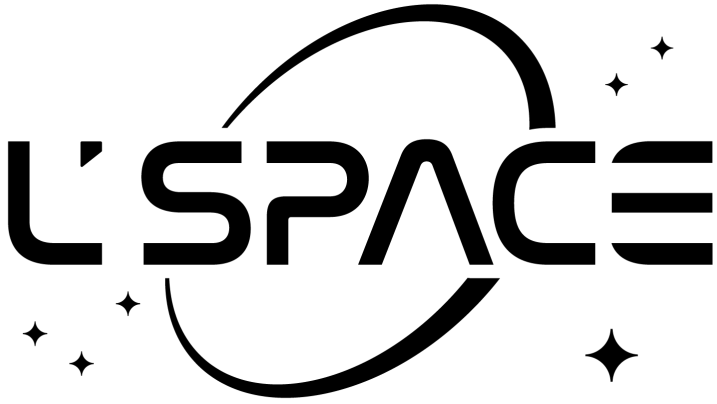
6. Conclusion

The “Aeris” aerobot mission aims to enhance the understanding of Venus’ atmosphere and surface interactions through high-resolution data collection. With the deployment of an aerobot that is able to descend through the atmosphere, the aerobot will analyze the atmospheric processes, chemical interactions, and potential signs of active volcanism, which can contribute to valuable insights on the planet's evolution and climate tendencies. Despite significant challenges surrounding Venus’ environmental conditions and NASA’s strict mission constraints, the team is believed to have developed an achievable plan that balances the scientific objectives with the technical limitations.

A PDR (Preliminary Design Review) is intended to “evaluate the completeness/consistency of the planning, technical, cost, and schedule baselines developed during Formulation. To assess compliance of the preliminary design with applicable requirements and to determine if the project is sufficiently mature to begin Phase C.” (NASA 2007), as defined by NASA Procedure Requirement (NPR) 7120.5.

Within this PDR, the team reviewed the established mission science objectives, and presented an overview of the proposed spacecraft as well as all of its subsystems. The team has aimed to provide a more mature spacecraft and mission design that is suitable for further development within Phase C. The team’s proposed schedule and cost estimate was revised to better suit recent changes and requirements from the stakeholders and a plan for scope management was included to better understand the team’s internal and external process for adaptability to changes.

As the team continues in the mission planning process, the engineers will further refine their subsystems and the programmatics division will work alongside them to finalize their cost analysis in order to prepare the mission for the next deliverable, the Critical Design Review, and transition into Phase C, where fabrication of the materials can begin.



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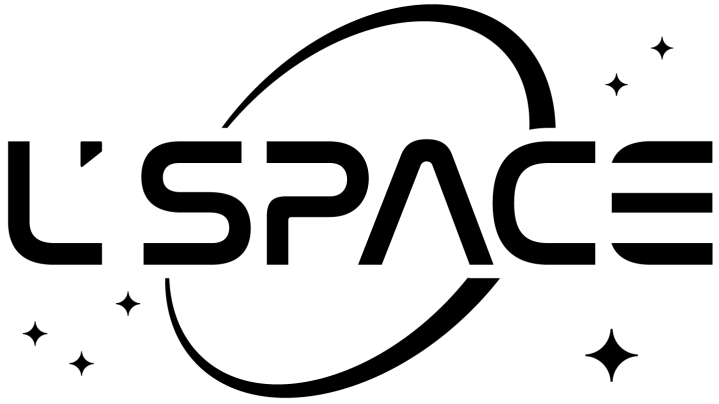
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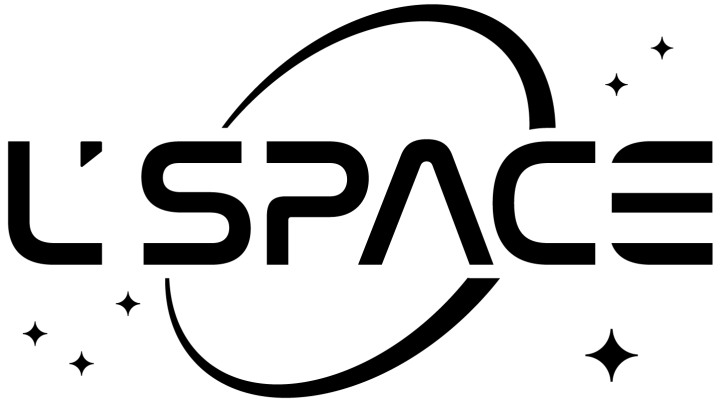
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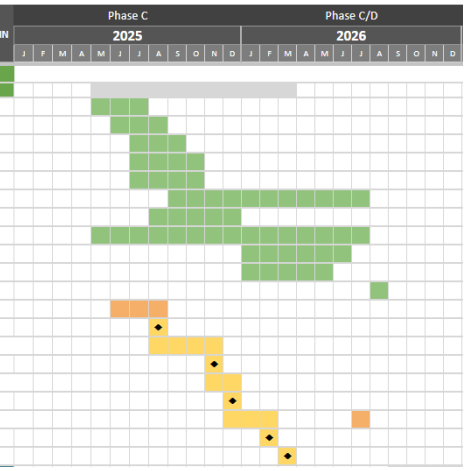
8. Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the team used ChatGPT in order to meticulously track costs for the Budget Template and locate data specifics for instrumentations. After using this tool/service, the team reviewed and edited the content as needed and took full responsibility for the content of the deliverable.

9. Appendix

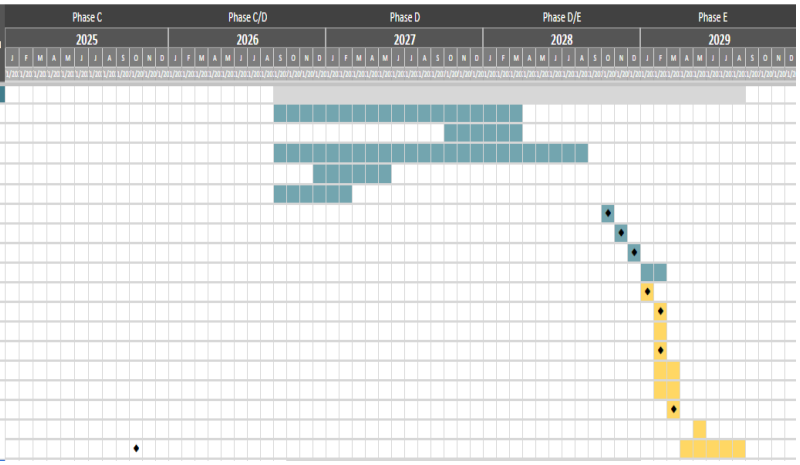
Aeris Gantt Chart - Phase C: Final Design & Fabrication

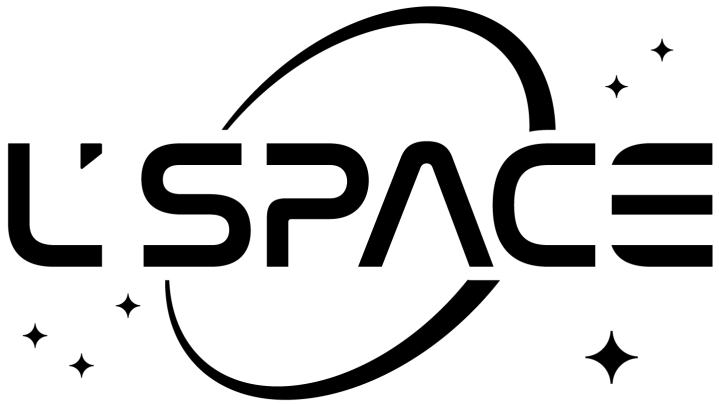
ID#	TASK	ASSIGNED TO	PROGRESS	START	END	DAYS	MARGIN
1	**Final Design & Fabrication**		0%	5/1/25	3/6/26	310	270
1.1	Updated Cost & Schedule Estimate Review	Programmatic	Not complete	5/1/25	7/1/25	62	
1.2	Interface Control Documentation	Engineering	Not complete	6/1/25	8/1/25	62	
1.3	Subsystem Characteristic Finalization	Engineering	Not complete	7/1/25	9/1/25	63	
1.4	Science Objectives Finalization	Science	Not complete	7/1/25	10/1/25	93	
1.5	Develop Decommision and Disposal Plan	Engineering	Not complete	7/1/25	10/1/25	93	
1.6	Perform and archive trade studies	Engineering	Not complete	9/1/25	7/1/26	304	
1.7	Develop Operations plans	Science	Not complete	8/1/25	12/1/25	123	
1.8	Identify and update risks	Programmatic	Not complete	5/1/25	7/1/26	427	
1.9	Refine Operations plans	Science	Not complete	1/1/26	6/1/26	152	
1.A	Establish a Bill of Materials	Programmatic	Not complete	1/1/26	5/1/26	121	
1.B	Outreach Event: Introduction	Programmatic	Not complete	8/20/26	8/20/26	1	
1.C	Schedule Margin			6/1/25	8/1/25	62	
1.D	◆ Submission of Critical Design Review (CDR)		Not complete	8/4/25	8/4/25	1	
1.E	Schedule Margin			8/1/25	11/22/25	114	
1.F	◆ Submission of Production Readiness Review (PRR)		Not complete	11/1/25	11/1/25	1	
1.G	Schedule Margin			11/1/25	12/1/25	31	
1.H	◆ Submission of System Integration Review (SIR)		Not complete	12/1/25	12/1/25	1	
1.I	Schedule Margin			12/1/25	2/1/26	63	
1.J	◆ Safety Review		Not complete	2/1/26	2/1/26	1	
1.K	◆ Key Decision Point D (KDP D)		Not complete	3/1/26	3/6/26	6	



Aeris Gantt Chart - Phase D: System Assembly, Integration, Test & Launch

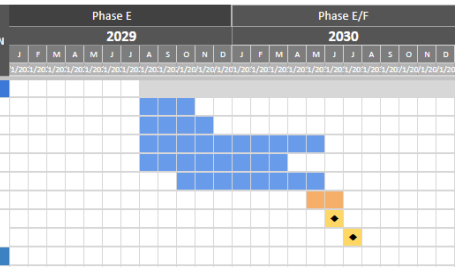
ID#	TASK	ASSIGNED TO	PROGRESS	START	END	DAYS	MARGIN
2	**System Assembly, Integration, Test & Launch**		0%	9/12/26	8/1/29	1055	1
2.1	Perform verification and validation on assemblies	Engineering / Technicians	Not complete	9/12/26	3/30/28	566	
2.2	Confirm telemetry validation and ground data processing	Engineering / Technicians	Not complete	10/30/27	3/2/28	125	
2.3	Instrumentation Testing & Payload Calibration	Engineering / Technicians	Not complete	9/12/26	8/1/28	690	
2.4	Perform planned on-orbit operational verification(s) and	Engineering / Technicians	Not complete	12/26/26	5/5/27	131	
2.5	Refine & Update CDR and SIR	Interdisciplinary	Not complete	9/12/26	2/12/27	153	
2.6	◆ System Integration Date		Not complete	10/1/28	10/1/28	1	
2.7	◆ Submission of System Acceptance Review (SAR)		Not complete	11/1/28	11/1/28	1	
2.8	◆ Standing Review Board		Not complete	12/1/28	12/1/28	1	
2.9	Schedule Margin		Not complete	1/1/29	2/1/29	32	
2.A	◆ Operation and Readiness Review (ORR)		Not complete	1/15/29	1/15/29	1	
2.B	◆ Standing Review Board		Not complete	2/1/29	2/1/29	1	
2.C	Schedule Margin		Not complete	2/1/29	2/15/29	14	
2.D	◆ Mission Readiness Review (MRR)		Not complete	2/15/29	2/15/29	1	
2.E	Schedule Margin		Not complete	2/1/29	3/1/29	29	
2.F	Travel to Cape Canaveral, Florida		Not complete	2/26/29	3/3/29	6	
2.G	◆ Launch		Not complete	3/1/29	3/1/29	1	
2.H	Outreach Event: STEM Engagment	Programmatic	Not complete	5/5/29	5/31/29	27	
2.I	Schedule Margin		Not complete	4/1/29	8/1/29	123	




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Aeris Gantt Chart - Phase E: Operations & Sustainment

ID#	TASK	ASSIGNED TO	PROGRESS	START	END	DAYS	MARGIN
3	**Operations and Sustainment**		0%	8/1/29	7/1/30	335	32
3.1	Conduct launch vehicle performance assessment	Engineering / Technicians	Not complete	8/1/29	10/1/29	62	
3.2	Prepare for deactivation, disassembly, decommissioning	Engineering / Technicians	Not complete	8/1/29	11/1/29	93	
3.3	Document Lessons Learned	Interdisciplinary	Not complete	8/1/29	5/1/30	274	
3.4	Perform analyses of the returned data	Scientists	Not complete	8/1/29	3/1/30	213	
3.3	Develop final mission report	Interdisciplinary	Not complete	10/1/29	5/1/30	213	
3.6	Schedule Margin			5/1/30	6/1/30	32	
3.8	◆ Critical Events Readiness Review (CERR)		Not complete	6/1/30	6/1/30		
3.9	◆ KDP E		Not complete	7/1/30	7/1/30	1	
4	**Task group 4**		0%	4/10/23	10/5/23	179	28



Aeris Gantt Chart - Phase F: Closeout

4	**Closeout**		0%	8/1/30	11/1/31	458	116
4.1	Archive data	Scientists	Not complete	8/1/30	10/1/30	62	
4.2	Dispose of the system and supporting processes	Engineering / Technicians	Not complete	10/1/30	2/1/2031	124	
4.3	Outreach Event	Programmatic	Not complete	2/1/31	5/1/31	90	
4.4	Outreach Event: STEM Engagment	Programmatic	Not complete	6/1/31	6/8/2031	8	
4.5	Schedule Margin			6/8/2031	10/1/31	116	
4.6	◆ Disposal Readiness Review		Not complete	11/1/31	11/1/31	1	

